IEEE Open Journal of
Intelligent Transportation
Systems

Received 15 December 2025; revised 5 February 2026; accepted 14 February 2026. Date of publication 17 February 2026; date of current version 23 February 2026.

Digital Object Identifier 10.1109/OJITS.2026.3665677

LLM and AI Agents for Autonomous Systems: A Survey of Applications, Datasets, and Security Challenges

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This work was supported in part by United Arab Emirates University through the Program for Advanced Research (UPAR) under Grant G00005016 and in part by the 6G Research Center (6GRC), Khalifa University of Science and Technology.

ABSTRACT The rapid integration of Large Language Models (LLMs) into autonomous systems marks a significant transition from modular, rule-based approaches to reasoning-driven, agent-based, and multimodal intelligence. LLM reasoning enables adaptive decision-making, context-aware planning, and human-aligned interaction, while AI agents extend these capabilities into structured autonomy pipelines that coordinate perception, reasoning, and control. These advancements are particularly critical in safety-sensitive domains such as autonomous driving (AD) and unmanned aerial vehicles (UAVs). This survey provides a comprehensive review of LLM reasoning and AI agents across scenario generation, decision-making, multimodal perception, cooperative V2X interactions, and UAV swarm autonomy. We examine the role of simulation platforms and datasets, including CARLA, Apollo ADS, AirSim, nuScenes, DriveLM, and emerging synthetic environments, in supporting reproducible evaluation and benchmarking. In addition, we analyze pressing security and robustness challenges, including adversarial prompt injection, data poisoning, multimodal perturbations, privacy leakage, and vulnerabilities in cooperative agent communication. Finally, we propose future research directions including adversarially robust pipelines, hybrid symbolic LLM planning, secure multimodal fusion, privacy-preserving human alignment, distributed trust mechanisms for swarm autonomy, and optimized Drone-LLM deployment across on-drone, edge, and cloud environments. By unifying applications, datasets, benchmarks, reasoning, agents, and security, this survey establishes a roadmap for developing robust, trustworthy, and secure LLM-enabled autonomous systems.

INDEX TERMS Large language models, AI agents, UAV communication, autonomous driving, benchmarks, security.

I. INTRODUCTION

THE rapid evolution of Large Language Models (LLMs) has transformed artificial intelligence by enabling systems to perform reasoning, problem-solving, and multimodal integration at unprecedented scales [1], [2]. Unlike earlier modular or rule-based approaches in autonomous systems, which often suffer from error propagation, rigidity, and poor generalization to novel situations [3], [4], LLMs offer context-aware reasoning, adaptability to unseen inputs, and interactive capabilities that align with human communication. These characteristics have made LLMs increasingly attractive for safety-critical domains such as autonomous driving (AD) and unmanned aerial vehicles (UAVs), where decision-making under uncertainty, robustness to corner cases and human–machine alignment remain open challenges [5], [6]. As a result, the integration of LLMs into autonomous systems has become a research frontier that requires both technological innovation and rigorous evaluation [7], [8].

A growing body of work explores the applications of LLMs in AD and UAV systems. In autonomous driving, LLMs have been applied to scenario generation [9], [10], interpretability

The review of this article was arranged by Associate Editor Junfeng Zhao.

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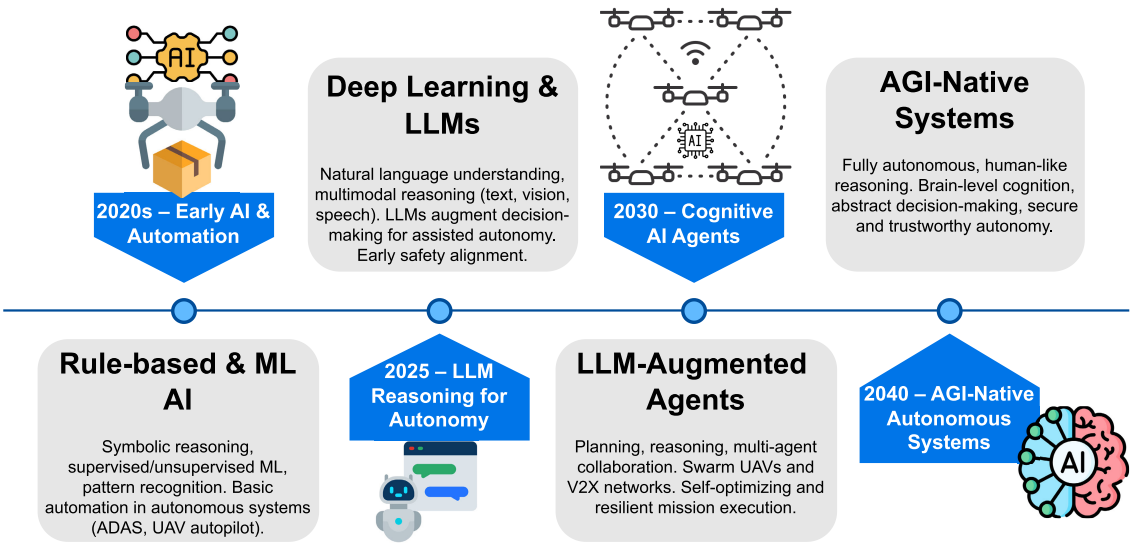


FIGURE 1. The evolution of AI, LLM reasoning, and AI agents in autonomous systems, from rule-based and machine learning approaches in the 2020s to AGI-native autonomous systems projected by 2040.

pipelines [11], planning and control [12], [13], and multimodal reasoning with integration of vision language [14], [15], [16]. Similarly, cooperative vehicle-to-vehicle (V2V) and vehicle-to-infrastructure (V2I) studies leverage LLMs for negotiation, fusion, and communication-aware optimization [17], [18], [19]. UAV research has followed parallel trajectories, with LLMs enabling swarm intelligence [20], [21], aerial routing and scheduling [22], [23], and mission-oriented autonomy under dynamic conditions [24], [25]. These applications demonstrate the potential of LLMs to unify perception, reasoning, and action in autonomous systems, bridging natural language intent with executable tasks [26].

Progress in this domain is inseparable from advances in simulation tools and datasets, which provide the backbone for training, benchmarking, and safety validation. Driving simulators such as CARLA [27], Apollo ADS [28], and InterDrive [18] have become central to testing LLM-driven control pipelines, while UAV evaluations often rely on AirSim [29], SITL [30], and wildfire monitoring testbeds [23]. Complementary datasets, including nuScenes [31], [32], Waymo Open [33], and targeted benchmarks such as STSBench [34], AD^2-Bench [35], DriveLM [36], UAV-ON [37], and BEDI [38], have enabled systematic evaluation of reasoning, perception, and planning. More recent platforms, such as DriveDreamer-2 [39], OmniDrive [40], and AeroVerse [41], extend the evaluation to synthetic environments that support counterfactual reasoning, embodied cognition, and agent-based navigation. Despite these advances, most resources remain simulation-heavy, lacking real-world interactivity, multi-agent negotiation, and robust transferability, which motivates the need for hybrid evaluation frameworks [42], [43], [44].

Figure 1 illustrates a clear progression in the development of intelligent autonomous systems. The 2020s mark the era of rule-based and machine learning methods, enabling basic automation in applications such as ADAS and UAV autopilots. By 2025, advances in deep learning and large language models (LLMs) will introduce natural language understanding and multimodal reasoning, supporting more sophisticated decision-making for assisted autonomy. The next milestone, around 2030, envisions LLM-augmented cognitive agents capable of multi-agent collaboration, swarm intelligence, and self-optimizing mission execution. Looking ahead to 2040, the trajectory culminates in AGI-native autonomous systems that integrate human-like reasoning, brain-level cognition, and secure, trustworthy autonomy. Therefore, hallucinations in reasoning pipelines may lead to misaligned control actions [50], [51], [52], while adversarial attacks on perception modules can undermine robustness in complex urban or aerial environments [53], [54]. UAVs face additional risks, such as GPS spoofing, energy–security trade-offs, and adversarial communication threats, in swarm deployments [46], [55], [56]. Autonomous driving systems must also ensure compliance with traffic regulations and ethical expectations to maintain safety and trustworthiness [57], [58]. Existing reviews often acknowledge these risks, but do not analyze strategies, particularly with respect to multimodal reasoning failures, secure communication, and resilience under adversarial conditions [6], [47].

A. KEY CONTRIBUTIONS AND NOVELTY

The main contributions of this study are:

- We provide the first comprehensive survey that jointly examines LLM reasoning models and AI

TABLE 1. Comparative analysis of related survey papers on LLMs for UAVs and autonomous driving.

Work	Year	Domain	Focus	Applications	Datasets / Benchmarks	Security Challenges	LLM Reasoning	AI Agents
Yang *et al.* [45]	2023	Driving	LLMs for Autonomous Driving; reasoning, open-world understanding	◐	◐	○	○	○
Javaid *et al.* [25]	2024	UAVs	LLM architectures and UAV applications for decision-making, sensing, emergency response	◐	◐	○	○	○
Cui *et al.* [5]	2024	Driving	Multimodal LLMs (MLLMs) for driving, transportation, and mapping	◐	◐	○	○	○
Fourati *et al.* [6]	2024	Driving	Cross-modal LLMs (XLMs) integrating sensory data and language	◐	◐	○	○	○
Chen *et al.* [2]	2025	UAVs	Empirical analysis of academic vs. industry integration; nine UAV task types	◐	◐	○	○	○
Yang *et al.* [46]	2025	UAVs	Safety and security threats in UAVs; role of LLMs in secure UAV operations	◐	◐	●	○	○
Zhu *et al.* [47]	2025	Driving	Comprehensive AI + LLM survey for AD; modular vs. end-to-end integration	◐	◐	○	○	○
Stefanidou *et al.* [48]	2025	Driving	Foundation models across the full perception–planning–control pipeline; PRISMA-based systematic review	◐	◐	◐	◐	○
Anuar *et al.* [49]	2026	Driving	Vision-based transformer architectures for distracted driving detection; multimodal spatiotemporal modeling	◐	◐	○	○	○
Our Work	2026	UAVs & Driving	Comprehensive survey of LLM reasoning and AI agents in autonomous systems; applications, benchmarks, and security	●	●	●	●	●

Not Considered = ○; Partial discussion = ◐; Considered = ●.

agent frameworks in autonomous systems, spanning autonomous driving and UAV domains. Unlike prior ITS-oriented surveys that focus on isolated tasks or single platforms, this work offers a unified, cross-domain perspective covering scenario generation, decision-making, multimodal perception, cooperative driving, and UAV swarm intelligence.

- We explicitly distinguish between reasoning-centric LLMs and agent-based autonomy, and analyze how LLM reasoning enhances high-level planning, multimodal integration, regulatory compliance, and human–machine alignment, while AI agents enable persistent goals, memory, tool use, closed-loop interaction, and adaptive control.
- We systematically review simulation platforms and benchmarks (e.g., CARLA, AirSim, Apollo ADS) together with modern datasets (e.g., nuScenes, DriveLM, UAV-ON, UAVBench), framing them as enablers of reasoning and agent evaluation.
- We present a pipeline-wide security and robustness analysis tailored to LLM-enabled autonomous systems, covering adversarial risks in scenario generation, hallucinations and vulnerabilities in reasoning-driven planning, multimodal perception attacks, privacy leakage in human alignment, and threats in cooperative driving and UAV swarm coordination—attack surfaces that are largely absent from traditional ITS security models.
- We outline a forward-looking research roadmap for trustworthy LLM reasoning and AI agents, identifying open challenges in secure reasoning validation, adversarially robust multimodal fusion, privacy-preserving human alignment, distributed trust mechanisms for swarm autonomy, and efficient Drone-LLM deployment across on-device, edge, and cloud environments.

B. Paper ORGANIZATION

Figure 2 provides a high-level overview of the survey structure, highlighting the guiding questions, the thematic scope, and the logical progression of the paper from motivation and related work to applications, evaluation platforms, security challenges, and concluding insights. Section II provides a comprehensive comparative analysis of existing survey papers that explore the role of LLMs in unmanned aerial vehicles (UAVs) and autonomous driving (AD). By positioning our work within this body of literature, we highlight both the overlaps and the distinctive gaps that motivate our study. Section III then reviews the applications of LLMs across AD and UAV systems, emphasizing their contributions to reasoning, decision-making, and adaptability in real-world scenarios. Section IV surveys the simulation platforms, datasets, and benchmarks that underpin LLM-driven autonomy, examining their unique strengths, limitations, and potential to support large-scale evaluations. In Section V, we turn to the security landscape, analyzing the vulnerabilities that arise when integrating LLMs into safety-critical domains and outlining promising avenues for future research. Finally, Section VI concludes the paper by summarizing the key insights, drawing attention to persistent challenges, and highlighting open problems that remain to be addressed.

II. RELATED SURVEYS

The integration of large language models (LLMs) into autonomous systems has attracted growing scholarly attention, particularly in unmanned aerial vehicle (UAV) and autonomous driving (AD) domains. Existing surveys vary significantly in scope, emphasis, and methodological depth, ranging from conceptual visions to technology-centric and safety-oriented analyses.

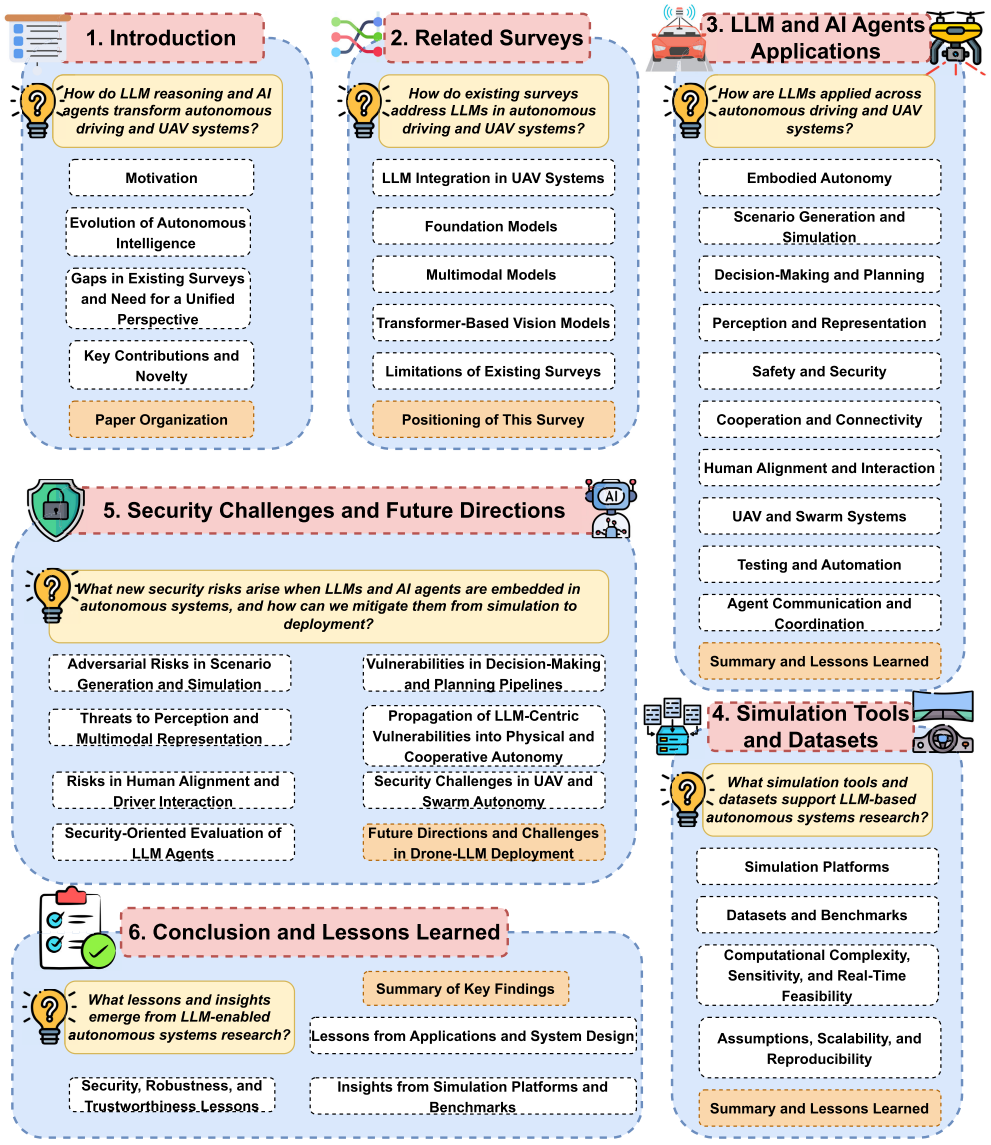


FIGURE 2. Overall structure of the survey and its guiding research questions.

A. LLM INTEGRATION IN UNMANNED AERIAL VEHICLE SYSTEMS

Early surveys on LLM-enabled UAV systems primarily focus on understanding how language models can enhance autonomy, decision-making, and mission adaptability. Chen et al. [2] conducted one of the first empirical studies bridging academic and industrial perspectives on LLM–UAV integration. By reviewing 74 academic papers and 56 open-source projects, the authors identified nine UAV task categories and revealed a clear mismatch between research priorities: academic work largely emphasizes theoretical modeling and isolated tasks, whereas industry prioritizes operational robustness, flight control, and efficiency. Their inclusion of a practitioner survey further highlighted real-world barriers, such as performance constraints, safety certification, and deployment costs, making this work particularly valuable for practical adoption.

From a more technology-driven viewpoint, Javaid et al. [25] examined the role of LLMs in enhancing autonomous data processing, spectral sensing, and decision-making capabilities in UAV systems. Their review emphasized potential applications in emergency response, infrastructure monitoring, and network restoration. However, the study remained largely conceptual, with limited discussion of system-level integration challenges or validation in real-world deployments. Complementing this perspective, Yang et al. [46] focused on safety and security threats in UAV operations. By reviewing traditional machine learning and deep learning approaches, the authors analyzed how LLMs could mitigate limitations in areas such as air traffic management,

agriculture, and emergency operations. While providing valuable insights into threat mitigation, their survey did not aim to provide a comprehensive overview of UAV–LLM applications.

B. FOUNDATION MODELS IN AUTONOMOUS DRIVING

Beyond UAVs, a substantial body of surveys has emerged around the adoption of LLMs and foundation models (FMs) in autonomous driving systems. Stefanidou et al. [48] presented a comprehensive and systematic review of foundation models across the full perception–planning–control pipeline. Following a PRISMA-inspired methodology, the authors screened 446 records from IEEE Xplore, Scopus, and arXiv, ultimately selecting 100 FM-centric studies published between 2022 and 2025. Their review classified over 70 representative models according to architecture, modality, task, and input–output structure, grouping them into vision foundation models, large language foundation models, and diffusion-based foundation models. Quantitative analysis showed that most applications focus on perception and scene understanding (40.2%), followed by reasoning and decision-making (33.3%), dataset generation (14.9%), and motion prediction and planning (11.5%). The survey also highlighted critical challenges related to computational cost, real-time latency constraints, robustness under adverse conditions, interpretability, and compliance with functional safety standards such as ISO 26262.

Focusing specifically on LLMs, Yang et al. [45] introduced one of the earliest systematic surveys on LLMs for autonomous driving (LLM4AD). The authors argued that LLMs, when integrated with vision foundation models, can overcome the limitations of rule-based and end-to-end driving systems by enabling reasoning, open-world understanding, and few-shot adaptation. Although the survey clearly articulated key challenges, it remained primarily conceptual. Similarly, Zhu et al. [47] provided a broader synthesis of artificial intelligence in autonomous driving, positioning LLMs as a promising solution to long-tail scenarios and extending the discussion toward artificial general intelligence (AGI) and societal concerns such as safety and security. However, this work was largely forward-looking and lacked empirical grounding.

C. MULTIMODAL MODELS FOR DRIVING AND MAPPING

Several surveys have further narrowed the focus to multimodal and cross-modal large models for driving and mapping applications. Cui et al. [5] reviewed multimodal large models for autonomous driving, detailing their technical foundations, available tools, datasets, benchmarks, and outcomes from the first LLVM-AD workshop. While the survey offered a structured overview of multimodal learning opportunities, it provided limited analysis of robustness, security, or failure modes. Along similar lines, Fourati et al. [6] explored cross-modal LLMs (XLMs), emphasizing their ability to fuse language with heterogeneous sensory data such as LiDAR and radar. Their work proposed a detailed taxonomy of architectures and deployment frameworks, but focused primarily on feasibility rather than systematic evaluation or security considerations.

D. TRANSFORMER-BASED VISION MODELS FOR DRIVING SAFETY AND BEHAVIOR ANALYSIS

In parallel with LLM-centric surveys, transformer-based vision architectures have been extensively studied for driving safety applications. Anuar et al. [49] provided a comprehensive and systematic review of vision-based transformer models for distracted driving behavior detection. Using a structured literature review methodology, the authors screened 96,291 publications from IEEE Xplore, Web of Science, Scopus, and arXiv (2020–2025), narrowing them to 168 relevant studies. Their analysis categorized transformer-based solutions into image-based, video-based, lightweight and efficient architectures, pose- and keypoint-aware transformers, and hybrid feature-fusion frameworks. The survey also examined multimodal and multi-view fusion approaches integrating RGB, infrared, depth, skeletal, and physiological signals, highlighting their effectiveness under real-world driving conditions. Benchmark datasets and comparative evaluations were discussed, along with evidence that distracted driving behaviors account for approximately 80.60% of road accidents, underscoring the safety relevance of this research.

E. LIMITATIONS OF EXISTING SURVEYS AND MOTIVATION FOR A UNIFIED PERSPECTIVE

Collectively, these surveys demonstrate the rapid convergence of LLMs, foundation models, and transformers with UAV and autonomous driving systems, addressing technological opportunities, industrial adoption, multimodality, and long-term AGI-oriented visions. However, as summarized in Table 1, most existing works remain domain-specific and emphasize either conceptual opportunities or isolated technical components. The majority of prior surveys treat LLMs as modular or task-specific elements — such as perception, planning, or mapping — without explicitly addressing the shift toward reasoning-centric models and agent-based autonomy.

F. POSITIONING OF THIS SURVEY

In contrast, our survey provides a comprehensive, unified perspective across both the UAV and autonomous driving domains. We systematically categorize applications, datasets, and benchmarks, while placing particular emphasis on security and robustness challenges that have been acknowledged but not thoroughly explored in prior work. By explicitly distinguishing between LLM-based reasoning and AI agent frameworks, and by analyzing their interaction with simulation platforms, datasets, and security mechanisms, we demonstrate why existing ITS-oriented architectures and surveys cannot be trivially extended to support embodied, closed-loop, and multi-agent autonomous systems. Importantly, our work is the only survey that jointly addresses

applications, datasets, benchmarks, security challenges, LLM reasoning, and AI agents, thereby highlighting its novelty and contribution to the field.

III. LLM AND AI AGENTS APPLICATIONS IN AUTONOMOUS SYSTEMS

LLMs are increasingly integrated into autonomous systems, enabling advances that span scenario generation, decision-making, safety validation, cooperative driving, human alignment, and UAV swarm coordination. Using their reasoning abilities, adaptability, and interactive potential, LLMs extend traditional pipelines beyond perception and control to include explanation, negotiation, and resilience against adversarial risks. This section presents a comprehensive review of these applications, organized into thematic areas that illustrate how LLMs enhance the intelligence and reliability of autonomous systems across various domains. Figure 4 illustrates how LLMs and AI agents can enhance the functions of the Advanced Driver Assistance System (ADAS). Using inputs from various sensors such as radar, LIDAR, cameras, and ultrasound, the AI agent connects the planning, memory, and act/observe modules with external tools. This integration enables advanced features, including adaptive cruise control, emergency braking, pedestrian detection, collision avoidance, lane departure warning, and traffic sign recognition, ultimately improving safety, situational awareness, and autonomous decision-making. At the same time, integrating LLMs into autonomous systems introduces critical safety and security challenges. Figure 3 presents the taxonomy of LLM Reasoning and AI Agent frameworks for autonomous systems applications.

A. EMBODIED AUTONOMY IN DRIVING AND UAV SYSTEMS

Table 2 presents a comparative analysis of LLM reasoning and LLM agents across multiple autonomy dimensions in driving and UAV systems. The table highlights how reasoning-oriented LLMs excel in semantic understanding, descriptive analysis, and contextual interpretation but remain constrained by the absence of persistent goals, memory, and actuation capabilities. In contrast, agentic frameworks often integrated within simulators such as CARLA [9], [12], [27], Apollo ADS [10], [28], AirSim [20], [29], and SITL [21], [30] extend reasoning into executable autonomy through closed-loop control, real-time data fusion, and dynamic tool use. While reasoning LLMs provide low-latency, text-based insights suitable for explanation and planning [11], [35], agent systems demonstrate adaptive behaviors such as live trajectory optimization, swarm coordination, and cooperative goal refinement [2], [22], [24]. However, these enhanced capabilities come at the expense of higher computational cost, latency, and reliability concerns stemming from multi-module integration [20], [64]. Overall, the table highlights the complementary nature of both paradigms: LLM reasoning facilitates interpretability and semantic grounding, while LLM agents bridge perception, planning, and execution to achieve embodied intelligence in complex, safety-critical environments.

B. SCENARIO GENERATION AND SIMULATION

Table 3 presents a comparative review of scenario generation and simulation frameworks leveraging LLMs. Fu [9] and Tian [10] highlight reasoning-driven pipelines for CARLA and Apollo ADS, with partial agent integration. Mei's LLM-Attacker [59] fully combines reasoning and agents to generate adversarial multi-agent cases. Peng [42] emphasizes reasoning with partial agent coverage through diffusion-based adversarial scenarios, while Petrovic [11] focuses on automated test generation, relying exclusively on reasoning without agents. Collectively, the table shows that reasoning is universally embedded, while agent integration is strongest in adversarial and multi-agent setups.

1) DRIVING SCENARIOS

We categorize existing LLM-based methods for driving scenario generation into four main groups: CARLA-driven frameworks, evolutionary search methods, adversarial scenario design, and diffusion-based adversarial generation. Each category highlights different techniques for creating diverse, realistic, and safety-critical scenarios to evaluate autonomous driving systems.

a: CARLA SCENARIO GENERATION

Fu et al. [9] propose a framework that utilizes LLMs to generate code with few shots, thereby automatically synthesizing various and critical driving scenarios within the CARLA simulator [27]. Utilizing the simulator's flexibility in scripting and enforcement of physical dynamics, the system generates scenario scripts that control the behavior and placement of traffic participants, with a particular emphasis on collision-prone situations. The framework extends beyond simulation by incorporating a video generation pipeline that combines Cosmos-Transfer1 with ControlNet, transforming rendered scenes into realistic driving videos and thereby narrowing the gap between synthetic environments and real-world appearances. This approach enables the creation of rare yet critical edge cases, such as pedestrians crossing under occlusion or sudden vehicle cut-ins, which are essential for evaluating the robustness of autonomous driving systems. The main advantages of this work are its ability to automate scenario generation, its integration of realism-enhancing video pipelines, and its focus on safety-critical events, while its limitations include dependence on CARLA-specific infrastructure, potential computational overhead from video synthesis, and a lack of large-scale validation across different simulation platforms.

b: EVOLUTIONARY SCENARIO SEARCH

Tian et al. [10] propose LEADE, an LLM-enhanced framework for generating diverse and safety-critical scenarios to test autonomous driving systems (ADSs). The framework

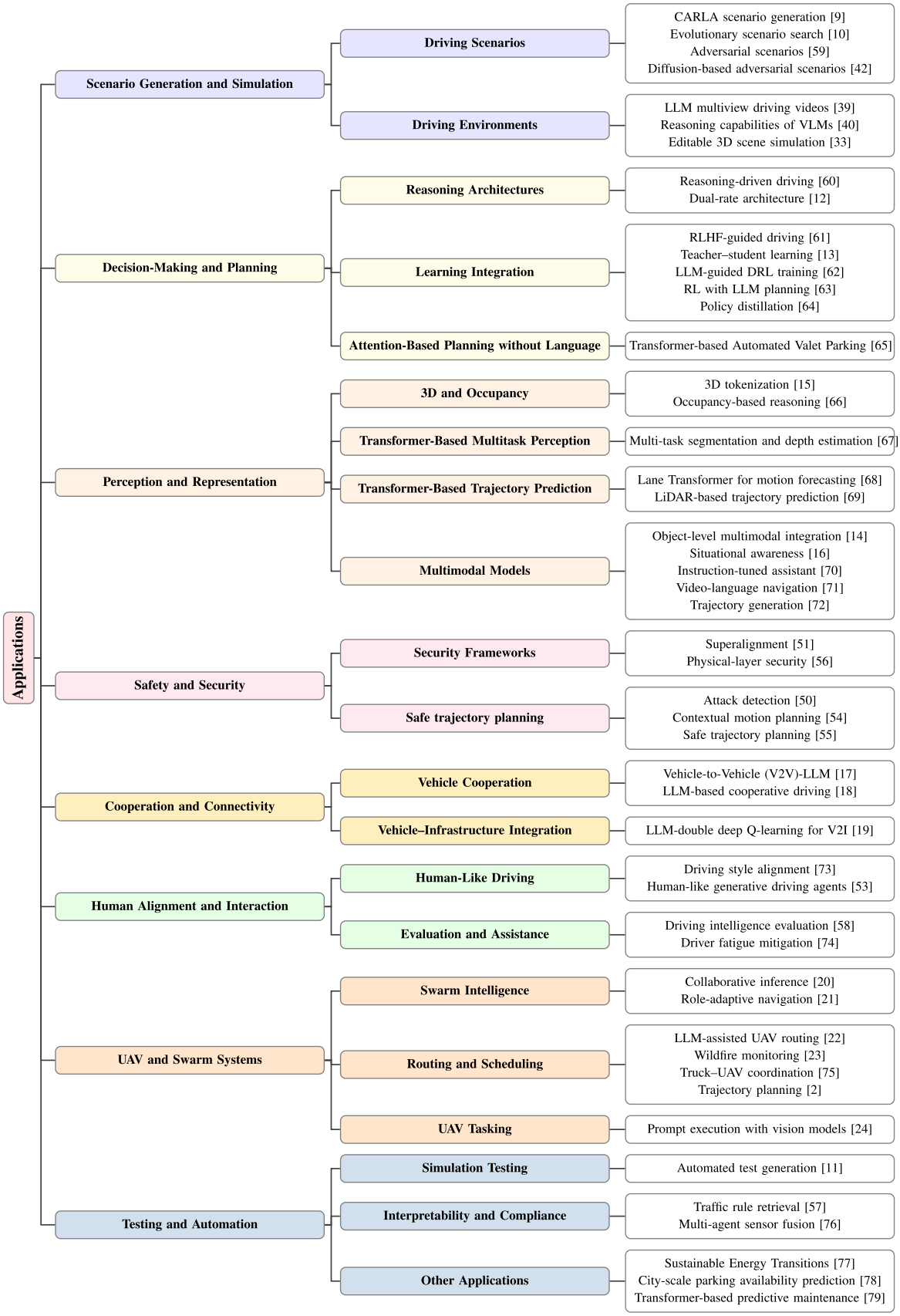


FIGURE 3. LLM reasoning and AI agents frameworks for autonomous systems applications.

TABLE 2. LLM reasoning vs. LLM agents in autonomous systems (Driving & UAV).

Dimension	LLM Reasoning	LLM Agents	Driving Example	UAV Example
Autonomy	○	●	The LLM interprets static traffic scenes and explains potential hazards but lacks real-time execution [9], [11], [12], [27], [28].	The model can answer mission or route queries but cannot autonomously adapt flight plans; agent-driven systems like AirSim or SITL demonstrate greater autonomy [20], [21], [24], [29], [30].
Goal Handling	◐	●	The LLM can produce routes or high-level goals when prompted, but only agents continuously monitor progress and refine goals dynamically [12], [13], [27], [35], [63].	UAV agents adapt waypoints in changing conditions and preserve objectives under constraints (e.g., energy or AoI) [2], [20]–[23].
Planning	◐	●	Generates static textual plans (e.g., overtaking, merging) without feedback; integrated agent frameworks link reasoning to trajectory planning and control loops [12], [39], [40], [42], [63].	UAV agents update flight paths in real time using wind, terrain, and environmental data, ensuring safe and efficient routing [2], [20], [24], [30], [55].
Action Execution	○	●	LLMs describe actions but cannot implement them; agentic systems execute maneuvers through APIs such as CARLA or Apollo ADS [9]–[11], [28], [64].	LLM-based UAV agents perform arm–takeoff–navigate tasks within AirSim or SITL, using ROS2 or feedback control for stability [20], [21], [24], [29], [30].
Tool Use (APIs)	◐	●	Reasoning models may call RAG or map APIs for contextual understanding; agents autonomously use CARLA or V2V-LLM APIs for continuous decision updates [10], [17], [18], [39], [40].	UAV agents exploit GNSS, weather, and swarm communication APIs to modify flight behavior and ensure mission safety [2], [20], [22], [24], [30].
Memory	○	●	Reasoning LLMs retain only conversational context, losing prior episodes; agents persist driving logs and environmental traces [13], [35], [36], [60], [61].	Persistent mission memory allows UAV agents to recall previous flights and environmental states for continual learning [20], [21], [23], [24].
Real-time Data	○	●	Static reasoning cannot adapt to sensor updates; agent systems fuse live LiDAR, camera, and radar data for responsive planning [33], [39], [40], [42], [76].	UAV agents utilize real-time telemetry to avoid no-fly zones and adjust altitude in turbulence [2], [20], [21], [24], [30].
Perception Support	◐	●	LLMs interpret textual scene descriptions but cannot refine detections; agents integrate perception pipelines (SLAM, VIO, CoT) [15], [32], [34], [36], [66].	UAV agents align visual–inertial odometry with semantic perception for accurate localization and reduced drift [21], [24], [37], [38], [41].
Safety Guardrails	◐	●	Reasoning models propose safety suggestions but depend on human oversight; agents enforce collision detection and rule-based safety through simulators [43], [50], [54], [57], [80].	UAV agents apply geo-fences, altitude limits, and emergency protocols under fault conditions [20], [22], [51], [55], [56].
Reliability	●	◐	LLM-only reasoning yields deterministic outputs; reliability improves via dataset-driven validation (e.g., nuScenes, Waymo) [11], [33], [39], [42].	Agents involve complex execution chains subject to latency and API errors; reliability enhanced by redundancy and safety checks [20], [21], [24], [30].
Latency/Cost	●	◐	Text-based reasoning is low-cost and fast to infer; simulation-driven agent loops increase compute and energy requirements [9], [11], [12], [27], [28].	UAV agent deployments incur real-time communication and computation overheads; lightweight or edge-assisted frameworks mitigate cost [2], [20]–[22], [24].

Symbols: ●= fully; ◐= partial; ○= none. **Abbr.:** CoT = Chain-of-Thought; RAG = Retrieval-Augmented Generation; VIO = Visual–Inertial Odometry; SLAM = Simultaneous Localization and Mapping; ROS2 = Robot Operating System 2; GNSS = Global Navigation Satellite System; CARLA/AirSim = simulation platforms.

combines LLM-based scenario understanding with an adaptive multiobjective genetic algorithm, reformulating the search process as a question–answer task in which LLMs provide quantitative reasoning to guide differential seeding. Implemented on Baidu Apollo [28], LEADE efficiently uncovers 10 distinct safety violations, four more than state-of-the-art search-based testing methods, demonstrating superior diversity and effectiveness. Despite its strengths, the approach still faces challenges related to high computational costs, reliance on LLM reasoning quality, and limited validation beyond the Apollo platform [81].

c: ADVERSARIAL SCENARIOS

Mei et al. [59] introduce LLM-attacker, a closed-loop adversarial scenario generation framework designed to enhance the safety and robustness of autonomous driving systems (ADS) by using LLMs. Unlike prior approaches that struggle with identifying the most effective adversarial participants in complex multi-agent environments, this framework coordinates multiple LLM agents to select optimal attackers and optimize their trajectories to induce safety-critical events. A key innovation is the iterative refinement process, where generated scenarios are continuously adapted based on ADS performance, creating a feedback loop that strengthens model robustness. Experimental results demonstrate that the LLM attacker generates more hazardous scenarios than existing methods, and that ADS trained on these scenarios achieves a collision rate that is half that of ADS trained on normal scenarios. The advantages of this work include its novel multi-agent language model design for attacker identification, its feedback-driven refinement mechanism, and its demonstrated effectiveness in improving ADS safety. Nevertheless, this work's limitations include reliance on simulation environments, potential computational overhead from iterative scenario generation, and the need for further validation under real-world driving conditions.

d: DIFFUSION-BASED ADVERSARIAL SCENARIOS

Peng et al. [42] introduce LD-Scene, a framework that integrates LLMs with latent diffusion models (LDMs) to generate user-controllable adversarial driving scenarios via natural language. By translating user queries into adversarial

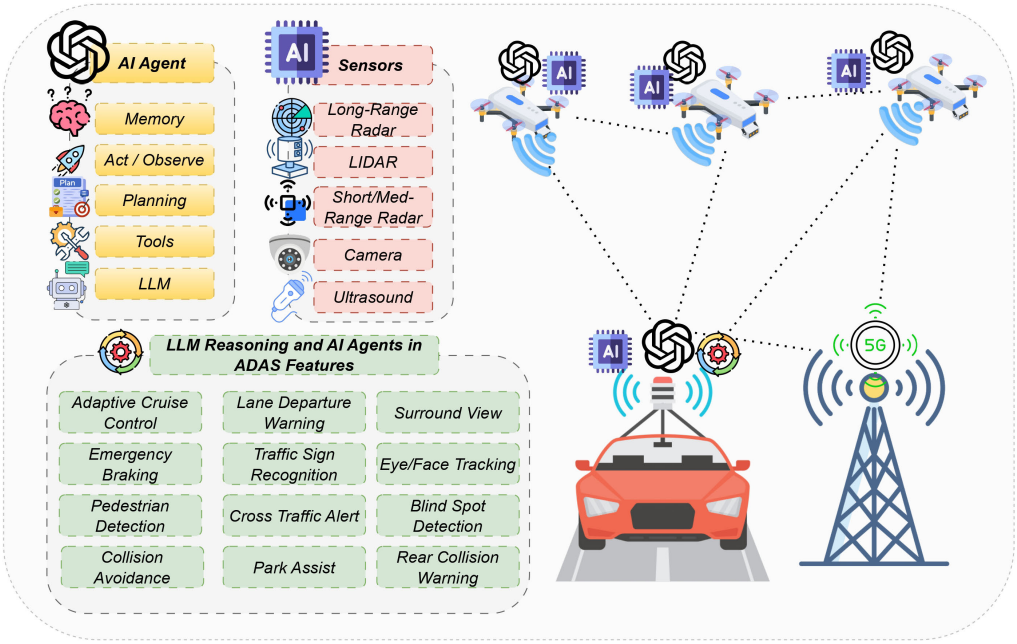


FIGURE 4. LLM reasoning and AI agents for ADAS integration.

TABLE 3. Comparative review of scenario generation and simulation methods leveraging LLMs.

Ref.	Year	Contribution Focus	LLM Mechanism	Platform / Dataset	Key Merits	Limitations	Quantitative Evaluation	LLM Reasoning	LLM Agents
Fu *et al.* [9]	2025	Automated CARLA scenario synthesis	CoT code generation + video realism	CARLA + Cosmos-Transfer1 + ControlNet	Automates critical edge-case generation; enhances realism	CARLA-specific; high compute; limited cross-platform	Edge-case coverage (qualitative); no aggregate metrics	●	◐
Tian *et al.* [10]	2024	LEADE: evolutionary search for diverse safety-critical scenarios	QA-guided seed generation; LLM reasoning	Baidu Apollo (industrial ADS) [28]	Exposes **10** safety violations in Apollo; uncovers **+4** new cases vs. SOTA	Computationally intensive; dependent on reasoning quality; single-platform	**10** safety violations (↑); **+4** novel cases (↑ diversity)	●	◐
Mei *et al.* [59]	2025	LLM-attacker: adversarial multi-agent scenarios	Multi-agent attacker coordination; closed-loop refinement	CARLA-like simulators	Generates stronger adversarial cases; improves ADS robustness	Simulation-only; iterative overhead; real-world validation lacking	ADS collision rate reduced by **50%** (↓)	●	●
Peng *et al.* [42]	2025	LD-Scene: diffusion-based adversarial scenario generation	LDM + LLM (CoT generator + debugger)	nuScenes dataset [31]	Fine-grained controllability; diverse, realistic adversarial scenarios	Synthetic-only setups; high compute; no closed-loop validation	SOTA realism/diversity (qual.); no numeric closed-loop scores	●	◐
Petrovic *et al.* [11]	2024	Automated test generation for ADS	Free-form test synthesis; code generation	CARLA (GPT-4, Llama3)	**10**× faster testing; reduced manual effort	Simulation-based only; possible LLM biases; no real-world validation	**10**× speedup; GPT-4 consistently > Llama3	●	○

Symbols: ●= fully covered, ◐= partially covered, ○= not covered.

loss functions, LD-Scene aligns scenario generation with user intent while leveraging an LDM to model realistic trajectory distributions. Its LLM-based guidance module, featuring chain-of-thought (CoT) code generation and debugging, enhances controllability and robustness. Evaluated on the nuScenes dataset [31], LD-Scene achieves state-of-the-art diversity and realism in adversarial scenarios. However, it remains limited by dependence on synthetic setups, computational overhead from diffusion models, and the lack of large-scale real-world validation.

2) DRIVING ENVIRONMENTS

We group LLM-driven driving environment frameworks into three main directions: (i) multiview video generation systems that create customized driving scenarios, (ii) vision-language reasoning frameworks that extend perception and counterfactual analysis to 3D tasks, and (iii) editable 3D scene simulation tools that enable natural language–based interaction and photorealistic rendering. These approaches collectively enhance the realism, depth of reasoning, and interactivity of simulated driving environments. Table 4 presents driving environments enriched by LLMs. Zhao's DriveDreamer-2 [39] and Wang's OmniDrive [40] emphasize reasoning for video synthesis and counterfactual reasoning, with only partial agent involvement. Wei's ChatSim [33] employs agent collaboration for scene editing, fully integrating both reasoning and agents. Yang's Traj-LLM [72] focuses exclusively on reasoning for text-to-trajectory generation without agent roles. Huang's DriVLMe [71] adopts a video–language agent, fully combining reasoning and agent dialogue.

a: LLM MULTIVIEW DRIVING VIDEOS

Zhao et al. [39] introduce DriveDreamer-2, a model framework that integrates LLMs to enable the generation of customized multiview driving videos for autonomous driving research. Addressing the limitations of existing world models, which struggle with user-defined customization, the proposed system incorporates a trajectory-generation function library to produce motion paths aligned with user descriptions, alongside an HDMap generator that translates trajectories into realistic road structures. To further enhance realism, the framework utilizes the Unified Multi-View Model (UniMVM), ensuring both spatial and temporal coherence across video outputs. DriveDreamer-2 is distinguished as the first world model capable of producing user-defined and uncommon driving scenarios, such as abrupt vehicle cut-ins, in a user-friendly way. The experimental evaluation demonstrates that the system not only enhances perception training tasks, such as 3D detection and tracking, but also outperforms state-of-the-art methods in video synthesis, achieving FID and FVD scores of 11.2 and 55.7, respectively, which reflect improvements of approximately 30% and 50%. The advantages of this work include its novel customization capability, strong performance gains in both video realism and utility for perception training, and its ability to generate rare driving events, while limitations concern the computational intensity of generating high-resolution multiview videos, reliance on controlled datasets for evaluation, and the need for further validation in large-scale or real-time applications.

b: REASONING CAPABILITIES OF VISION-LANGUAGE MODELS

Wang et al. [40] present OmniDrive, a holistic vision-language dataset and framework designed to extend the reasoning capabilities of vision-language models (VLMs) from 2D to 3D, thereby addressing a key limitation in current autonomous driving research. By incorporating counterfactual reasoning, OmniDrive enables models to evaluate potential scenarios and their outcomes, mirroring the way human drivers consider alternative courses of action. The framework introduces a synthetic data annotation process that generates large-scale, high-quality datasets with dense supervision signals, effectively linking planning trajectories with language-based reasoning. To further assess the interplay between vision-language alignment and 3D perception, the study explores two agent frameworks, Omni-L and Omni-Q, which provide valuable insights into designing effective LLM-based driving agents. Experimental results demonstrate substantial improvements on the DriveLM Q&A benchmark and nuScenes open-loop planning tasks [31], validating the utility of both the dataset and the proposed methods. The advantages of this work include its novel use of counterfactual reasoning for 3D driving tasks, the creation of a large-scale dataset with dense annotations, and its demonstrated performance gains on established benchmarks. However, its limitations include reliance on synthetic data that may not fully capture real-world variability, potential computational demands of large-scale counterfactual reasoning, and the need for further validation in closed-loop real-world driving environments.

Figure 5 illustrates the architecture of LLM-driven autonomous driving benchmarks, adopted by the OmniDrive framework. The design integrates diverse inputs, such as ego states, 3D objects, map features, and multi-view images, along with constraints such as traffic signals, drivable regions, and collision checks. These components are processed through LLM-based reasoning to produce predicted and ground-truth trajectories, as well as question–answer pairs that contribute to dataset construction. The framework enables the evaluation of critical capabilities, including scene understanding, traffic law awareness, 3D environment alignment, counterfactual reasoning, and path-planning decision-making.

c: EDITABLE 3D SCENE SIMULATION

Wei et al. [33] propose ChatSim, a novel system for generating editable photorealistic 3D driving scene simulations through natural language interaction, addressing the inefficiencies and limitations of prior scene simulation approaches. Unlike existing methods that struggle with user interaction efficiency, multi-camera rendering, and the integration of external assets, ChatSim introduces an LLM-based agent collaboration framework that enables flexible editing via natural language commands. The system further advances simulation quality by employing a multi-camera neural radiance field method for photo-realistic rendering and a multi-camera lighting estimation approach to ensure consistent integration of external digital assets. Experimental validation on the Waymo Open Dataset [82] demonstrates that ChatSim can

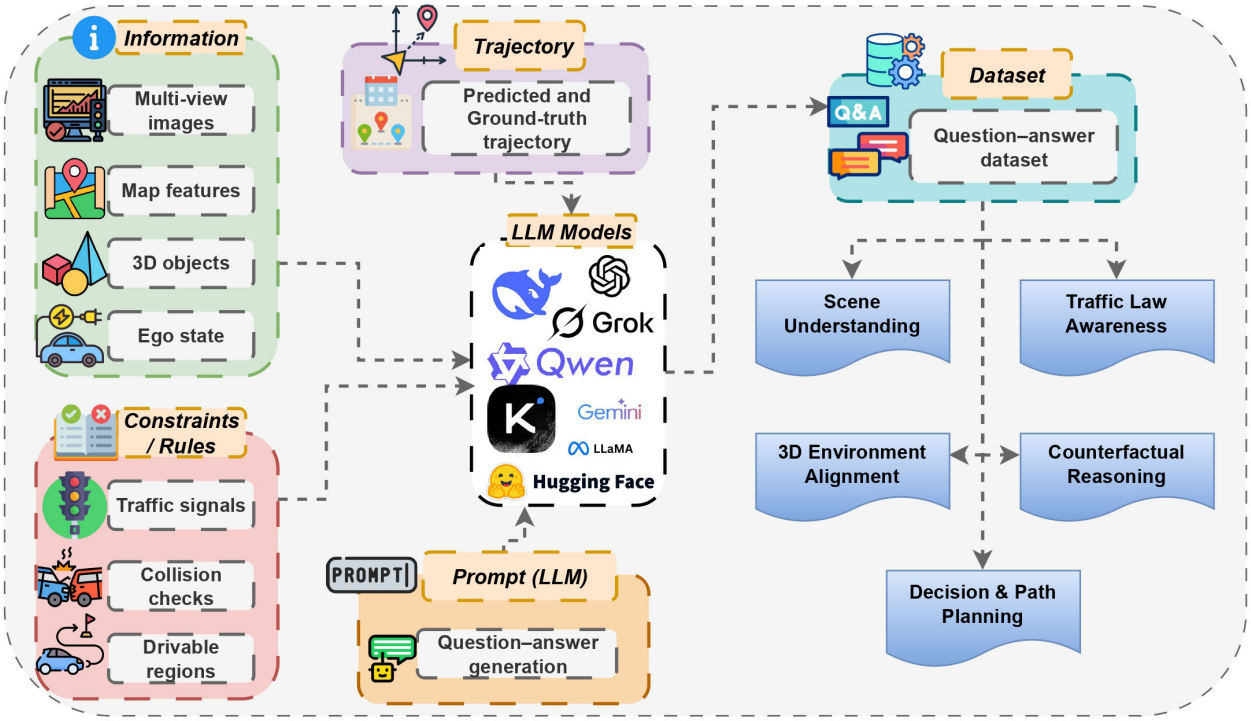


FIGURE 5. Architecture of LLM-powered autonomous driving benchmarks.

accurately interpret complex commands and generate coherent, photorealistic video sequences. The main advantages of this work lie in its natural-language-driven editing capabilities, high-quality rendering, and seamless asset integration, while its limitations include the computational intensity of neural rendering, potential latency in real-time applications, and reliance on pre-existing datasets for evaluation.

C. DECISION-MAKING AND PLANNING

In this subsection, we review decision-making and planning approaches for autonomous driving that employ attention-based models, large language models (LLMs), and hybrid learning frameworks. These methods aim to enhance interpretability, adaptability, and robustness in complex and safety-critical driving scenarios. We distinguish between three main categories: (i) reasoning architectures, which embed LLMs directly into planning pipelines to provide high-level semantic understanding and commonsense reasoning, (ii) learning integration frameworks, which combine LLMs with reinforcement learning to improve efficiency and safety alignment, and (iii) attention-based planning models without explicit language grounding, which serve as important precursors to LLM-driven planners.

Table 5 presents architectures that embed reasoning at the core of autonomous driving. Fu [60] and Zhang's LeAD [12] adopt reasoning-driven integration in custom and CARLA-based environments, with partial agent coverage. Sun [61] uniquely integrates human-in-loop agents, achieving full reasoning–agent synergy. Xu's TeLL-Drive [13] and LEAD [62], as well as Li's hierarchical RL [63], rely on reasoning with partial agent intervention. Wu's RAPID [64] leverages reasoning-only distillation without agents.

1) REASONING ARCHITECTURES

We classify reasoning-centric approaches that reframe autonomous driving as a reasoning-driven task. These include closed-loop reasoning systems and dual-rate architectures that combine high-frequency perception with low-frequency LLM reasoning to enhance decision-making in complex and edge-case scenarios.

a: REASONING-DRIVEN DRIVING

Fu et al. [60] present an innovative study on the use of LLMs for autonomous driving with the aim of achieving human-like comprehension of complex environments. Unlike traditional modular and optimization-based systems, which often fail in rare corner cases, the proposed approach emphasizes reasoning, interpretation, and memorization as core abilities for robust performance. By integrating an LLM into a closed-loop system, the study demonstrates its ability to interact with the driving environment and to solve atypical scenarios that challenge conventional methods. The results highlight the adaptability of LLMs in reasoning through complex cases, providing valuable insights for the development of human-like autonomous driving systems. The advantages of this work include its pioneering reframing of autonomous driving (AD) as a reasoning-driven task and its demonstration of common-sense capabilities, while limitations remain in terms

TABLE 4. Comparative analysis of LLM-enabled driving environments.

Ref.	Year	Contribution Focus	LLM Mechanism	Platform / Dataset	Key Merits	Limitations	Quantitative Evaluation	LLM Reasoning	LLM Agents
Zhao *et al.* [39]	2025	DriveDreamer-2: user-defined multi-view driving video synthesis	Trajectory library + HDMap generator; LLM-guided scene-to-video	UniMVM + custom data	Generates rare events; boosts perception training utility; spatial–temporal coherence	Heavy compute; dataset dependency; unclear real-time feasibility	FID **11.2** (↓), FVD **55.7** (↓); **∼30%/50%** improvements vs. baselines	●	◐
Wang *et al.* [40]	2024	OmniDrive: 3D counterfactual reasoning dataset + framework	LLM-enabled counterfactual QA and planning reasoning	DriveLM Q&A; nuScenes (open-loop) [31]	Extends 2D→3D reasoning; dense synthetic annotations	Synthetic-heavy; compute-intensive; limited closed-loop validation	Substantial gains on DriveLM Q&A and nuScenes planning (↑); exact figures not specified	●	◐
Wei *et al.* [33]	2024	ChatSim: editable photorealistic 3D scenes via natural-language commands	LLM agent collaboration for scene editing	Waymo Open Dataset [82]	Natural-language editing; multi-camera NeRF rendering; external asset integration	Computationally expensive; potential latency; reliance on pre-existing datasets	High visual fidelity/coherence (qual.); no aggregate numeric metrics reported	◐	●
Yang *et al.* [72]	2025	Traj-LLM: text-to-trajectory generation of interactive vehicles	LLM maps interaction text → trajectories (L2T)	L2T dataset (240K pairs)	Large-scale synthetic pairs; improves downstream trajectory predictors	Synthetic bias risk; real-world generalization not fully established	Consistent improvements across public prediction benchmarks (↑); exact numbers not provided	●	○
Huang *et al.* [71]	2024	DriVLMe: video–language agent for navigation and dialogue	LLM fused with embodied + human dialogue experience	Open-loop benchmarks + closed-loop human studies	Better long-horizon navigation; more natural human interaction	High inference latency; limited multi-turn dialogue capability; imbalanced training data	Competitive in both open/closed-loop (↑); supported by human studies; no single aggregate metric	●	●

Symbols: ●= fully covered, ◐= partially covered, ○= not covered.

of scalability, computational cost, and the need for validation beyond closed-loop experiments.

b: DUAL-RATE ARCHITECTURE

Zhang et al. [12] propose LeAD, a dual-rate autonomous driving architecture that integrates imitation learning-based end-to-end (E2E) frameworks with LLM augmentation to better address complex urban scenarios and edge cases. In this design, the high-frequency E2E subsystem ensures real-time perception, planning, and control, while the low-frequency LLM module enhances situational understanding through multi-modal perception fusion with HD maps and employs chain-of-thought (CoT) reasoning to generate more human-like decisions when baseline planners are insufficient. Experimental results in the CARLA simulator [27] demonstrate that LeAD outperforms conventional approaches, achieving 71 points on the Leaderboard V1 benchmark with a route completion rate of 93%, highlighting its robustness in unconventional driving contexts. The advantages of this work include its innovative dual-rate architecture, effective integration of CoT reasoning for improved decision-making, and strong empirical performance on established benchmarks. Its limitations, however, include reliance on simulation-based validation, potential latency introduced by low-frequency reasoning, and the need for further testing in large-scale real-world deployments.

2) LEARNING INTEGRATION

We classify learning integration approaches that embed LLMs into reinforcement learning pipelines. These methods leverage human feedback, teacher–student strategies, hierarchical planning, and policy distillation to accelerate training, improve safety alignment, and achieve more scalable and adaptive driving policies.

a: RLHF-GUIDED DRIVING

Sun et al. [61] present a hybrid framework that combines Reinforcement Learning from Human Feedback (RLHF) with LLMs to improve the safety and reliability of autonomous driving systems. Unlike conventional reinforcement learning approaches, which often fall short in capturing nuanced human preferences, or standard RLHF methods, which are typically restricted to fine-tuning with direct human input, this work introduces a more holistic integration. The framework begins with a pre-trained autonomous car agent and introduces multiple human-controlled agents, including vehicles and pedestrians, to simulate realistic traffic environments. Rather than directly controlling the autonomous car, humans provide both physical and physiological feedback signals, which are then incorporated into the training loop via LLM-based reasoning to guide the optimization process. The advantages of this study include its innovative integration of RLHF and LLMs, the use of

TABLE 5. Comparative analysis of reasoning-centric autonomous driving architectures integrating LLMs.

Ref.	Year	Contribution Focus	LLM Mechanism	Platform / Dataset	Key Merits	Limitations	Quantitative Evaluation	LLM Reasoning	LLM Agents
Fu *et al.* [60]	2024	Reasoning-driven driving: human-like comprehension of complex environments	Closed-loop integration; LLM for reasoning, interpretation, memorization	Custom driving environment	Human-like reasoning; addresses rare corner cases; reframes AD as reasoning task	Scalability concerns; compute cost; closed-loop validation only	Qualitative robustness in atypical cases; no benchmark scores reported	●	◐
Zhang *et al.* [12]	2025	LeAD: dual-rate E2E + LLM reasoning architecture for urban scenarios	CoT reasoning integrated into low-frequency LLM module + HD map fusion	CARLA Leaderboard V1 [27]	Robust in edge cases; innovative dual-rate design; effective multimodal reasoning	Simulation-based only; latency risk from low-frequency reasoning	**71 points** on Leaderboard V1; **93%** route completion (↑ vs. baselines)	●	◐
Sun *et al.* [61]	2024	RLHF-guided ADS decision-making with human preference signals	LLM processes human physiological + behavioral feedback	Custom simulator; human-in-loop agents	Aligns ADS with nuanced human feedback; pre-deployment validation	Complex feedback-to-action mapping; high resource demands	Demonstrated safety/reliability improvements qualitatively; no single aggregate metric	●	●
Xu *et al.* [13]	2025	TeLL-Drive: teacher–student hybrid learning with LLM reasoning	Teacher LLM generates CoT strategies; student DRL learns via self-attention	Simulated + real vehicle-in-loop platform	Faster policy convergence; robust cross-scenario decisions; real–sim fusion validation	Scalability limits; reliance on LLM accuracy; inference cost	Outperforms baselines in success rate + average return (↑); ablation confirms LLM role	●	◐
Xu *et al.* [62]	2025	LEAD: LLM-guided DRL training with dynamic intervention	High-level LLM injects strategies at critical points during DRL training	CARLA simulator [27]	Reduces collision rates; transitions smoothly to DRL autonomy	Simulation-only; compute overhead; scalability uncertain	Collisions reduced by **49.5%** (low density) and **59.4%** (high density); task completion ↑ up to **65.6%**	●	◐
Li *et al.* [63]	2025	Hierarchical RL + LLM high-level planner with reward shaping	LLM defines meta-actions, goals, reward functions; RL executes low-level actions	CARLA simulator [27]	Human-aligned planning; GGT mechanism accelerates learning; interpretable reasoning	Simulation reliance; inference delay from high-level LLM; complex reward alignment	Improved generalization + interpretability; exact percentage gains reported in study	●	◐
Wu *et al.* [64]	2024	RAPID: policy distillation from LLM → efficient RL policies	LLM generates offline data; distilled into mix-of-policy RL	Custom driving datasets	Faster inference; robust policies; scalable through distillation	Reliance on LLM-generated data quality; compute overhead	Demonstrated robust performance across tasks; no single benchmark metric given	●	○

Symbols: ●= fully covered, ◐= partially covered, ○= not covered.

diverse feedback modalities, and its design for realistic pre-deployment validation, while limitations include the complexity of aligning physiological feedback with actionable learning signals, the high resource demands of multi-agent simulations, and the uncertainty of scalability to large, heterogeneous traffic environments.

b: TEACHER–STUDENT LEARNING

Xu et al. [13] propose TeLL-Drive, a hybrid framework that unites deep reinforcement learning (DRL) with large language models (LLMs) to enhance autonomous driving decision-making. A Teacher LLM generates high-level driving strategies through chain-of-thought reasoning, integrating risk metrics, scenario history, and domain heuristics, while a Student DRL agent refines these strategies via a self-attention mechanism to accelerate learning and improve robustness. Experiments across diverse traffic scenarios show that TeLL-Drive outperforms both DRL and other LLM-based baselines in success rate, efficiency, and real-time performance. Further validation on a virtual–real fusion platform, including vehicle-in-loop testing, confirms its reliability. However, scalability to complex urban settings, dependence on LLM-generated strategy accuracy, and high computational cost remain key limitations.

c: LLM-GUIDED DRL TRAINING

Xu et al. [62] introduce LEAD, an LLM-Enhanced Autonomous Driving framework that embeds LLM reasoning within deep reinforcement learning (DRL) to improve decision-making. While LLMs provide strong reasoning but are costly to deploy, DRL agents are efficient yet suffer from poor exploration. LEAD combines their strengths via a dynamic intervention mechanism: early training uses LLM-based expert guidance at critical decisions, while later stages let the DRL agent refine policies autonomously. The LEAD-PPO model cuts collision rates by up to 59.4 % and boosts task completion by 9.6–65.6% over baselines. Strengths include its hybrid LLM-DRL integration, adaptive

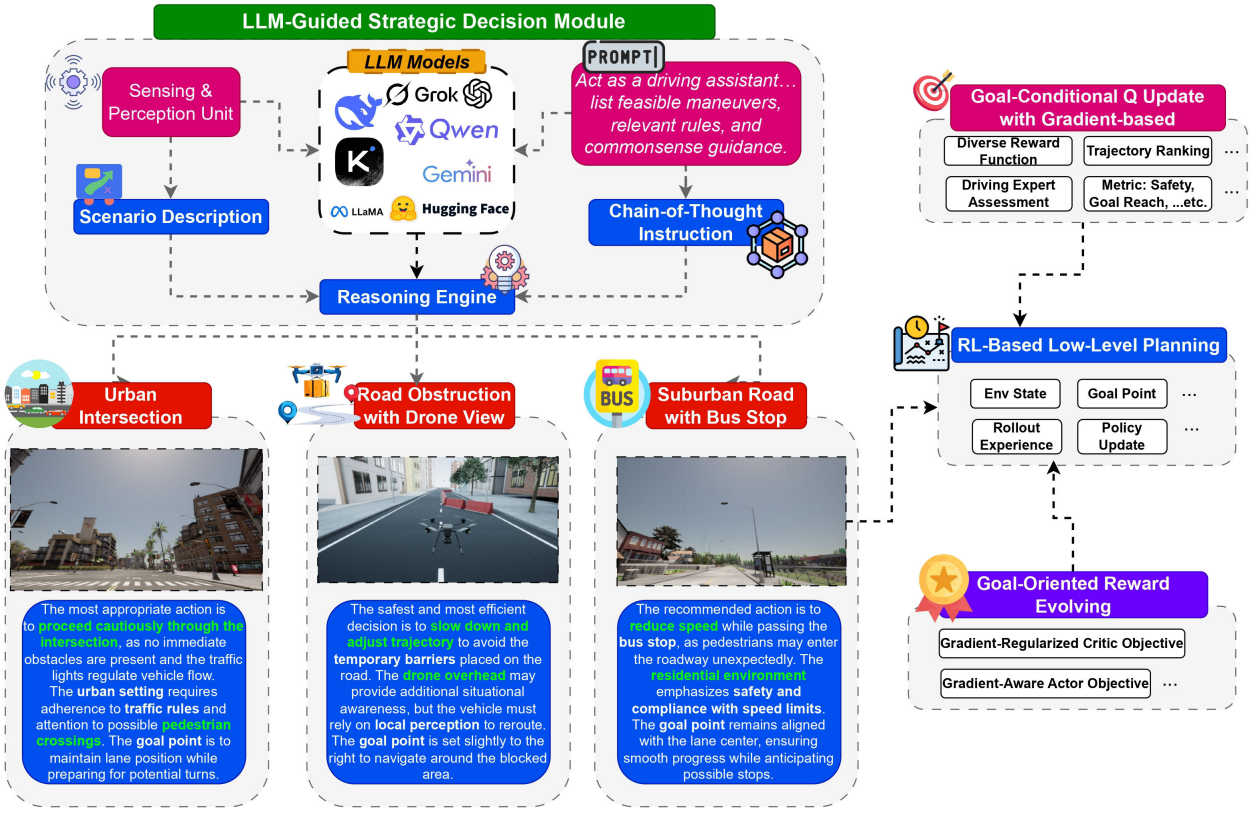


FIGURE 6. Framework for decision-making and planning in autonomous driving.

intervention, and robust performance gains. Limitations include simulation-based evaluation, increased computational requirements during training, and limited real-world validation.

d: HIERARCHICAL RL WITH LLM PLANNING

Li et al. [63] present a hierarchical reinforcement learning (RL) framework enhanced with an LLM to improve decision-making in safety-critical driving scenarios. The LLM functions as a high-level planner, interpreting textual descriptions to set long-term goals and meta-actions, while a low-level RL agent handles real-time control such as acceleration and steering. A Goal Gradient-based Transfer (GGT) mechanism aligns the two levels by embedding gradients toward LLM-defined goals, expediting learning. Additionally, an LLM-driven, human-aligned reward design optimizes candidate reward functions through expert feedback. Experiments in CARLA [27] show improved generalization, interpretability, and safety alignment. Despite its strengths, the framework faces limitations in simulation-only validation, potential inference latency from the LLM planner, and challenges in scaling human-in-the-loop reward tuning to real-world systems.

Figure 6 presents an integrated framework proposed by Li et al. [63], which combines LLMs with reinforcement learning (RL) to handle a diverse range of driving scenarios. The high-level decision module leverages an LLM-guided reasoning engine, which interprets scenario descriptions, feasible maneuvers, traffic rules, and commonsense guidance to generate meta-actions and long-term goals. Example cases include urban intersections, roads with temporary obstructions, and suburban streets with bus stops, each requiring distinct considerations of safety, efficiency, and compliance with traffic regulations.

At the low level, RL-based planning refines the vehicle's actions using environmental states and goal points, supported by rollout experiences and policy updates. A goal-oriented reward evolution mechanism introduces multiple evaluation metrics, including safety, efficiency, smoothness, and goal achievement. Expert assessments and trajectory ranking further enhance learning reliability, while gradient-based goal-conditional updates stabilize the actor and critic objectives. Collectively, this framework demonstrates how LLM reasoning augments RL policy learning to achieve more adaptive and explainable autonomous driving behavior.

e: POLICY DISTILLATION

Wu et al. [64] introduce RAPID, a Robust Adaptive Policy Infusion and Distillation framework that fuses LLM knowledge with reinforcement learning (RL) to address slow

inference and limited real-time adaptability in LLM-based driving agents. RAPID combines LLM-synthesized offline data with online adaptation, featuring three core components: LLM-to-RL knowledge distillation for efficient inference, robust distillation to maintain performance under uncertainty, and a mix-of-policy structure with a policy adapter for joint decision decoding. Online fine-tuning enhances adaptability while retaining distilled LLM knowledge. Experiments show that RAPID achieves robust, scalable performance across a range of driving tasks. Nonetheless, its effectiveness depends on the quality of LLM-generated data and faces scalability and computational overhead challenges in complex real-world settings.

3) ATTENTION-BASED PLANNING WITHOUT LANGUAGE

Li and Zhang [65] present a dual-branch conditional Transformer framework for Automated Valet Parking (AVP) based on imitation learning that explicitly addresses low-speed vehicle–obstacle interactions and the integration of multi-modal behaviors within constrained parking environments. The method introduces a conditional intention interaction module that leverages multi-head attention to jointly model behavior-level and trajectory-level interactions, using the ego vehicle's deterministic intent as an anchor to suppress irrelevant obstacle modalities. Spatial map semantics and ego task features are processed in parallel residual branches to mitigate feature degradation, enabling unified handling of cruising and parking behaviors. The approach is evaluated on the real-world Dragon Lake Parking (DLP) dataset, comprising approximately 3.5 hours of 4K aerial video over a $140\,\text{m} \times 80\,\text{m}$ parking lot, reorganized into nearly one million frames at 5 Hz. Quantitative results demonstrate consistent improvements over ParkingE2E and TransFuser, reducing L2 distance (0.017 vs. 0.026), Hausdorff distance (0.037 vs. 0.058), and Fourier descriptor difference (0.244 vs. 0.295) while maintaining real-time inference at $\sim$40 ms. Open- and closed-loop evaluations further confirm higher no-collision rates (up to 97.3%) and robust execution across interactive cruising and precise parking scenarios, validating the effectiveness of conditional attention for AVP trajectory planning.

D. PERCEPTION AND REPRESENTATION

In this subsection, we organize recent perception and representation approaches for autonomous driving that leverage transformers, large language models (LLMs), and multi-modal learning paradigms. These works address the challenge of bridging low-level sensor data (e.g., cameras, LiDAR, radar, and occupancy grids) with high-level semantic understanding and reasoning. We group the literature into three complementary directions: (i) 3D and occupancy-based perception methods that provide structured spatial representations, (ii) transformer-based multitask and trajectory perception models that enable efficient and long-horizon prediction, and (iii) multimodal perception frameworks that integrate heterogeneous sensory inputs with language-based reasoning.

1) 3D AND OCCUPANCY-BASED PERCEPTION

We classify perception works into 3D tokenization and occupancy-based reasoning approaches, both of which aim to provide richer spatial and dynamic representations for downstream planning and decision-making.

a: 3D TOKENIZATION

Bai et al. [15] propose Atlas, a vision–language framework that enhances end-to-end autonomous driving by incorporating 3D spatial reasoning into LLM-based systems. Unlike prior 2D vision–language models, Atlas employs DETR-style 3D perceptrons as tokenizers linked to an LLM via a lightweight linear projector, embedding rich 3D geometric priors into the planning process. This design enables accurate spatiotemporal modeling and multi-view perception, improving both 3D detection and ego-vehicle planning. Experiments on the nuScenes dataset [31] demonstrate superior performance over 2D tokenized baselines, highlighting the importance of 3D-aware tokenization for robust autonomous driving. While Atlas is efficient and reliable, it still relies on large annotated 3D datasets and requires broader real-world validation.

b: OCCUPANCY-BASED REASONING

Xu et al. [66] propose Occ-LLM, the first occupancy-based large language model that integrates occupancy representations into LLM reasoning for robotic and autonomous driving. To handle occupancy encoding and category imbalance, they introduce the Motion Separation Variational Autoencoder (MS-VAE), which isolates dynamic objects from static scenes before applying a tailored VAE, enabling precise trajectory focus and accurate static reconstruction. Validated on tasks such as 4D occupancy forecasting, ego planning, and scene question answering, Occ-LLM surpasses state-of-the-art methods with gains of about 6% in IoU and 4% in mean IoU. While offering strong performance and innovative design, its limitations include higher computational costs, reliance on the quality of occupancy data, and limited validation beyond controlled benchmarks.

2) TRANSFORMER-BASED MULTITASK PERCEPTION

Bavirisetti et al. [67] propose a unified multi-task Vision Transformer architecture for autonomous driving that jointly performs semantic segmentation and monocular depth estimation from a single RGB image, aiming to reduce computational overhead while maintaining competitive accuracy. The method integrates a shared hierarchical Transformer encoder based on SegFormer with two lightweight task-specific decoders, namely an MLP-based segmentation head and a GLPDepth-inspired depth decoder, enabling both tasks to be executed in a single forward pass. Extensive experiments are conducted on four datasets: Cityscapes (5,000 annotated frames), KITTI-360 (61,280 images), Apollo Synthetic (45,235 selected frames from an original set of 273,000), and a proprietary NAPLab dataset to evaluate performance,

generalization, and efficiency. Quantitative results demonstrate that the proposed multi-task model achieves accuracy comparable to single-task counterparts while significantly reducing total inference time, despite performing two dense prediction tasks simultaneously. Backbone scaling experiments using SegFormer variants (B0, B2, B4) reveal a clear trade-off between accuracy and real-time performance, with B0 achieving approximately 56–61 FPS, B2 operating near real-time at 18–19 FPS, and B4 providing the highest accuracy at 11–12 FPS. Furthermore, pre-training on synthetic data is shown to substantially improve monocular depth estimation metrics, particularly absolute relative error, when real-world labeled data is limited, while preserving segmentation performance. The results highlight the effectiveness, efficiency, and generalization capability of Vision Transformer–based multi-task learning as a viable alternative to separate perception models in resource-constrained autonomous vehicle systems.

3) TRANSFORMER-BASED TRAJECTORY PREDICTION

Wang et al. [68] introduce the *Lane Transformer*, a high-efficiency trajectory prediction model designed for real-time autonomous driving applications with limited onboard computational resources. The proposed approach replaces the graph convolutional network (GCN) backbone of prior vector-based models, such as LaneGCN, with multi-head attention–based Transformer blocks, enabling efficient fusion of agent trajectories and high-definition lane map information while preserving geometric structure. The model encodes vectorized trajectories and road segments using attention-enhanced encoders and performs hierarchical local and global interaction modeling via stacked Transformer blocks, followed by a multi-modal MLP decoder that predicts $K = 6$ future candidate trajectories with associated confidence scores. Extensive experiments on the Argoverse motion forecasting dataset, comprising 323,557 scenes from Pittsburgh and Miami, demonstrate that the Lane Transformer achieves prediction accuracy comparable to or slightly better than LaneGCN (e.g. minADE $= 0.86$ and minFDE $= 1.31$ for $K = 1$), while delivering a dramatic efficiency gain of $10\times$ to $25\times$ in inference speed. In particular, the model achieves an inference latency of only 7 ms when deployed with TensorRT, the fastest among publicly available methods, and reduces the parameter count to approximately 1.55 million, substantially lower than that of competing GCN-based models. Ablation studies confirm the importance of attention-based hierarchical fusion and trajectory smoothing, while strict adherence to TensorRT-compatible coding practices ensures seamless deployment on real vehicles, highlighting the practical viability of attention-based trajectory prediction in safety-critical autonomous driving systems.

Abyaneh and Jang [69] present a transformer-based framework for trajectory prediction using real-world 3D LiDAR data to enhance situational awareness in complex urban environments. A large-scale dataset was collected using an Ouster OS1 LiDAR sensor at a busy urban intersection in West Palm Beach, Florida, capturing more than 12,390 trajectories of vehicles, pedestrians, and bicycles over a 37-minute period at 10 Hz. After object detection and rigorous data curation to remove noise from glass reflections, near-stationary agents, and physically implausible motions, a Transformer model was trained to learn spatio-temporal dependencies from relative displacement trajectories. Systematic hyperparameter tuning identified an embedding dimension of 32, eight attention heads, and two encoder–decoder layers as the optimal configuration. Quantitative evaluation demonstrates that the proposed model achieves an Average Displacement Error (ADE) of 1.44 m and a Final Displacement Error (FDE) of 2.55 m, corresponding to 15.24% and 12.4% improvements in ADE and FDE, respectively, over an LSTM baseline. Additional metrics show reduced miss rate (23.4% vs. 31.8%) and higher endpoint prediction accuracy (mAP of 76.6%), confirming the effectiveness of attention-based models for reliable multi-agent trajectory prediction in noisy, real-world urban settings.

4) MULTIMODAL MODELS

We classify multimodal perception approaches into object-level multimodal integration, situational awareness frameworks, instruction-tuned assistants, video-language navigation systems, and trajectory generation methods, and highlight their roles in combining diverse inputs to enable more adaptive and interpretable autonomous driving.

a: OBJECT-LEVEL MULTIMODAL INTEGRATION

Chen et al. [14] present LLM-driver, an object-level multimodal LLM architecture that fuses vectorized numeric modalities with a pre-trained language model to improve contextual understanding in autonomous driving. They develop a dataset of 160k QA pairs across 10k driving scenarios, combining RL-collected control commands with GPT-3.5–generated question–answer samples. A novel pretraining strategy aligns numeric vectors with LLM representations via vector captioning, enabling seamless reasoning across structured and textual inputs. A new Driving QA metric is also introduced to evaluate performance. Results show superior interpretability and decision-making compared to behavioral cloning. However, the approach depends on synthetic QA data, entails high computational costs for multimodal alignment, and requires further validation in real-world autonomous driving settings.

b: SITUATIONAL AWARENESS

Luo et al. [16] introduce SenseRAG, a framework that enhances situational awareness in autonomous driving by integrating real-time multimodal sensor data with LLM-based contextual reasoning. Moving beyond static, label-driven perception systems, it constructs an LLM-readable knowledge base for dynamic scene interpretation. To mitigate latency and modality constraints, the framework employs Retrieval-Augmented Generation (RAG) with

chain-of-thought prompting, enabling faster, context-aware reasoning. Evaluations on real-world V2X datasets show marked improvements in perception and prediction, boosting safety and adaptability in complex driving environments. Despite its promise, SenseRAG faces challenges related to sensor data synchronization, computational overhead from real-time RAG operations, and limited large-scale closed-loop validation.

c: INSTRUCTION-TUNED ASSISTANT

Park et al. [70] present VLAAD, a multi-modal LLM driving assistant designed to advance interpretable decision-making in autonomous driving through natural language integration. To support this system, they constructed a novel instruction-tuning dataset that enables language models to learn from visual instructions in a wide variety of driving scenarios. The dataset is structured around three key tasks: conversation, detailed description, and complex reasoning, ensuring that the model can handle both straightforward and nuanced interpretive demands. By fine-tuning on this dataset, VLAAD demonstrates strong capabilities in interpreting and reasoning about diverse driving situations, thereby enhancing transparency and human-aligned understanding in autonomous decision-making. The advantages of this work include its unique focus on interpretability, the creation of a dedicated multi-modal instruction dataset, and the effective integration of language reasoning into driving contexts, while its limitations concern the lack of large-scale benchmarking against existing interpretable AD models, potential dataset bias in scenario coverage, and the absence of validation in real-world driving conditions.

d: VIDEO-LANGUAGE NAVIGATION

Huang et al. [71] propose DriVLMe, a video language model-based agent designed to enhance natural communication and navigation capabilities in autonomous driving by combining embodied experiences from simulated environments with social experiences derived from human dialogue. Unlike previous work with simplified settings, DriVLMe is evaluated on both open-loop benchmarks and closed-loop human studies, demonstrating competitive performance in long-horizon navigation tasks and interactive dialogue. The framework seeks to address the underexplored question of whether foundation models (FMs) can manage unexpected events, environmental dynamics, and task changes while maintaining natural communication with human users. The advantages of this work include its novel integration of video-language modeling for autonomous navigation, its dual training strategy combining simulation and human dialogue, and its comprehensive evaluation across multiple settings, while its limitations include its high inference latency, imbalanced training data, limited visual understanding, difficulties in handling multiturn dialogue, simplified language generation from robotic experiences, and reduced effectiveness in managing unexpected real-time environmental changes.

e: TRAJECTORY GENERATION

Yang et al. [72] propose Trajectory-LLM (Traj-LLM), a framework that uses large language models to generate realistic and diverse vehicle trajectories from concise textual descriptions of driving interactions. Through an interaction–behavior–trajectory process, Traj-LLM aligns natural-language prompts with plausible motion trajectories, reducing the need for costly real-world data collection. The authors also release the Language-to-Trajectory (L2T) dataset, comprising 240K paired samples of textual interactions, map topologies, and trajectory segments, enabling adaptability across varied road layouts. Experiments demonstrate that Traj-LLM not only produces high-quality synthetic trajectories but also improves the performance of downstream prediction models. Despite its strengths, the framework's dependence on textual inputs, potential biases in synthetic data, and limited real-world validation remain key challenges.

E. SAFETY AND SECURITY

Table 6 presents LLM-based frameworks addressing safety, security, and robustness. Kong [51] introduces multi-agent superalignment for secure inference, fully covering both reasoning and agent aspects. Zheng's HeMARL [56] improves secrecy in UAV swarms with reasoning and partial agent use. Song [50] employs Hudson reasoning agents for adversarial robustness with partial agent integration. Chen's HCRMP [54] and Gong's DRL–LLM hybrid [55] apply reasoning to mitigate hallucinations and ensure compliance, but rely only partially on agents. This shows that safety frameworks remain primarily reasoning-centric, with selective agent integration.

1) SECURITY FRAMEWORKS

We categorize security-focused approaches into frameworks that ensure data privacy, trust alignment, and physical-layer protections. These works highlight methods such as superalignment strategies to safeguard sensitive driving data and physical-layer security mechanisms to protect communication in UAV and autonomous driving systems.

a: SUPERALIGNMENT

Kong et al. [51] introduce a multi-agent LLM-based security framework to enhance data protection in LLM- and MLLM-driven autonomous driving systems. Recognizing that such systems rely on sensitive inputs such as vehicle location, sensor data, and environmental conditions, transmitted to cloud-based models, the framework implements layered defenses to filter irrelevant queries, verify inference outputs, and enforce alignment with traffic laws and human values. Empirical evaluation across eleven LLM-driven driving signals demonstrates notable improvements in privacy, security, and cost efficiency, supported by quality assurance tests. While the framework effectively safeguards sensitive data and ensures trustworthy outputs, its limitations include dependency on controlled evaluation settings, limited

TABLE 6. LLM-based frameworks for safety, security, and robustness in autonomous driving.

Referei	Year	Threat / Safety Focus	LLM / Hybrid Approach	Evaluation Setting	Security Strengths	Limitations / Risks	Quantitative Analysis	LLM Reasoning	LLM Agents
Kong *et al.* [51]	2024	Data privacy and secure inference	Multi-agent LLM superalignment with layered filters	11 AD signals (privacy + QA tests)	Proactive filtering of sensitive inputs; layered trustworthiness checks	Scalability issues for fleet deployment; limited adversarial testing	Validated on 11 signal types; privacy leakage ↓; cost efficiency confirmed	●	●
Zheng *et al.* [56]	2025	Physical-layer secrecy in UAV swarms	Hierarchical optimization with LLM-guided HeMARL	Simulated HetUAV networks	Improves secrecy rate + energy efficiency; integrates expert LLM heuristics	Relies on synthetic UAV setups; scalability uncertain in large swarms	Secrecy rate ↑; energy savings reported; robust across seeds/swarm sizes	●	◐
Song *et al.* [50]	2024	Robustness against ODT attacks	DSL-driven Hudson reasoning agent with LLM	Simulated adversarial driving (ODT attacks)	Detects adversarial inconsistencies; cross-LLM validation	Performance varies across LLMs; simulation-only evaluation	Attack detection acc: GPT-4 (83.3%), LLaMA (63.6%), Gemma (73.6%)	●	◐
Chen *et al.* [54]	2025	Reducing hallucinations in motion planning	HCRMP: LLM-hinted contextual RL motion planner	CARLA benchmark (multi-density traffic) [27]	Semantic hints improve reliability; contextual anchor boosts stability	Some latency from semantic cache; simulation dependence	Success rate ↑80.3%; collisions ↓11.4% across densities	●	◐
Gong *et al.* [55]	2025	UAV safe + regulatory trajectory planning	DRL + LLM hybrid for semantic reasoning	Low-altitude UAV scenarios (regulatory)	Ensures compliance with airspace rules; balances cost + safety	High computation for real-time LLM inference; limited field testing	Outperforms baselines on 5 KPIs (data rate, collision avoidance, landings, compliance, efficiency)	●	◐

Symbols: ●= fully covered, ◐= partially covered, ○= not covered.

scalability to large fleet deployments, and the need for robustness validation under adversarial conditions.

b: PHYSICAL-LAYER SECURITY

Zheng et al. [56] propose a hierarchical optimization framework to enhance physical layer security (PLS) in heterogeneous UAV networks (HetUAVNs) while accounting for propulsion energy constraints and varying UAV capabilities. The inner layer employs an SDR-based S2DC algorithm combining penalty functions and difference-of-convex programming to optimize secrecy precoding for fixed UAV positions. The outer layer introduces LLM-HeMARL, a heuristic multi-agent reinforcement learning approach guided by LLM-generated expert strategies, enabling UAVs to learn energy-efficient, secure trajectories without real-time LLM inference overhead. Simulations demonstrate significant gains in secrecy rate, energy efficiency, and robustness across different swarm configurations. The framework's strengths lie in its hierarchical design and effective integration of LLM heuristics, though it remains limited by simulation-based validation, scalability challenges in dynamic environments, and reliance on the quality of LLM-generated heuristics.

2) ROBUSTNESS

We categorize robustness-oriented approaches into methods that enhance resilience against attacks, improve motion planning under uncertainty, and ensure safe trajectory generation. These works include attack detection frameworks, contextual motion planning strategies, and trajectory planning systems that integrate LLM reasoning with reinforcement learning to strengthen autonomous driving and UAV safety.

a: ATTACK DETECTION

Song et al. [50] present Hudson, a driving reasoning agent designed to strengthen autonomous driving systems against object detection and tracking (ODT) attacks that exploit vulnerabilities in LLM-integrated frameworks. Their analysis of four LLM agents reveals a 63.26% attack success rate, exposing weaknesses in memory handling, prompt robustness, and reliance on perception data. Hudson mitigates these risks by instrumenting the driving stack to capture real-time perception and scene context, formalizing this information into a domain-specific language (DSL) that is converted into natural language with embedded attack-detection cues. This allows the LLM to identify inconsistencies and make safer control decisions. Experiments with GPT-4, Llama, and Gemma yield average attack detection accuracies of 83.3%, 63.6%, and 73.6%, respectively, and safe decision rates up to 86.4%. Hudson's strengths include its innovative DSL-based reasoning approach, robust multi-LLM validation, and measurable safety improvements, though it remains limited by simulation-based testing, inter-model performance variability, and the absence of large-scale real-world validation.

b: CONTEXTUAL MOTION PLANNING

Chen et al. [54] introduce HCRMP, an LLM-Hinted Contextual Reinforcement Learning Motion Planner that reduces over-reliance on LLMs in autonomous driving. Unlike prior LLM-dominated RL methods, where hallucinations yield

only 57.95% non-hallucination rates, HCRMP uses the LLM solely to provide semantic hints for state augmentation and policy optimization, while the RL agent independently refines motion planning. The architecture integrates three modules: an Augmented Semantic Representation Module to expand state space, a Contextual Stability Anchor Module to ensure reliable multi-critic weighting via knowledge bases, and a Semantic Cache Module to balance low-frequency LLM guidance with high-frequency RL control. Experiments in CARLA [27] show up to 80.3% task success and an 11.4% reduction in collisions under dense traffic. HCRMP's strengths lie in its balanced LLM–RL integration, semantic augmentation, and demonstrated safety gains, though it remains limited by simulation-only testing, potential latency from semantic processing, and a lack of large-scale real-world validation.

c: SAFE TRAJECTORY PLANNING

Gong et al. [55] propose a UAV trajectory planning framework that integrates deep reinforcement learning (DRL) with LLM reasoning to address the unique challenges of the rapidly expanding low-altitude economy. While UAV deployment in urban environments continues to grow, existing approaches often overlook critical aspects, such as regulatory airspace constraints and economic efficiency, which limit their practicality. In this framework, DRL provides adaptive decision-making for trajectory optimization, while the LLM contributes semantic reasoning to ensure safety, regulatory compliance, and cost-effectiveness in path planning. Experimental evaluations show that the proposed method significantly outperforms baseline approaches across several key metrics, including data collection rate, collision avoidance, successful landing, regulatory adherence, and energy efficiency. The advantages of this work include its novel integration of LLM reasoning with DRL for trajectory planning, its explicit focus on economic and regulatory considerations in low-altitude environments, and its strong empirical results across multiple performance indicators. This research is subject to several limitations, including reliance on simulation-based validation, potential computational overhead from incorporating LLM reasoning into real-time planning, and the need for large-scale field testing to confirm effectiveness in practical deployments.

F. COOPERATION AND CONNECTIVITY

In this subsection, we classify cooperation and connectivity approaches in autonomous driving into two main directions: (i) vehicle cooperation, where LLMs enable collaborative perception, negotiation, and decision-making among connected vehicles, and (ii) vehicle–infrastructure integration, where LLMs are combined with reinforcement learning to optimize communication reliability and joint driving policies. Together, these works highlight how LLMs can strengthen safety, coordination, and efficiency in connected transportation ecosystems.

1) VEHICLE COOPERATION

a: VEHICLE-TO-VEHICLE (V2V)-LLM

Chiu et al. [17] present V2V-LLM, a baseline framework that integrates multimodal LLMs into cooperative autonomous driving to enhance joint planning and decision-making beyond perception tasks. Unlike traditional vehicle-to-vehicle (V2V) methods focused on detection or tracking, V2V-LLM introduces the Vehicle-to-Vehicle Question-Answering (V2V-QA) dataset and benchmark, enabling LLMs to fuse multimodal perception data from connected autonomous vehicles and answer complex driving-related queries on object grounding, identification, and planning. Experiments show that V2V-LLM surpasses existing fusion techniques, establishing a foundation for cooperative reasoning in autonomous driving. Nonetheless, it faces challenges such as reliance on reliable V2V communication, latency and computation in real-time fusion, and the need for large-scale real-world validation.

b: LLM-BASED COOPERATIVE DRIVING

Liu et al. [18] present CoLMDriver, the first full-pipeline LLM-based cooperative driving framework that advances vehicle-to-vehicle (V2V) collaboration beyond the constraints of traditional rule-based protocols. Unlike earlier cooperative methods, which often lack flexibility and fail to generalize to unseen interactions, CoLMDriver introduces a language-driven negotiation mechanism that enables adaptive, context-aware cooperation among autonomous vehicles. The system employs a parallel driving pipeline comprising two main modules: an LLM-powered negotiation component, built on an actor–critic paradigm, that iteratively refines cooperation policies using feedback from all vehicles; and an intention-guided waypoint generator that translates negotiation results into precise driving commands. To evaluate the framework, the authors design InterDrive, a CARLA-based benchmark featuring 10 complex interactive driving scenarios. Experimental findings highlight that CoLMDriver outperforms existing baselines, achieving an 11% improvement in success rate across challenging V2V cooperation tasks. This work offers several strengths, such as its innovative integration of negotiation-driven decision-making into cooperative driving, the introduction of a dedicated benchmark for interactive V2V scenarios, and empirical evidence of superior performance. Nonetheless, its limitations include potential inference latency from LLM-driven negotiation, reliance on simulation-based validation, and the need for further assessment in large-scale real-world deployments.

2) VEHICLE–INFRASTRUCTURE INTEGRATION

a: LLM-DOUBLE DEEP Q-LEARNING FOR V2I

Yan et al. [19] present a hybrid framework that integrates LLMs with double deep Q-learning (DDQN) to jointly optimize vehicle-to-infrastructure (V2I) communications and autonomous driving (AD) policies. In this design, LLMs are deployed for AD decision-making, leveraging Euclidean

distance to retrieve prior driving experiences and refine decision quality by learning from both successful and unsuccessful past actions. These LLM-informed AD decisions are then incorporated into the state space of the V2I optimization problem, where DDQN is applied to maximize data rates and minimize frequent handovers. The AD and V2I modules are iteratively optimized until convergence, enabling deeper integration between LLM-driven reasoning and reinforcement learning. Simulation results show that this hybrid LLM-DDQN framework converges faster and achieves higher average rewards than conventional DDQN approaches, demonstrating its potential to enhance safety, efficiency, and communication reliability in vehicular networks. The strengths of this study can be summarized as follows: the coupling of LLM reasoning with reinforcement learning for joint AD and V2I optimization, its iterative design that balances safety and communication efficiency, and its demonstrated performance improvements in simulations. The study has limitations, including reliance on simulated validation without real-world deployment, potential computational overhead from iterative optimization, and the challenge of scaling to highly dynamic traffic and network environments.

G. HUMAN ALIGNMENT AND INTERACTION

In this subsection, we categorize works that align autonomous driving agents with human drivers and enhance human–AI interaction. The first direction focuses on human-like driving, where LLM agents are aligned with human driving styles and thought processes. The second direction focuses on evaluation and assistance, where LLMs are utilized to assess driving intelligence and offer real-time support for driver well-being.

1) HUMAN-LIKE DRIVING

a: DRIVING STYLE ALIGNMENT

Yang et al. [73] propose a multi-alignment framework that seeks to align LLM-powered driver agents with human driving styles, addressing the current lack of research and data in this area. To overcome the scarcity of high-quality natural language data, they construct a dataset of human driver behaviors derived from naturalistic driving experiments and post-driving interviews, providing rich demonstrations for LLM alignment. The framework leverages these demonstrations and feedback to better capture diverse driving styles and is validated through simulation experiments in the CARLA urban traffic simulator, with additional support from human evaluations. The results highlight the potential of this approach for designing driver agents that more closely mimic human-like driving patterns. The advantages of this work include its innovative focus on aligning LLM agents with human driving styles, the development of a dedicated natural language dataset, and its dual validation through simulation and human assessment. The study is not without limitations, including reliance on the representativeness of the collected dataset, potential biases in subjective human feedback, and the need for large-scale real-world testing to fully establish robustness and generalizability.

b: HUMAN-LIKE GENERATIVE DRIVING AGENTS

Jin et al. [53] propose a framework for constructing human-like generative driving agents by leveraging post-driving self-report ‘‘driving-thinking’’ data from human drivers as both demonstrations and feedback. Addressing the limitation that LLMs, while strong in reasoning and world knowledge, lack embodiment, the framework captures high-quality natural language data through urban driving experiments in which drivers verbalize their thought processes under different conditions. These verbalized thoughts serve as chain-of-thought prompts and demonstration examples for training the LLM-based agent. The framework is validated through simulations and human assessments, with results showing an 81.04% reduction in collision rates and a 50% improvement in perceived human likeness compared to baseline LLM agents. The notable features of this work are the use of natural-language self-reports as expert demonstrations, the effective reduction in collisions, and the substantial gains in human-like driving behavior. Its limitations, however, include dependence on the quality and diversity of collected verbalized data, reliance on simulation-based validation with limited real-world testing, and potential scalability challenges in capturing large volumes of expert driver thought processes.

2) EVALUATION AND ASSISTANCE

a: DRIVING INTELLIGENCE EVALUATION

You et al. [58] propose an evaluation framework for assessing driving behavior intelligence in complex traffic environments, addressing the absence of comprehensive methods for measuring the level of autonomous driving intelligence. To support this framework, they construct a natural language evaluation dataset collected from professional drivers and passengers through naturalistic driving experiments and post-driving interviews. Leveraging this dataset, the framework integrates LLMs to evaluate autonomous driving behaviors, providing a more human-like and interpretable assessment process. The framework’s effectiveness is validated through simulated experiments in the CARLA urban traffic simulator and further corroborated by human evaluation, demonstrating its reliability and practical relevance. The advantages of this work include its novel focus on evaluating driving intelligence through natural language datasets, its integration of LLMs for interpretable assessments, and its dual validation via simulation and human feedback. Its limitations, however, include reliance on the scope and representativeness of the collected dataset, potential bias from subjective human evaluations, and the need for broader validation across large-scale real-world driving conditions.

b: DRIVER FATIGUE MITIGATION

Huang et al. [74] investigate the role of voice assistants based on LLMs in mitigating passive driving fatigue and

TABLE 7. LLM-based frameworks for UAV swarms, routing, scheduling, and aerial tasking.

Refere	Year	UAV / Swarm Focus	LLM-Enhanced Strategy	Testbed / Simulation	Aerial Strengths	Deployment Challenges	Quantitative Analysis	LLM Reasoning	LLM Agents
Han *et al.* [20]	2025	Swarm inference + command execution	SwarmChain: tensor-parallel CoLLM + adaptive load scheduling	AirSim + Raspberry Pi UAV sim [29]	Collaborative inference across UAVs; reduced latency; parses NL → UAV commands	Vulnerable to LLM hallucinations; tested in simulation only	Latency ↓33–52%; parsing accuracy 88.9%; throughput ↑2.3×	●	●
Wang *et al.* [21]	2025	Role-adaptive swarm navigation	RALLY: LLM semantic reasoning + MARL RMIX policies	Multi-Agent Particle Env. + SITL [30]	Flexible role switching; scalable to swarm tasks; semantic collaboration	Processing overhead from NL comms; limited real-world validation	Superior task coverage + faster convergence vs MARL baselines	●	●
Wei *et al.* [22]	2025	UAV routing in sensor networks	LAURA: LLM-assisted evolutionary crossover operators	UAV-assisted WSN simulation	Reduces age of information (AoI); scales to dense deployments	High compute for LLM-enhanced search; simulation only	Max AoI significantly reduced; scalable in large networks	◐	○
Emami *et al.* [23]	2025	Wildfire monitoring UAV scheduling	FRSICL: in-context LLM scheduling + velocity control	Wildfire UAV monitoring sim	Real-time scheduling; avoids costly retraining; adapts to env. feedback	Strict latency requirements; limited to simulation	Outperforms PPO + NN in AoI reduction; efficient real-time updates	●	◐
Zhou *et al.* [75]	2025	Truck–UAV delivery coordination (mFSTSP)	LLM-QL: prompt-guided Q-learning for routing	Logistics + delivery sim	Faster convergence; avoids local optima; improves UAV–truck synergy	Prompt sensitivity; computational overhead	1.35× runtime improvement; UAV utilization ↑; completion time ↓	●	◐
Chen *et al.* [2]	2025	UAV trajectory planning + resource allocation	LLM-CRDT: critic-regularized decision transformer + LoRA fine-tuning	IoT UAV data collection sim	Energy-efficient planning; stable offline RL + LLM reasoning	Dataset quality sensitive; transformer compute heavy	Energy efficiency ↑36.7% vs SOTA DT; robust offline RL	●	◐
Hu *et al.* [24]	2025	UAV tasking with vision + NL prompts	LLVM-Drone: structured prompts + lightweight vision models	UAV hardware + aerial mapping	Zero-shot task execution; mitigates hallucinations; deployed on UAVs	Limited by lightweight models; structured prompts required	Supports tasks (detection, tracking, navigation); hardware feasibility confirmed	●	●

Symbols: ●= fully covered, ◐= partially covered, ○= not covered.

enhancing overall safety and performance. Using a driving simulator and the LLM-based assistant ''Driver Mate,'' the study empirically evaluates the effects of conversation complexity and frequency on driver alertness and behavior, drawing on electroencephalogram (EEG) data, simulator outputs, and self-reported measures. Results demonstrate that low-complexity, high-frequency interactions are most effective in sustaining driver alertness and acceptance, while low-complexity, low-frequency interactions yield significant improvements in driving performance. This work offers novel insights into how conversational AI can be optimized for in-vehicle settings while balancing engagement and safety. The advantages of this study include its empirical validation using multimodal data sources, clear evidence of performance and alertness benefits, and practical recommendations for designing future in-vehicle voice assistants, while its limitations include reliance on simulated driving environments, a narrow focus on passive fatigue rather than broader distraction risks, and the need for validation across diverse driver populations and real-world conditions.

H. UAV AND SWARM SYSTEMS

Table 7 presents a survey of LLMs enhanced UAV swarm frameworks. Han's SwarmChain [20], Wang's RALLY [21], and Hu's LLVM-Drone [24] integrate both reasoning and agent workflows, enabling cooperative inference and swarm navigation. Wei's LAURA [22] and Emami's FRSICL [23] emphasize reasoning through routing and scheduling, with agents playing only partial roles. Zhou's LLM-QL [75] and Chen's LLM-CRDT [2] adopt reasoning-driven reinforcement learning while selectively incorporating agents. These differences highlight how agent integration is strongest in swarm coordination, whereas reasoning dominates in optimization-focused UAV frameworks.

Figure 7 presents the task execution cycle of UAV agents as a continuous loop of perception, decision, and action, where each stage is enhanced by diverse cognitive abilities. Perception combines multiple layers of intelligence, ranging from semantic and spatial understanding to more advanced capabilities such as temporal awareness, multimodal fusion, environmental sensing, contextual reasoning, predictive insights, social interpretation, anomaly detection, and self-awareness. Decision-making, in turn, encompasses abilities such as motion control, tool utilization, mission planning, collaboration, risk assessment, adaptation, resource management, communication strategies, continuous learning, and ethical compliance. These processes are further strengthened by integrating LLMs, which enable contextual reasoning and adaptive decision-making. By leveraging LLM-based reasoning, UAV agents can align perception with mission objectives, interpret complex scenarios, and gener-

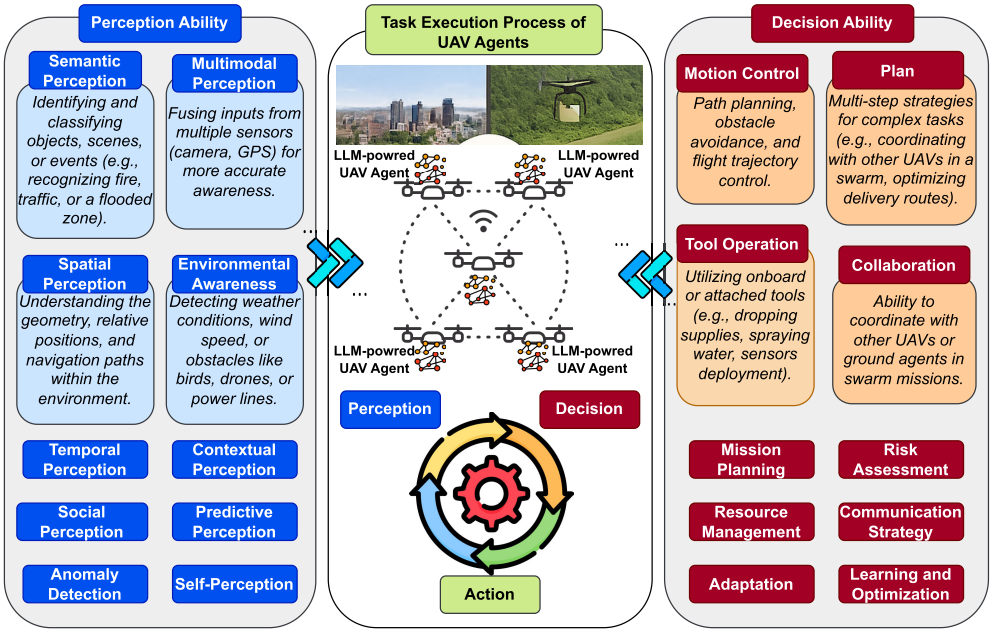


FIGURE 7. Task execution cycle and cognitive abilities of UAV agents with LLM reasoning.

ate intelligent responses in real time. This synergy among enriched perception, structured decision-making, and LLM reasoning provides a comprehensive foundation for robust, adaptive, and trustworthy autonomy in real-world environments.

1) SWARM INTELLIGENCE

a: COLLABORATIVE INFERENCE

Han et al. [20] propose SwarmChain, a collaborative LLM inference system designed to enhance UAV swarm control in harsh environments without reliance on ground command centers. While LLMs offer strong reasoning and communication capabilities for real-world tasks, their deployment on small UAVs is hindered by limited onboard computational resources and issues such as uncertainty and hallucination. To address these challenges, SwarmChain introduces CoLLM, a tensor-parallelism-based collaborative reasoning framework that distributes inference workloads across multiple devices, and a resource-aware, adaptive load-scheduling algorithm that balances computational demands. The system leverages LLMs to parse natural-language instructions into UAV API commands, ensuring accurate execution of mission requirements. Experiments conducted with StableLM and Llama2 on GPU-less devices such as Raspberry Pis, with UAV flight states simulated in AirSim [29], show that SwarmChain achieves 1.9–2.3× performance improvement over Llama.cpp MPI, reduces inference latency by 33.33%–52.38%, and reaches an instruction parsing accuracy of 88.9%. This research is characterized by the following strengths: collaborative inference design for resource-constrained UAV swarms, significant improvements in efficiency and latency reduction, and demonstrated ability to translate natural language into actionable UAV commands. Despite its merits, the approach has certain drawbacks: reliance on simulated UAV environments, potential vulnerability to LLM hallucinations in mission-critical scenarios, and the challenge of scaling the framework to larger or more heterogeneous UAV swarms in real-world deployments.

b: ROLE-ADAPTIVE NAVIGATION

Wang et al. [21] introduce RALLY, a Role-Adaptive LLM-Driven Yoked navigation algorithm designed to enhance intelligent control of UAV swarms through improved collaboration, adaptability, and scalability. Traditional Multi-Agent Reinforcement Learning (MARL) methods, while adaptive, often suffer from rigid homogeneous role structures and purely numerical communication, which limits semantic understanding and task generalization. Similarly, prior LLM-based control frameworks leverage strong reasoning abilities but depend heavily on static priors, lacking effective online learning and exploration. RALLY addresses these shortcomings by combining the strengths of both approaches: it employs an LLM-driven semantic decision framework for structured natural language communication and collaborative reasoning, introduces a dynamic role-heterogeneity mechanism that supports adaptive role switching and personalized decision-making, and integrates an RMIX-based role assignment strategy to align LLM offline priors with MARL online policies, enabling semi-offline training for role selection. Experimental evaluations in the Multi-Agent Particle Environment (MPE) and a Software-in-the-Loop (SITL) platform [30] show that RALLY achieves superior task coverage,

faster convergence, and stronger generalization compared to conventional baselines. The advantages of this work include its novel integration of LLM semantic reasoning with MARL policies, its dynamic role-heterogeneity mechanism for flexible coordination, and its demonstrated performance gains across simulation environments. Its limitations, however, include reliance on structured natural language communication, which may introduce processing overhead; potential challenges in scaling to highly complex, real-world swarm missions; and the need for further testing on physical UAV platforms beyond simulation.

2) ROUTING AND SCHEDULING

a: LLM-ASSISTED UAV ROUTING

Wei et al. [22] present LAURA, an LLM-assisted UAV routing algorithm designed to minimize the age of information (AoI) in UAV-assisted wireless sensor networks, a critical task given the rising demand for real-time data collection in the low-altitude economy. Unlike traditional approaches that struggle to efficiently optimize UAV flight paths in large-scale sensor deployments, LAURA integrates an evolutionary optimization framework with LLM-guided intelligent crossover operators, enabling more effective exploration of the solution space. Simulation results demonstrate that LAURA significantly reduces the maximum AoI compared to benchmark methods, particularly in scenarios with large numbers of sensor nodes. The advantages of this work include the use of LLMs to enhance evolutionary optimization, its demonstrated effectiveness in reducing AoI, and its scalability to dense sensor networks. The present work faces several challenges, including reliance on simulation-based validation, potential computational costs associated with LLM-assisted optimization, and the need for further validation in dynamic real-world UAV deployments.

b: WILDFIRE MONITORING

Emami et al. [23] propose FRSICL, an online flight resource allocation scheme that takes advantage of LLM-enabled in-context learning to optimize UAV-assisted wildfire monitoring (UAWM). In such systems, minimizing the Age of Information (AoI) is crucial to ensure timely updates from ground sensors. However, existing deep reinforcement learning (DRL)-based methods suffer from low sampling efficiency, issues with simulation-to-reality transfer, and complex training requirements, rendering them unsuitable for time-sensitive applications. FRSICL addresses these limitations by dynamically generating data-collection schedules and velocity controls from natural-language task descriptions, combined with real-time environmental feedback, eliminating the need for costly retraining. Simulation results demonstrate that FRSICL outperforms the Proximal Policy Optimization (PPO) and Nearest-Neighbor baselines in reducing average AoI, validating its effectiveness for real-time wildfire monitoring. The advantages of this work include its novel application of LLM-enabled in-context learning for UAV control, its ability to jointly optimize flight and data collection in real-time, and its superior empirical performance compared to strong baselines. Nevertheless, the limitations of this work include reliance on simulation validation rather than real-world deployment, potential inference latency in LLM-based decision-making under strict time constraints, and the need for further assessment in large-scale wildfire environments with highly dynamic conditions.

c: TRUCK–UAV COORDINATION

Zhou et al. [75] tackle the Multiple Flying Sidekicks Traveling Salesman Problem (mFSTSP), where UAVs collaborate with trucks to improve delivery efficiency, by introducing LLM-QL, a Large Language Model Enhanced Q-Learning approach. Traditional scheduling methods for large-scale mFSTSP often suffer from high computational costs and risk converging to local optima due to excessive exploration. LLM-QL addresses these challenges by combining the local exploration strengths of Q-Learning with the global reasoning capabilities of LLMs, thereby improving both efficiency and scalability in path planning. A novel prompt strategy is introduced to reframe problem modeling into a format more accessible to LLMs, guiding exploration and accelerating convergence. The study also provides proof of convergence for LLM-QL, reinforcing its theoretical foundation. Experimental results demonstrate that LLM-QL achieves a $1.35\times$ improvement over state-of-the-art methods in total completion time, algorithm runtime, and UAV utilization. The advantages of this work include its innovative integration of LLM reasoning with Q-Learning, its tailored prompt strategy for efficient problem modeling, and its strong empirical improvements across multiple performance metrics. The drawbacks of this approach include reliance on problem-specific prompt engineering, potential computational overhead from combining Q-Learning with LLM inference, and the need for further validation in large-scale, real-world delivery operations.

d: TRAJECTORY PLANNING

Chen et al. [2] propose LLM-CRDT, a critic-regularized decision transformer framework enhanced with LLMs for intelligent UAV trajectory planning and resource allocation in IoT data collection tasks. Recognizing the limitations of reinforcement learning (RL) methods, such as high costs and risks in real-world UAV training and the instability of offline RL when dependent on expert-quality datasets, the study formulates a joint optimization problem that maximizes energy efficiency. The resource allocation component is reformulated as a linear programming problem and solved optimally with polynomial-time complexity, while UAV trajectory control is handled by LLM-CRDT. In this framework, critic networks provide value-based regularization during training, ensuring more stable policy learning. A pre-trained LLM backbone with LoRA fine-tuning provides strong sequence modeling while reducing data and computational demands. The simulation results demonstrate that LLM-CRDT achieves

up to 36.7% higher energy efficiency compared to state-of-the-art decision transformer approaches, outperforming both online and offline RL baselines. The advantages of this work include its effective integration of LLM reasoning into offline RL, its innovative use of critic-regularized decision transformers with LoRA fine-tuning, and its substantial empirical improvements in energy efficiency. However, its limitations include reliance on simulation-based validation without large-scale field trials, potential sensitivity to the quality of available datasets, and computational overhead associated with transformer-based training despite the use of parameter-efficient techniques.

3) UAV TASKING

a: PROMPT EXECUTION WITH VISION MODELS

Hu et al. [24] introduce LLVM-Drone, a framework that integrates LLMs and lightweight task-specific vision models to enhance UAV capabilities while addressing the challenges of uncertainty and hallucination in natural language processing. Direct deployment of LLMs in UAVs is risky because they tend to misinterpret ambiguous instructions and generate unreliable executable code, particularly in safety-critical vision-based tasks. To mitigate these issues, LLVM-Drone employs the Domain-Guided Structured Prompt Execution Framework (DGSPEF), which leverages structured prompts and domain knowledge to translate user intent into precise commands, while lightweight vision models provide real-time perceptual validation. This separation of language understanding from visual processing enables zero-shot execution of UAV vision tasks without additional training. Extensive evaluations across eight state-of-the-art LLMs demonstrate the framework's effectiveness in aerial missions, including object detection, localization, tracking, and vision-language navigation. Furthermore, LLVM-Drone has been successfully deployed on UAV hardware for basic tasks, demonstrating practical feasibility and applicability to real-world use cases, including autonomous mapping, disaster response, precision agriculture, and infrastructure inspection. The advantages of this work include its innovative combination of DGSPEF with vision models to mitigate hallucinations, its zero-shot capability for diverse UAV vision tasks, and its demonstrated deployment on real UAV hardware. The proposed method is not without shortcomings, including its reliance on carefully engineered, structured prompts, potential constraints posed by lightweight vision models in highly complex tasks, and the need for further validation in large-scale, real-world operations under dynamic environmental conditions.

I. TESTING AND AUTOMATION

In this subsection, we classify testing and automation approaches for autonomous driving into two main directions: (i) simulation testing, where LLMs are leveraged to automate test generation and accelerate evaluation within simulated environments, and (ii) interpretability and compliance, where LLM-based frameworks integrate traffic rule retrieval and multi-agent sensor fusion to ensure transparency, legal compliance, and robust decision-making.

1) SIMULATION TESTING

a: AUTOMATED TEST GENERATION

Petrovic et al. [11] investigate the use of LLMs for automated test generation in the automotive domain, with a focus on generating free-form textual descriptions of autonomous driving features. They develop a prototype and evaluate the approach within the CARLA open-source simulation environment, comparing the performance of two pre-trained LLMs, GPT-4 and Llama3. The results show that GPT-4 consistently outperforms Llama3, while the proposed method significantly accelerates testing by more than 10x and reduces cognitive load by automating code generation and leveraging a flexible simulation environment for rapid evaluation. The advantages of this work include effective automated test generation, substantial improvements in test efficiency, and reduced reliance on manual effort. However, its limitations include reliance on simulation-based validation, potential biases in LLM-generated test cases, and the lack of large-scale real-world testing to confirm generalizability.

2) INTERPRETABILITY AND COMPLIANCE

a: TRAFFIC RULE RETRIEVAL

Cai et al. [57] present an interpretable decision-making framework for autonomous vehicles that comprehensively integrates traffic regulations, norms, and safety guidelines while ensuring adaptability across different regions. Unlike traditional rule-based approaches that struggle to capture the full scope of traffic rules, the proposed framework introduces a Traffic Regulation Retrieval (TRR) Agent built on Retrieval-Augmented Generation (RAG) to automatically extract relevant rules and records tailored to the ego vehicle's context. To address the semantic complexity of the retrieved information, a reasoning module powered by a LLM is employed to interpret the rules, distinguish between mandatory regulations and safety guidelines, and evaluate planned actions for legal compliance and safety assurance. Importantly, this reasoning process is designed to be interpretable, thereby improving both transparency and reliability. The framework demonstrates robust performance across simulated and real-world scenarios, with strong adaptability to varying regional contexts. The advantages of this work include its novel integration of RAG for traffic rule retrieval, its LLM-based reasoning for interpretability and compliance assessment, and its demonstrated ability to generalize across regions. Despite these advantages, the work has limitations, including reliance on the availability and quality of regulatory documents, potential latency in retrieving and processing complex rule sets in real-time, and the need for further validation in large-scale real-world deployments.

b: MULTI-AGENT SENSOR FUSION

Hou et al. [76] introduce DriveAgent, a multi-agent autonomous driving framework that combines LLM reasoning with multimodal sensor fusion to improve situational awareness and decision-making. The framework integrates diverse sensor modalities such as camera, LiDAR, GPS, and IMU within an agent-based pipeline composed of four specialized modules: a descriptive analysis agent for identifying critical events, vehicle-level analysis agents using LiDAR and vision for assessing conditions and movements, environmental reasoning and causal analysis agents for interpreting contextual changes, and an urgency-aware decision-generation agent for prioritizing insights and proposing maneuvers. This modular design enables the LLM to coordinate perception and reasoning across agents, producing cohesive, interpretable insights in complex scenarios. Experimental results on challenging datasets show that DriveAgent outperforms baseline approaches across multiple metrics, validating the effectiveness of the proposed multi-agent LLM framework. The advantages of this work include its novel, modular design that integrates multimodal sensor fusion with LLM reasoning, its interpretable multi-agent coordination for complex driving scenarios, and its superior empirical performance. On the other hand, its limitations include potential computational overhead from coordinating multiple agents, reliance on high-quality multimodal sensor data, and the need for further validation in large-scale real-world deployments.

J. AGENT COMMUNICATION AND COORDINATION IN LLM-ENABLED AUTONOMOUS SYSTEMS

Coordination in autonomous multi-agent systems fundamentally depends on how agents communicate, negotiate, and align their actions under uncertainty [83]. As LLM-based agents move beyond isolated reasoning toward embodied autonomy, communication mechanisms increasingly shift from low-level state exchange to structured, protocol-aware, and semantically grounded interaction [84]. This transition is particularly salient in autonomous driving and UAV swarm systems, where safety, scalability, and real-time adaptability require agents to exchange not only observations but also intents, constraints, and coordination outcomes [85].

Recent advances have introduced explicit communication abstractions that formalize agent interaction. One prominent direction is the use of structured tool-interaction protocols, such as the Model Context Protocol (MCP) [86], which standardize how an agent queries system state, invokes capabilities, and receives verifiable outcomes. By enforcing schema validity and deterministic execution semantics, MCP-style communication reduces ambiguity between reasoning and actuation, enabling LLM agents to operate as reliable controllers rather than free-form text generators [87]. In autonomous systems, this paradigm supports closed-loop interaction with sensing, planning, and control modules while preserving auditability and safety guarantees [88].

Complementary to agent–tool interaction, Agent-to-Agent (A2A) communication protocols [89] explicitly model coordination among multiple autonomous entities. Rather than relying on centralized orchestration or implicit message passing, A2A-style interaction enables agents to issue coordination requests, share situational summaries, and acknowledge cooperative actions through structured messages [90]. This form of interaction is especially relevant in multi-vehicle traffic scenarios and UAV swarms, where collision avoidance, formation control, and distributed task allocation must be resolved under partial observability and dynamic network conditions. By decoupling coordination from local control execution, A2A communication allows agents to reason collectively while retaining autonomy at the individual level.

At a broader level, these protocol-driven approaches align with emerging semantic-aware communication paradigms. Semantic communication emphasizes the transmission of task-relevant meaning rather than raw data, prioritizing efficiency, robustness, and contextual relevance. For LLM-based agents, this enables the exchange of high-level intents (e.g., navigation goals, safety alerts, or coordination states) instead of dense sensor streams, reducing communication overhead and improving resilience under bandwidth or latency constraints. Surveys on semantic communication networks [91] highlight their suitability for AI-native systems, where reasoning models can explicitly interpret and generate meaning-aware messages.

These developments resonate with system-level visions such as the Internet of Agents [92] and the Agentic Web [93], which conceptualize autonomous agents as interoperable network entities that interact through standardized, machine-interpretable protocols. In such architectures, vehicles, UAVs, edge nodes, and cloud services function as cooperating agents that dynamically discover capabilities, negotiate responsibilities, and coordinate actions. From this perspective, MCP- and A2A-style protocols can be viewed as concrete instantiations of broader agentic communication principles, bridging LLM reasoning with networked autonomy.

Despite their promise, agent communication mechanisms introduce new challenges. Ensuring protocol compliance, preventing hallucinated or unauthorized actions, and maintaining trust under adversarial or degraded communication conditions remain open problems. Moreover, as agent populations scale, coordination overhead and semantic drift may undermine efficiency and safety. Addressing these challenges requires integrating secure protocol design, semantic consistency checks, and policy enforcement into agent communication stacks. Overall, advances in protocol-aware and semantic communication represent a critical enabler for robust coordination in LLM-enabled autonomous systems and constitute a key research frontier at the intersection of AI agents, networking, and autonomous control.

K. SUMMARY AND LESSONS LEARNED

Across the surveyed literature, LLMs emerge as a unifying abstraction for enhancing autonomous systems, extending traditional pipelines beyond perception and control toward

TABLE 8. Comparative overview of simulation and test platforms employed in LLM-based autonomous systems research.

Simulation Tool / Platform	Domain	Representative Tasks	Representative Works	Strengths	Limitations	Adoption Level	LLM Reasoning	LLM Agents
CARLA [27]	Driving (urban traffic)	Scenario generation, planning, human alignment	Fu *et al.* [9], Zhang *et al.* [12], Xu *et al.* [13], Li *et al.* [63], Wu *et al.* [64]	Open-source, flexible APIs, supports closed-loop testing	Realism gap vs real-world, compute-intensive rendering	High	●	◐
Apollo ADS (Baidu) [28]	Driving (industrial-grade ADS)	Evolutionary scenario search, safety validation	Tian *et al.* [10]	Full-stack ADS pipeline, industrial realism	Proprietary dependencies, less accessible to academia	Medium	◐	○
Waymo Open Dataset [82]	Driving (perception)	3D detection, video synthesis, planning	Wei *et al.* [33], Zhao *et al.* [39]	High-quality labeled real-world data, diverse scenes	Limited to pre-recorded logs, licensing restrictions	High	◐	○
nuScenes [31]	Driving (multimodal)	Trajectory generation, adversarial scenarios, 3D tokenization	Peng *et al.* [42], Bai *et al.* [15], Xu *et al.* [66]	Rich multimodal benchmark, widely adopted	Dataset bias, limited rare corner cases	High	●	◐
AirSim (Microsoft) [29]	UAVs, aerial autonomy	Swarm inference, routing, tasking	Han *et al.* [20], Hu *et al.* [24]	Realistic physics for UAVs, Unity-based 3D scenes	GPU-heavy, limited real-world fidelity	Medium	◐	●
SITL (Software-in-the-Loop) [30]	UAV swarm control	Role-adaptive navigation, MARL integration	Wang *et al.* [21]	Firmware-level realism, integration with UAV autopilots	Scaling overhead in large swarms, setup complexity	Emerging	◐	●

Symbols: ●= fully covered, ◐= partially covered, ○= not covered.

Adoption Level: indicates how suitable and adaptable the platform is for LLM-based research (High = highly adaptable and widely leveraged in LLM contexts, Medium = moderately adaptable with some LLM integration, Emerging = early or limited adaptability for LLM research).

semantic reasoning, explanation, and human-aligned interaction. In both driving and UAV domains, LLM reasoning strengthens situation understanding, task decomposition, and high-level decision-making, while LLM agents connect these capabilities to embodied autonomy through closed-loop control, tool use, and multi-agent coordination. Yet most current systems for autonomous driving and aerial robotics remain simulation-centric, incur nontrivial latency and computational overhead, and adopt only partial agent integration, limiting robustness and scalability in real-world autonomous systems. A recurring lesson is that effective autonomy increasingly hinges on structured interaction via APIs, simulators, and communication protocols rather than unconstrained language generation alone. Overall, the reviewed applications indicate that future autonomous systems will require co-designed reasoning, agent execution, communication, and safety mechanisms, supported by standardized benchmarks and validation pipelines that move beyond isolated tasks toward holistic, deployment-oriented evaluation.

IV. SIMULATION TOOLS AND DATASETS IN LLM-BASED AUTONOMOUS SYSTEMS RESEARCH

In this section, we structure the key simulation tools and datasets that underpin research on LLM-driven autonomous systems. We review simulation platforms for both driving and UAV domains, synthetic data generation tools, cooperative and connectivity benchmarks, and LLM-based dataset generation pipelines. We also survey recent datasets and benchmarks that evaluate LLM reasoning, perception, and decision-making in autonomous driving and UAV settings.

A. SIMULATION PLATFORMS IN LLM-DRIVEN AUTONOMOUS SYSTEMS

In this subsection, we categorize the simulation platforms utilized in LLM-driven autonomous systems, encompassing driving simulators, synthetic data generation tools, UAV and swarm frameworks, cooperative connectivity platforms, and simulation-based dataset generation pipelines. These platforms form the foundation for evaluating reasoning, safety, and collaboration in both driving and aerial autonomy. Table 8 presents a comparative view of simulation and test platforms in the research of autonomous systems based on LLM. CARLA [27], used in works such as Fu [9], Zhang [12], Xu [13], Li [63], and Wu [64], achieves strong reasoning coverage with partial agent integration. Apollo ADS [28], adopted by Tian [10], demonstrates partial reasoning support but no explicit agent role. Dataset-driven environments such as Waymo Open [82] and nuScenes [31], used in Wei [33], Zhao [39], Peng [42], Bai [15], and Xu [66], show high reasoning adaptability with limited agent integration. In the UAV domain, AirSim [29] (Han [20], Hu [24]), and SITL [30] (Wang [21]) emphasize cooperative agent tasks, with reasoning only partially supported.

1) DRIVING SIMULATION PLATFORMS

CARLA has emerged as the most prominent driving simulator, frequently adopted in LLM-based scenario generation [9], dual-rate architectures [12], hierarchical RL planning [63], and interpretability frameworks [11]. Its modularity and scripting flexibility make it especially suitable for collision-prone and edge-case testing. In parallel,

industrial-grade platforms such as Apollo ADS [28] offer higher fidelity for evaluating evolutionary search-based frameworks [10], thereby bridging the gap between academic simulations and production-level deployments. Datasets such as nuScenes [40], [42] and the Waymo Open Dataset [33] complement these simulators by enabling large-scale benchmarking of perception and reasoning tasks, particularly in 3D-aware or counterfactual reasoning settings.

Across driving simulators, a clear trade-off emerges between controllability and realism. CARLA dominates LLM-centered research due to its programmability and tight integration with reasoning-driven planners, making it well-suited for stress-testing decision logic and edge cases. In contrast, Apollo ADS offers higher system fidelity and industrial relevance, but at the expense of reduced flexibility for agent-centric experimentation. Dataset-driven environments such as nuScenes and Waymo Open shift the emphasis from closed-loop control to large-scale, perception- and reasoning-focused benchmarking, revealing that no single platform simultaneously satisfies requirements for realism, agent autonomy, and reasoning evaluation. This fragmentation motivates cross-platform validation strategies in LLM-based autonomous driving research.

2) SYNTHETIC DATA GENERATION TOOLS

Recent advances also leverage synthetic data generation tools. DriveDreamer-2 [39] combines trajectory synthesis with multi-view video generation, while OmniDrive [40] and ChatSim [33] push the boundaries of vision-language alignment and editable scene rendering. These platforms demonstrate how LLMs can move beyond static datasets toward dynamic environments that integrate realism-enhancing pipelines, photorealistic rendering, and counterfactual reasoning.

Synthetic data generation tools extend simulators by prioritizing controllable realism and semantic alignment rather than control execution. DriveDreamer-2 emphasizes photorealistic multi-view synthesis for perception transfer, whereas OmniDrive and ChatSim explicitly target counterfactual reasoning and interactive scene manipulation. Compared to classical simulators, these tools better support language-conditioned variation and long-tail scenario generation, but remain largely disconnected from closed-loop safety validation. As a result, synthetic generators are best viewed as complementary to simulators, enriching training and evaluation data rather than replacing executable autonomy testbeds.

3) UAV AND SWARM SIMULATION FRAMEWORKS

In the UAV and swarm intelligence domain, Microsoft AirSim [29], and SITL [30] have proven indispensable. Works such as SwarmChain [20] and RALLY [21] exploit AirSim [29] and SITL [30] to evaluate collaborative inference and role-adaptive navigation strategies under resource-constrained swarm settings. For routing and scheduling, wildfire monitoring frameworks [23] and UAV-assisted data collection systems [2], [22] rely on tailored simulation setups to capture dynamic environmental conditions, including trade-offs between Age of Information (AoI) and energy efficiency. These tools enable researchers to integrate LLM reasoning with aerial autonomy, offering physics-aware, mission-oriented evaluation pipelines.

UAV simulation frameworks differ fundamentally from driving platforms in that they prioritize coordination, energy constraints, and communication over lane-level control. AirSim and SITL provide sufficient physical fidelity to evaluate LLM-assisted swarm behaviors, yet their support for high-level reasoning remains indirect and task-specific. Compared to driving simulators, UAV platforms expose a larger gap between symbolic LLM reasoning and real-time control, particularly under dynamic network and energy constraints. This highlights the need for tighter integration between reasoning agents and low-level flight controllers in aerial autonomy benchmarks.

4) COOPERATIVE AND CONNECTIVITY PLATFORMS

Cooperative and connectivity research benefits from Vehicle-to-Vehicle (V2V) and Vehicle-to-Infrastructure (V2I) benchmarks. V2V-LLM [17] introduces a novel cooperative planning dataset for LLM-driven perception fusion, while CoLMDriver [18] builds on CARLA-based InterDrive benchmarks for negotiation-driven V2V coordination. Similarly, hybrid frameworks such as LLM-DDQN for V2I optimization [19] highlight the importance of communication-aware testbeds that jointly assess data rates, handover management, and safety performance. These studies emphasize how communication simulators complement perception and decision-making simulators in advancing connected autonomy.

Connectivity-oriented platforms reveal that communication is not merely an auxiliary component but a first-class variable in LLM-enabled autonomy. The V2V- and V2I-focused benchmarks introduce negotiation, bandwidth constraints, and partial observability, which are absent in standalone simulators. Compared to perception-centric platforms, these environments better expose failure modes related to coordination, delayed reasoning, and trust among agents. However, their reliance on abstracted network models limits fidelity, underscoring the need for joint simulation of communication, perception, and control in future LLM-based cooperative autonomy research.

5) SIMULATION AND LLM-BASED DATASET GENERATION

Algorithm 1 outlines an iterative pipeline that combines LLM reasoning with driving and UAV simulators to generate safety-critical datasets. Structured prompts guide the LLM to generate scenario scripts that describe environments, agents, missions, and events. These scripts are executed in simulators such as CARLA [9], [12], [27], [63] and Apollo ADS [10], [28] for driving, or AirSim [20], [24], [29] and SITL [21], [30] for UAVs, yielding multimodal traces and logs. Domain-specific safety metrics such as time-to-collision, collisions,

Algorithm 1 LLM-Powered Dataset Generation for Driving and UAV Simulation

Require: **Domain** $\in$ {Driving, UAV}; **Simulator**; **LLM**; **PromptTemplate**; **FewShotExamples**; **SafetyRules**; **CoverageTarget**; **RandomSeed**

Ensure: **Dataset** of records: {ScenarioID, ScenarioScript, SensorTraces, Metrics, SafetyLabels}

Init: SetSeed(**RandomSeed**); Dataset $\leftarrow \emptyset$; Coverage $\leftarrow$ InitCoverage()
while not ReachedCoverage(**CoverageTarget**, Coverage) **do**
  Prompt $\leftarrow$ BuildPrompt(**PromptTemplate**, **FewShotExamples**, **Domain**, **SafetyRules**, Coverage)
  ScenarioScript $\leftarrow$ **LLM.GenerateScenario**(Prompt) ▷ defines map/weather/agents/goals/events
  (SensorTraces, RunLog) $\leftarrow$ **Simulator.Run**(ScenarioScript) ▷ e.g., trajectories, events, video
  Metrics $\leftarrow$ ComputeMetrics(**Domain**, SensorTraces) ▷ Driving: TTC, collisions, red-light/stop-line compliance; UAV: separation, stability, geofence
  SafetyLabels $\leftarrow$ ApplySafetyRules(Metrics, **SafetyRules**) ▷ e.g., Safe/Warning/Unsafe
  if not (IsScenarioValid(ScenarioScript) **and** PassQualityGates(Metrics, SafetyLabels)) **then**
    Prompt $\leftarrow$ **LLM.Refine**(Prompt, FindGaps(Metrics, SafetyLabels, Coverage))
    continue ▷ retry with improved prompt
  end if
  Dataset $\leftarrow$ Dataset $\cup$ { NewRecord(ScenarioScript, SensorTraces, Metrics, SafetyLabels) }
  Coverage $\leftarrow$ UpdateCoverage(Coverage, SafetyLabels)
end while
return Dataset

and traffic-rule compliance in driving [11], [33], or stability, separation, and geofence adherence in UAVs [23], [25] are computed and used to label outcomes. An iterative regeneration step ensures diversity and systematic coverage of the full incident lifecycle (pre-incident, hazard, conflict, incident, recovery) [15], [16], enriching the dataset with edge cases while preserving consistent annotation across domains.

Table 9 complements Algorithm 1 by mapping six core states: environment, agent, sensor, mission/rules, safety, and event to domain-specific safety metrics. Driving datasets such as nuScenes [31], [42], [66] and Waymo Open [33], [39], [82] emphasize layouts, traffic density, and perception, allowing LLMs to generate counterfactual scenarios for testing TTC thresholds and collision avoidance. UAV platforms like AirSim [29] and SITL [30] provide turbulence, swarm coordination, and geofence challenges, enabling stability and cooperative reasoning evaluation [20], [21], [24]. Embedding

TABLE 9. LLM-guided mapping of simulation states to safety metrics in driving and UAV domains.

State	Driving Metrics	UAV Metrics
Environment	Road layout, traffic density, weather, lighting (LLM prompts)	Wind/turbulence, terrain, visibility (LLM scripts)
Agent	Vehicle dynamics (speed, braking), lane keeping, driver intent	UAV dynamics (altitude, velocity), swarm coordination, role adaptation
Sensor	Camera/LiDAR/radar/GPS, V2X accuracy, LLM-based fusion	Camera/LiDAR/IMU/GPS, telemetry reliability, LLM-driven multisensor cases
Mission/Rules	Traffic signals, right-of-way, stop/yield, speed limits	Flight path, mission objectives, geofence enforcement
Safety	Time-to-collision, gap distance, lane discipline, collision type	Separation, altitude stability, geofence violation, energy margins
Event	Near-misses, hazard emergence, violations, crash/recovery	Hazard emergence, geofence breach, mid-air conflict, recovery

LLM reasoning across states ensures benchmarks remain adaptive, safety-oriented, and reproducible, offering a unified methodology for simulation-driven dataset generation.

From a cross-domain perspective, simulation-driven dataset generation pipelines reveal a unifying pattern: LLMs act most effectively as scenario composers and semantic validators, while simulators enforce physical and safety constraints. Driving platforms emphasize collision risk and rule compliance, whereas UAV simulators foreground stability, separation, and mission continuity. Despite domain differences, both settings converge on iterative LLM–simulation loops as the dominant methodology for producing scalable, safety-critical datasets. This convergence suggests that future benchmarks should prioritize standardized state abstractions and cross-domain safety metrics to enable reproducible evaluation of reasoning-driven autonomy.

6) LIMITATIONS AND OPEN CHALLENGES

Despite their versatility, simulation platforms still face limitations. Heavy reliance on synthetic setups raises concerns about transferability to real-world driving or flight conditions, as demonstrated in works that highlight the gap between diffusion-based adversarial scenarios and real-world deployments [42], or between multimodal reasoning agents and closed-loop testing [40]. Moreover, while tools like CARLA and AirSim [29] offer flexibility, they also impose high computational demands, especially when integrated with LLM-guided pipelines that require iterative reasoning or real-time feedback. Bridging these gaps through hybrid validation frameworks combining large-scale simulation, hardware-in-the-loop, and controlled field trials remains a key research direction.

B. DATASETS AND BENCHMARKS

In this subsection, we group recent datasets and benchmarks that assess LLM reasoning in autonomous systems. We categorize them into driving knowledge and theory exams, perception-centric QA, structured spatio-temporal and chain-of-thought reasoning, HD map-grounded QA,

TABLE 10. Review of recent datasets and benchmarks for LLMs in autonomous driving and UAV systems.

Work	Year	Dataset / Benchmark Details	Main Evaluation Results	LLM Reasoning	LLM Agents
Yao *et al.* [41] (AeroVerse)	2024	AerialAgent-Ego10k + CyberAgent-Ego500k; 5 downstream tasks; SkyAgentEval	Integrated 10+ VLMs; GPT-4 evaluation	Multi-task reasoning: spatial, planning, motion	Full UAV-agent pipeline, but simulation-driven
Wang *et al.* [103] (OpenUAV + UAV-Need-Help)	2024	12k UAV VLN trajectories; assistant-guided benchmark	UAV-LLM outperforms baselines but lags humans	Multimodal reasoning (multi-view + instructions)	UAV LLM-agent for hierarchical trajectories
Tang *et al.* [95]	2024	500+ driving MCQs; latency + cost analysis (CAVs)	GPT-4: >90% (MCQ + multimodal); Ernie 85%; smaller models fail	Reasoning + cost-efficiency trade-offs	No interactive agents; static evaluation
Ma *et al.* [99] (LaMPilot)	2024	LaMPilot-Bench: instruction → program synthesis	Off-the-shelf LLMs handle simple tasks; fail on complex instructions	Instruction-following reasoning	LaMPilot-Agent generates driving primitives
Qian *et al.* [32] (NuScenes-QA)	2024	34k scenes; 460k QA pairs (images + LiDAR)	Baselines fall short in temporal reasoning	Multimodal spatio-temporal reasoning	No agents; offline VQA focus
Marcu *et al.* [97] (LingoQA)	2024	28k short videos; 419k QA pairs	GPT-4V: 59.6% truthfulness vs humans 96.6%	Human-level truthfulness evaluation	No agents; only QA
Cao *et al.* [98] (MAPLM-QA)	2024	HD map QA dataset (point cloud + images + text)	CLIP + LLaMA-2/Vicuna baselines show gaps	Cross-modal spatial reasoning	No agent-in-loop yet
Sima *et al.* [36] (DriveLM)	2024	Graph VQA + DriveLM-Data; nuScenes + CARLA	DriveLM-Agent competitive with SOTA driving models	Graph-structured multi-step reasoning	DriveLM-Agent baseline supports E2E control
Ferrag *et al.* [104] (UAVBench / UAVBench_MCQ)	2025	50k validated UAV flight scenarios (JSON schema) + 50k MCQs	32 LLMs evaluated; strong in perception/policy reasoning	Multi-style reasoning (10 styles)	Agentic UAV reasoning
Lebioda *et al.* [43]	2025	Abstract requirements → CARLA code (simulation generation)	LLMs partially succeed; require human supervision	Translation of standards into simulation code	No agents; CARLA-only
Chiu *et al.* [17] (V2V-QA)	2025	Cooperative driving QA benchmark; multi-vehicle fusion	V2V-LLM > alternative fusion approaches	Multi-agent reasoning for planning	Cooperative agents (CAVs) tested
Yao *et al.* [102] (AGENTS-LLM)	2025	Scenario augmentation via LLMs; expert validation	Small LLMs achieve robust scenario enhancement	Natural language reasoning for scenario editing	Agent-based augmentation framework
Pei *et al.* [94]	2025	700+ MCQs + 54 hazard videos (UK Driving Theory + DVSA)	GPT-4o: 88.21% MCQ pass; all failed hazard perception	Tests factual + situational reasoning; lacks dynamic hazard interpretation	No agent modeling; limited to static Q&A
Zhao *et al.* [96]	2025	1.5k drone videos → 5.2k MCQs (recall, perception, reasoning, navigation)	17 Video-LLMs; strong on recall/navigation, weak on causal/counterfactual reasoning	Embodied cognition; fragmented multimodal reasoning	No agent interactivity; lacks planning/feedback
Wei *et al.* [35] (AD^2-Bench)	2025	5.4k annotated Chain-of-Thought (CoT) instances	SOTA MLLMs <60% accuracy	Fine-grained multi-step reasoning (text/point/region level)	No adaptive planning; no multi-agent negotiation
Xiao *et al.* [37] (UAV-ON)	2025	14 UE5 envs; 1,270 annotated target objects	Baselines fail in long-horizon navigation	Spatial/semantic grounding for ObjectNav	Single-agent Aerial ObjectNav Agent (AOA) tested
Fruhwirth-Reisinger *et al.* [34] (STSBench)	2025	43 multi-view scenarios; 971 MCQs (NuScenes)	VLMs weak on spatio-temporal dynamics	Multi-perspective temporal reasoning	Not interactive; lacks agent-in-loop
Guo *et al.* [38] (BEDI)	2025	UAV benchmark; 5 sub-skills: perception, control, planning, tool use	SOTA VLMs weak in coordinated UAV subtasks	Modular reasoning across UAV subtasks	UAV-EAs (embodied agents), but no real-time interaction
Zhou *et al.* [80]	2025	Scenario extraction/classification (motorway ADS)	GPT-4-1106-preview highest accuracy; others 40–60% errors	Semantic classification; robust to textual variation	Limited to text; no multimodal replay
Ferrag *et al.* [100] (α^3-Bench)	2026	113k conversational UAV mission episodes derived from UAVBench	17 LLMs evaluated	Conversational, safety-, network-, and efficiency-aware reasoning across 6 pillars	Fully agentic UAV controllers
Ferrag *et al.* [101] (α^3-SecBench)	2026	20k+ security-overlay attack scenarios	23 LLMs evaluated; overall scores 12.9–57.1%	Security-aware conversational reasoning	Fully agentic UAV control

scenario extraction and augmentation pipelines, instruction-following benchmarks, cooperative CAV/V2V datasets, and UAV-embodied intelligence benchmarks. Table 10 presents a comparative review of recent benchmarks and datasets for LLMs in autonomous driving and UAV systems. Early works emphasize rule-based knowledge and MCQ evaluations [94], [95], while newer contributions focus on multimodal reasoning, spatio-temporal dynamics, and Chain-of-Thought analysis [32], [34], [35], [97]. UAV-oriented benchmarks extend evaluation to aerial domains, highlighting challenges in long-horizon navigation and embodied intelligence [37], [38], [41], [103]. More recent frameworks, such as DriveLM [36], LaMPilot [99], and V2V-QA [17], integrate LLM reasoning with agent-based decision-making, marking a shift toward interactive and multi-agent autonomy. Nevertheless, most benchmarks remain simulation-heavy and limited to static or text-based settings, lacking real-world validation, adaptive decision-making, and multi-agent negotiation [43], [96], [102]. Figure 8 illustrates a taxonomy of datasets and benchmarks, organizing existing works into categories such as driving theory, perception-centric QA, structured reasoning, HD map QA, scenario pipelines, instruction synthesis, cooperative CAV reasoning, and UAV intelligence."

Across driving and UAV domains, existing datasets and benchmarks reveal a clear methodological progression: early evaluations prioritize rule-based knowledge and static correctness, whereas recent benchmarks increasingly emphasize multimodal grounding, spatio-temporal dependencies, and explicit reasoning traces. However, despite this evolution, most benchmarks remain decoupled from closed-loop autonomy, offering limited support for adaptive decision-making, multi-agent coordination, or real-time feedback. This gap suggests that current benchmarks are well-suited for diagnosing reasoning competence, but insufficient for evaluating end-to-end autonomy, motivating the need for agent-in-the-loop and interaction-aware benchmarks.

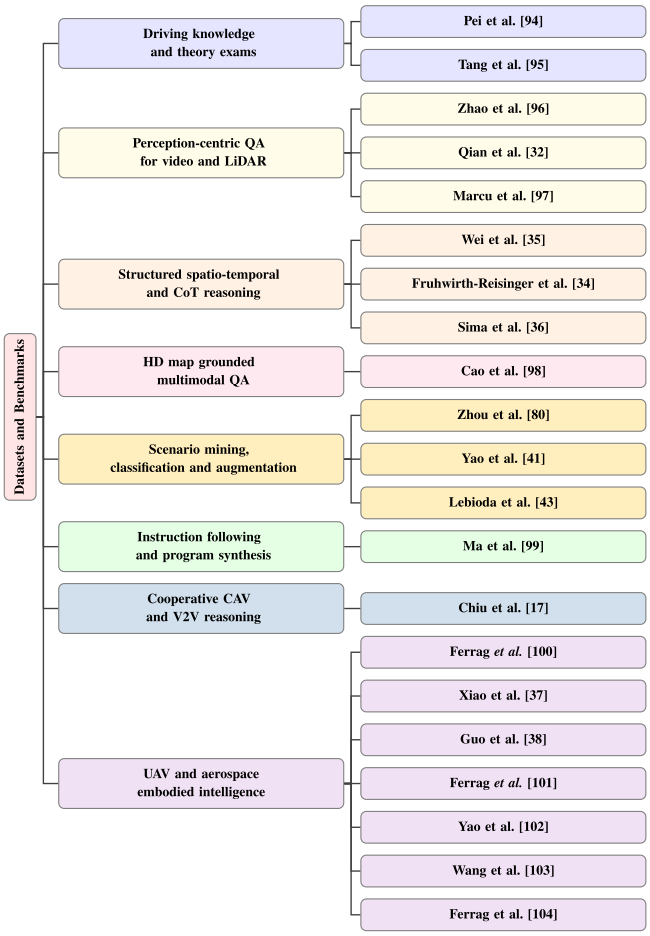


FIGURE 8. Taxonomy of datasets and benchmarks for autonomous driving and UAV applications.

1) DRIVING KNOWLEDGE AND THEORY EXAMS

Pei et al. [94] introduce the first benchmark to assess LLMs on driving theory and hazard perception, combining over 700 MCQs and 54 video tests from the UK driving exam with DVSA question sets. While GPT-4o achieved 88.21% accuracy on MCQs, all models failed the hazard perception task, exposing limits in dynamic reasoning. Tang et al. [95] extend this line of work by evaluating proprietary and open-source models on 500 theory questions, incorporating accuracy, cost, and latency for edge-assisted CAVs. Their results highlight GPT-4's strong performance but high cost, with GPT-4o excelling in multimodal reasoning (96%) and lightweight models lagging behind. Together, these studies establish important baselines but remain limited to static tests, underscoring the need for benchmarks with interactive, multimodal, and cooperative evaluation to better guide LLM reasoning in autonomous driving.

Driving theory benchmarks provide valuable diagnostics for rule comprehension and hazard awareness, but their reliance on static MCQs and predefined answers fundamentally limits their ability to assess situational reasoning under uncertainty. Compared to multimodal and spatiotemporal benchmarks, these exams primarily serve as lower-bound sanity checks rather than predictors of autonomous driving competence, underscoring their complementary yet insufficient role in evaluating LLM-powered driving systems.

2) PERCEPTION-CENTRIC QA FOR VIDEO AND LiDAR

Zhao et al. [96] introduce a benchmark of 1.5k drone-captured video clips transformed into 5.2k questions to evaluate Video-LLMs on recall, perception, reasoning, and navigation, revealing strong Sim-to-Real transfer but limited support for interactive decision-making. Qian et al. [32] develop NuScenes-QA, a large-scale multimodal VQA benchmark built on camera and LiDAR data with 460k QA pairs, advancing spatial and temporal reasoning evaluation though constrained by template-based questions. Marcu et al. [97] propose LingoQA, comprising 28k driving video scenarios and 419k annotations, with a novel Lingo-Judge metric closely aligned with human judgment, exposing large gaps between human (96.6%) and GPT-4 (59.6%) truthfulness. Collectively, these benchmarks expand multimodal evaluation for autonomous driving but remain limited by static or template-based designs, underscoring the need for interactive, agent-in-the-loop, and real-time reasoning tasks to better test LLM-powered driving agents.

Perception-centric QA benchmarks significantly advance multimodal evaluation by incorporating video and LiDAR data, yet they largely frame reasoning as retrospective question answering rather than prospective decision-making. While these datasets outperform theory-based exams at testing spatial and temporal understanding, their static, template-driven nature limits the assessment of interactive behaviors, suggesting that perception QA alone cannot fully characterize agent-level reasoning in autonomous systems.

3) STRUCTURED SPATIO-TEMPORAL AND CHAIN-OF-THOUGHT REASONING

Wei et al. [35] propose AD^2-Bench, the first CoT benchmark for autonomous driving, featuring 5.4k fine-grained reasoning traces that expose current MLLMs' weaknesses (sub-60% accuracy) in multi-step reasoning under adverse conditions. Fruhwirth-Reisinger et al. [34] introduce STSBench, a scenario-driven framework with 971 human-validated MCQs from NuScenes, highlighting spatio-temporal reasoning gaps in dynamic traffic interactions. Sima et al. [36] present DriveLM, combining Graph VQA, a structured benchmark (DriveLM-Data), and a baseline agent (DriveLM-Agent) to link perception, prediction, and planning, showing competitive results with strong zero-shot generalization. Together, these benchmarks move beyond static theory tests by foregrounding reasoning transparency, spatio-temporal dependencies, and structured planning, yet all remain limited by the absence of interactive, agent-in-the-loop, or cooperative tasks crucial for realistic LLM-powered autonomous driving.

Benchmarks that emphasize structured reasoning and Chain-of-Thought represent a critical shift toward

interpretability and transparency of reasoning. Compared to perception-centric QA, these datasets better expose multi-step failures and causal inconsistencies in LLM outputs. Nevertheless, their focus on offline reasoning traces limits their ability to capture how reasoning evolves under continuous interaction, underscoring the need to extend CoT-based benchmarks toward closed-loop and cooperative settings.

4) HD MAP GROUNDED MULTIMODAL QA

Cao et al. [98] present MAPLM-QA, a vision–language benchmark designed to support cross-modal reasoning for traffic and high-definition (HD) map systems, an area underrepresented in existing datasets. Unlike prior VQA benchmarks, such as NuScenes-QA and LingoQA, which focus on perception and reasoning from driving scenes, MAPLM-QA integrates point clouds, images, and language pairs, enabling models to align multimodal spatial information with linguistic instructions. The benchmark is built by annotating large-scale, broad-coverage HD map data, creating an instruction-following dataset suitable for finetuning domain-specific foundation models. Using CLIP in conjunction with LLaMA-2/Vicuna as baselines, the authors demonstrate that vision–language instruction-tuning can yield meaningful map-grounded representations, although performance evaluations reveal considerable gaps between current methods and the desired reasoning capabilities. In contrast with V2V-QA, which emphasizes cooperative perception, and LingoQA, which stresses truthfulness in QA, MAPLM-QA contributes a map-centric multimodal dimension, directly relevant to navigation, localization, and planning. However, the benchmark remains limited in scope to offline QA pairs and lacks support for interactive spatial reasoning or real-time map updates, which are critical in dynamic driving environments. Future extensions could integrate agent-in-the-loop HD map reasoning, incremental map updates, and planning-driven QA tasks, making MAPLM-QA a more powerful tool for assessing LLM and AI agent reasoning in mapping-intensive autonomous driving systems.

Map-grounded QA benchmarks uniquely target the alignment between spatial representations and linguistic reasoning, a capability largely absent from perception-only datasets. Compared to cooperative V2V benchmarks, MAP-centric datasets emphasize localization and navigation fidelity over inter-agent reasoning. However, their offline formulation constrains temporal adaptation, suggesting that HD map QA should be integrated with agent-based planning tasks to fully evaluate reasoning in dynamic environments.

5) SCENARIO MINING, CLASSIFICATION AND AUGMENTATION PIPELINES

Zhou et al. [80] benchmark LLMs for motorway scenario extraction and classification, showing that GPT-4 variants outperform others in robustness to linguistic variation, positioning LLMs as semantic intermediaries for ADS safety validation, though limited to textual inputs. Yao et al. [41] propose AeroVerse, a large-scale UAV benchmark ecosystem integrating real and synthetic datasets, downstream tasks, and a unified evaluation pipeline, advancing embodied aerial intelligence beyond prior UAV benchmarks but remaining simulation-centric. Lebioda et al. [43] explore translating abstract automotive requirements into CARLA simulation code, highlighting both the potential and limitations of LLMs in bridging regulatory language with executable scenarios. Collectively, these works broaden the scope of evaluation from semantic scenario extraction to aerospace autonomy and requirement-to-code translation, yet all lack multimodal, interactive, and cross-simulator integration needed for robust LLM-powered autonomous systems.

Scenario mining and augmentation pipelines shift evaluation from answering questions to generating and classifying safety-critical situations. Compared to QA-based benchmarks, these pipelines more directly support safety validation and regulatory compliance, yet their reliance on textual or simulator-specific abstractions limits generalization. Bridging these methods with multimodal and cross-simulator benchmarks remains essential for scalable and robust evaluation of LLM reasoning in autonomous systems.

6) INSTRUCTION FOLLOWING AND PROGRAM SYNTHESIS FOR DRIVING

Ma et al. [99] propose LaMPilot, a framework that integrates LLMs into autonomous driving (AD) pipelines, enabling vehicles to interpret and execute spontaneous natural language instructions through program synthesis. Unlike conventional AD benchmarks that focus on perception, planning, or scenario reasoning, LaMPilot addresses the challenge of instruction-following, where user commands such as ‘‘overtake the car ahead’’ are translated into executable code built from predefined driving primitives. To support systematic evaluation, the authors introduce LaMPilot-Bench, the first dataset explicitly designed to measure the effectiveness of LLM-generated driving programs across a diverse set of driving instructions and scenarios. Experiments with off-the-shelf LLMs show promising results, with models demonstrating competence in mapping instructions to functional actions, though performance varies with the complexity of the driving tasks. Compared with earlier works such as STSBench, which assesses spatio-temporal reasoning, and the driving theory benchmarks, which evaluate rule-based knowledge, LaMPilot-Bench provides a complementary dimension by testing instruction grounding and program synthesis for driving actions. Its main limitation lies in its abstraction: the framework operates through predefined functional primitives rather than open-ended control, meaning it does not yet evaluate whether LLMs can generate novel strategies or adapt to unstructured conditions. Extending LaMPilot-Bench toward interactive closed-loop simulations and multi-turn instruction handling would significantly increase its relevance for benchmarking reasoning-driven AI agents in autonomous driving systems.

Instruction-following benchmarks introduce an important interaction dimension by testing whether LLMs can

ground natural language commands into executable actions. Unlike spatio-temporal reasoning benchmarks, which assess descriptive reasoning, program synthesis benchmarks probe actionable understanding. However, their reliance on predefined primitives limits behavioral diversity, suggesting that future benchmarks should jointly evaluate instruction grounding, adaptive planning, and real-time execution.

7) COOPERATIVE CAV AND V2V REASONING

Chiu et al. [17] propose V2V-QA, a benchmark designed to explore cooperative autonomous driving through vehicle-to-vehicle (V2V) communication mediated by multimodal LLMs. Unlike conventional cooperative perception approaches that emphasize object detection or tracking, V2V-QA extends the scope to planning-oriented reasoning, evaluating how information exchanged between connected autonomous vehicles (CAVs) can enhance joint decision-making. The benchmark defines tasks across grounding, notable object identification, and cooperative planning, while the baseline model, V2V-LLM, fuses perception data from multiple vehicles and answers structured driving-related questions. Experimental results show that V2V-LLM outperforms alternative fusion approaches, indicating the potential of LLM-based architectures to unify perception and planning under a cooperative paradigm. Compared to earlier works such as STSBench, which tests spatio-temporal reasoning from ego-vehicle perspectives, and AD2-Bench, which focuses on CoT reasoning under adverse conditions, V2V-QA uniquely positions reasoning within a multi-agent communication framework, addressing occlusion and sensor-failure scenarios that individual-vehicle datasets cannot capture. While the benchmark provides a compelling proof of concept, it currently abstracts away critical real-world aspects such as communication latency, bandwidth constraints, and adversarial V2V signals. Future work should incorporate realistic communication models and adversarial robustness testing to enhance the applicability of this approach for benchmarking LLM reasoning and AI agents in cooperative autonomous driving ecosystems.

Cooperative CAV benchmarks uniquely position reasoning within a multi-agent context, exposing limitations that ego-centric datasets cannot reveal. Compared to individual-vehicle benchmarks, V2V-focused datasets better capture occlusion handling and shared situational awareness, but they abstract away communication uncertainty and adversarial behavior. Incorporating realistic networking constraints is therefore critical for advancing cooperative LLM-based autonomy.

8) UAV AND AEROSPACE EMBODIED INTELLIGENCE

Xiao et al. [37] introduce UAV-ON, a large-scale benchmark for aerial Object Goal Navigation, highlighting challenges in long-horizon semantic exploration across 14 complex environments. Guo et al. [38] propose BEDI, a modular framework decomposing UAV intelligence into perception, control, and planning sub-skills, offering a systematic evaluation platform but lacking multi-agent adaptability. Yao et al. [102] present an LLM-driven framework for scenario augmentation in ADS testing, enabling scalable, intent-aligned test enrichment, albeit with limitations to text-based generation. Wang et al. [103] advance UAV Vision-Language Navigation with OpenUAV, a 12k-trajectory dataset, and UAV-Need-Help, integrating assistant-guided navigation, yet performance gaps to humans remain. Collectively, these works expand UAV and ADS benchmarks from semantic navigation to embodied intelligence and scenario generation; however, all still fall short in real-world, multimodal, and resource-constrained settings, which are essential for robust LLM-powered agents.

UAVBench [104] introduces a large-scale, physically grounded benchmark for systematically evaluating reasoning and agentic intelligence in autonomous aerial systems. The dataset comprises 50,000 validated UAV flight scenarios generated via taxonomy-guided LLM prompting and multi-stage safety validation, each encoded in a structured JSON schema capturing mission objectives, vehicle configuration, environment, airspace, and risk annotations. Building on this foundation, UAVBench_MCQ provides an additional 50,000 structured multiple-choice questions spanning ten reasoning styles, including aerodynamics, navigation, energy management, ethics, and multi-agent coordination. A comprehensive evaluation of 32 state-of-the-art LLMs reveals strong performance in perception and policy reasoning, but persistent weaknesses in ethics-aware, resource-constrained, and cooperative decision-making. By unifying physical realism, structured reasoning, and interpretable risk labeling, UAVBench advances beyond prior UAV benchmarks that focus on isolated tasks or low-fidelity simulations, offering a reproducible foundation for benchmarking reasoning-driven UAV autonomy.

Ferrag et al. [100] introduce α^3-Bench, a large-scale benchmark for evaluating conversational LLM-based UAV agents operating under dynamic 6G network conditions. The benchmark comprises 113k multi-turn UAV mission episodes derived from UAVBench scenarios [104], modeling language-mediated control loops with strict safety, protocol, and schema constraints. α^3-Bench evaluates 17 state-of-the-art LLMs using a composite α^3 metric that integrates six pillars, including task outcome, safety policy compliance, network robustness, and communication efficiency. Experimental results show near-perfect mission success and safety adherence for frontier models, while revealing up to 30–40% degradation under adverse 6G conditions and more than 2× variance in efficiency-normalized performance. Building directly on this conversational and network-aware evaluation framework, Ferrag et al. [101] extend α^3-Bench to adversarial settings through α^3-SecBench, introducing a security-centric benchmark for LLM-based UAV autonomy under 6G conditions. The framework augments the original 113k conversational UAV missions with more than 20,000 validated security overlays that inject controlled attacks across seven autonomy layers, enabling systematic evaluation

of security, resilience, and trust. By evaluating 23 state-of-the-art LLMs using CWE-aligned threat taxonomies and time-bounded safe-degradation requirements, α^3-SecBench reveals a pronounced gap between accurate attack detection and reliable security-aware decision-making, with normalized overall performance ranging from 12.9% to 57.1%.

UAV and aerospace benchmarks extend autonomous reasoning evaluation beyond ground vehicles by introducing long-horizon navigation, energy constraints, and embodied interaction. Compared to driving benchmarks, UAV datasets place greater emphasis on exploration, mission continuity, and safety-aware operation under physical constraints. While recent benchmarks such as UAVBench are beginning to address multi-agent coordination and risk-grounded evaluation, most existing datasets remain limited in terms of real-world interactivity and closed-loop deployment. This convergence suggests that unified embodied intelligence benchmarks spanning driving and aerial domains are necessary to rigorously evaluate general-purpose LLM-based agents.

C. COMPUTATIONAL COMPLEXITY, SENSITIVITY, AND REAL-TIME FEASIBILITY

Table 12 provides a qualitative synthesis of sensitivity, scalability, computational complexity, and real-time feasibility trends reported across LLM-enabled autonomous systems. Reasoning-only LLM pipelines exhibit relatively high robustness and real-time feasibility but limited embodiment, while tightly coupled LLM-agent systems are more sensitive to traffic density, perception noise, and agent population size, particularly in multi-agent or swarm settings [119]. Computational cost and latency increase with model scale and interaction complexity, motivating dual-rate architectures, policy distillation, and selective LLM invocation [125]. These trends indicate that real-time feasibility is achievable primarily when LLM reasoning is decoupled from high-frequency control, underscoring the importance of sensitivity-aware evaluation and deployment-oriented benchmarking [20].

From a computational perspective, reasoning-centric LLMs typically introduce higher inference latency and memory overhead compared to conventional learning-based planners, especially when deployed with long-context prompts [124], chain-of-thought reasoning [121], or multimodal inputs [130]. To address real-time constraints, many systems adopt dual-rate architectures, policy distillation [131], lightweight agent designs [132], or hybrid edge–cloud execution [126], where high-frequency control remains model-based or learned, while low-frequency LLM reasoning is invoked only at critical decision points [85]. These findings collectively indicate that scalability, robustness to noise, and computational efficiency remain open challenges for LLM-enabled autonomy. As a result, sensitivity analysis, ablation studies, and real-time benchmarking are increasingly recognized as essential components of future evaluation frameworks, particularly for safety-critical autonomous driving and UAV swarm deployments.

D. ASSUMPTIONS, SCALABILITY, AND REPRODUCIBILITY

Beyond individual Large Language Models (LLMs), modern autonomous systems increasingly rely on heterogeneous system nodes including autonomous vehicles, UAVs, robotic platforms, and edge/cloud infrastructure coordinated through AI agent architectures that integrate reasoning, memory, tool use, and agent-to-agent (A2A) communication via structured protocols [2], [25], [45], [47], [83], [92], [123]. While these paradigms significantly expand system capabilities, they also introduce new challenges related to modeling assumptions, scalability, real-time operation, and experimental reproducibility. These challenges are summarized at a system level in Table 13.

- **Modeling Assumptions in LLM- and Agent-Based Pipelines.** Many LLM- and agent-enabled autonomous systems assume abstracted perception inputs (e.g., object-level or map-level representations), reliable tool execution, and well-defined interfaces between perception, reasoning, and control modules across autonomous vehicles, UAVs, and robotic platforms [2], [14], [24], [25], [40], [48], [116]. In agent-based frameworks, additional assumptions are often made regarding deterministic tool invocation, stable memory states, and consistent agent coordination through middleware or Model Context Protocol (MCP)-style interfaces [52], [83], [86], [88], [93]. Reported evaluations indicate that violations of these assumptions such as noisy perception, partial observability, tool execution delays, or inconsistent agent state synchronization can degrade reasoning consistency, planning reliability, and multi-agent coordination, particularly in dense traffic or complex swarm scenarios [20], [58], [76], [101], [119].
- **Context Management and Tool-Oriented Reasoning.** LLM-driven agents typically operate within fixed context windows, commonly ranging from 4k to 16k tokens, which must jointly encode environment state, agent memory, tool outputs, and A2A messages. As context length increases driven by richer perception, larger agent populations, or more frequent tool interactions inference latency and memory usage grow accordingly [3], [121], [124]. Recent work on prompt compression and efficient context management highlights both the potential and the limitations of long-context LLMs in autonomous systems [8], [44], [124]. Managing context truncation, summarization, and selective tool grounding therefore remains a critical challenge for scalable agent-based autonomy across both ground and aerial systems [90], [123], [132].
- **Scalability of Multi-Agent and A2A Coordination.** Most current evaluations consider small to moderate numbers of AI agents (typically 2–10), coordinated via explicit A2A communication or centralized planners [17], [76], [131]. As the number of autonomous vehicles, UAVs, or robotic agents increases, reasoning

TABLE 11. Taxonomy of security challenges and future directions in LLM-enabled autonomous systems.

Domain	Representative Systems	Key Security Challenges	Adversary Actions	Consequences	Future Directions
Scenario Generation and Simulation	CARLA-based generation [9], LEADE [10], LLM-Attacker [59], LD-Scene [42], DriveDreamer-2 [39], OmniDrive [40], ChatSim	Malicious scenario crafting, poisoning in diffusion pipelines, DoS from heavy rendering, distributional shifts, supply-chain risks	Craft hazardous scenarios, bypass filters with adversarial prompts, inject corrupted assets	Unsafe evaluation, biased datasets, degraded trust in simulators	Cross-platform validation; adversarially robust pipelines; secure data provenance; anomaly detection in outputs; automated red teaming for simulation pipelines [105]–[108]
Decision-Making and Planning	DriveGPT [60], LeAD [12], RLHF frameworks [61], TeLL-Drive [13], RAPID [64], AEIA [109]	Hallucinated reasoning, latency exploitation, feedback poisoning, reward manipulation, policy extraction, adversarial context injection	Manipulate reward models and corrupt preference data	Unsafe plans, delayed reactions, systemic bias propagation	Reasoning validators; secure human-in-the-loop alignment; adversarial training; hybrid symbolic–LLM planning; robust reward shaping [110]
Perception and Multimodal Representation	Atlas [15], Occ-LLM [66], SenserAG [16], Traj-LLM [72], multimodal fusion pipelines [14]	Adversarial point cloud attacks, occupancy poisoning, cross-modal inconsistencies, RAG poisoning, membership inference, trajectory poisoning	Tamper with LiDAR/RADAR streams, poison the datastore, inject unrealistic trajectories	Misleading perception, unsafe predictions, privacy breaches	Secure multimodal fusion; verifiable trajectory datasets; privacy-preserving RAG; multimodal security benchmarks with stress-testing [111]
Human Alignment and Driver Interaction	Driving style alignment [73], SurrealDriver [53], Driver Mate [74], evaluation frameworks [58]	Dataset poisoning, biased/verbalized driver reports, adversarial voice injection, evaluation prompt manipulation, privacy leakage	Corrupt demos, mislead conversational agents, extract sensitive data, bias scoring	Unsafe behavior alignment, distracted drivers, data breaches, ethical risks	Privacy-preserving alignment protocols; secure conversational AI; explainable evaluation; ethical governance for driver-agent interactions [112]
UAV and Swarm Autonomy	SwarmChain [20], RALLY [21], LAURA [22], FR-SICL [23], LLVM-Drone [24], federated UAV swarms [113]	Semantic poisoning, poisoned updates, DoS in low-power devices, adversarial routing, hallucination errors, prompt injection, contagion attacks	Exploit comm bottlenecks, corrupt role assignments, false routing signals, adversarial instructions	Swarm destabilization, unsafe missions, comm collapse, poisoning propagation	Distributed trust models; secure UAV communication; adversarially robust routing; resilient federated UAV learning; encrypted multi-agent workflows [114]
Benchmarks and Evaluation	Driving theory MCQs [94], AD^2-Bench [35], STSBench [34], V2V-QA [17], AeroVerse [41]	Lack of adversarial test cases, limited cooperative reasoning, no protocol-level stress tests	Evade benchmark checks, exploit abstraction gaps	Overestimate safety, unprepared systems	Security-centric benchmarks: adversarial MCQs; interactive red-team simulators; multimodal stress-tests; robustness scorecards; reproducible red-team suites [115]

TABLE 12. Sensitivity and real-time analysis of LLM-based autonomy.

Paradigm	Sensitivity	Scalability	Compute Cost	Real-Time
LLM Reasoning Only	Medium	High	Low–Medium	High
LLM + Single-Agent Control	Medium–High	Medium	Medium	Medium
LLM + Multi-Agent / Swarm	High	Low–Medium	High	Low
Dual-Rate / Distilled LLM Agents	Low–Medium	High	Low	High

latency, communication overhead, and coordination complexity grow rapidly due to expanded context, higher message exchange rates, and more complex negotiation or consensus mechanisms [20], [84], [120], [125]. These scalability limitations are particularly pronounced in cooperative driving, V2X-enabled autonomy, and UAV swarm systems that rely on continuous A2A interactions over constrained wireless links [47], [92], [100], [119].

- **Real-Time Constraints in Agent-Oriented Architectures.** In practice, LLM reasoning and agent-level planning are invoked at low frequency (commonly below 1–2 Hz), while high-frequency perception and control loops operate independently on autonomous vehicles and UAV platforms [24], [40], [87], [117]. Agent-based systems further introduce latency associated with tool calls, MCP-style orchestration, and inter-agent messaging [52], [83], [85], [88]. Empirical evidence from edge and multi-LLM orchestration studies suggests that tightly coupling LLM reasoning, tool execution, and A2A communication at higher rates can violate real-time constraints, motivating asynchronous, event-driven, or dual-rate agent architectures, often supported by edge or cloud offloading [8], [44], [84], [126], [133].
- **Reproducibility in LLM-, Agent-, and Protocol-Based Systems.** Reproducibility challenges extend beyond model architecture to include agent design choices, tool interfaces, MCP-style coordination logic, and A2A communication protocols deployed across autonomous system nodes [52], [83], [88], [113]. Critical parameters influencing reproducibility include prompt templates, context window size, agent memory policies, tool invocation frequency, message formats, communication latency assumptions, number of agents, traffic density, simulator configurations, and random seeds. Benchmarking and evaluation efforts in both autonomous driving and UAV domains such as Lampilot, STSBench, AD$\hat{2}$-Bench, DriveDreamer-2, Bench2ADVLM, AeroVerse, BEDI, UAV-ON, UAVBench, and the α^3 benchmark suites underscore the importance of explicitly documenting these choices

TABLE 13. Assumptions, scalability, and reproducibility across autonomous system nodes.

System Dimension	Common Assumptions	Scalability / Real-Time Impact	Reproducibility-Critical Parameters
Autonomous Vehicle Nodes	Abstracted perception and scene representations; low-frequency LLM or FM-based reasoning in the loop [40], [45], [47], [48], [116], [117]	Dense traffic and complex scenarios increase reasoning and coordination load, stressing latency budgets [58], [60], [76]	Traffic density, sensor noise model, control-loop frequency, benchmark configuration (e.g., nuScenes, STSBench, AD$\hat{2}$-Bench) [31], [32], [34], [35], [99]
UAV / Aerial Nodes	Reliable localization and communication; structured mission descriptions for LLM-guided planning [2], [24], [25], [104], [118]	Swarm size and mission complexity amplify latency and coordination overhead in agentic UAV control [20], [21], [101], [119], [120]	Swarm size, communication delay, update rate, scenario benchmarks (e.g., AeroVerse, BEDI, UAV-ON, UAVBench, α^3-Bench) [37], [38], [41], [100], [104]
Robotic / Embodied Agents	Stable task execution, tool availability, and consistent sensorimotor loops for LLM-guided agents [24], [71], [90], [90]	Task complexity and embodied interaction length increase reasoning cost and context requirements [41], [90], [96]	Task graphs, tool sets, execution latency, environment, and dataset configuration [37], [41], [96]
LLM Reasoning Modules	Fixed context windows (typically 4k–16k tokens); chain-of-thought or tool-augmented prompting [3], [121]–[123]	Inference latency and memory footprint grow with context length and prompt complexity; high call rates are difficult to sustain in real time [8], [26], [44], [124]	Context size, token budget, prompt templates, reasoning strategy (CoT, tool use) [3], [121], [122], [124]
AI Agent / MCP Orchestration	Deterministic tool and memory access; structured orchestration via MCP-style or agent frameworks [52], [83], [85], [86], [88], [93]	Tool and memory latency affect planning stability and responsiveness; more tools and skills increase orchestration overhead [85], [92], [93], [125]	Tool set, memory policy, invocation rate, orchestration logic, and security assumptions [52], [86], [88], [113]
A2A Communication	Reliable, synchronized messaging among agents and system nodes (vehicles, UAVs, edge servers) [83], [84], [89], [91], [92]	Delays on the order of 10–100 ms and packet loss $>$5–10% degrade cooperative coordination and safety margins [20], [119], [120], [125]	Latency, packet loss rate, message size, update frequency, V2X/A2A protocol configuration [83], [89], [91], [92]
Edge / Cloud Nodes	Stable offloading and connectivity for medium- to large-scale LLMs and multi-LLM orchestration [8], [44], [84], [126]	Network congestion and contention impact end-to-end response time and service-level guarantees [1], [26], [84], [126]	Offload policy, bandwidth, round-trip time, model placement and scaling configuration [8], [44], [126]
Reproducibility	Fixed simulators, datasets, and prompt sets for evaluation [34], [35], [99], [104], [127], [128]	Results are sensitive to simulation configuration, prompt design, and benchmark selection [41], [99]–[101], [129]	Prompts, seeds, simulator version, benchmark choice, scenario generation pipeline, and evaluation metrics [34], [99], [104], [127], [129]

for fair comparison and longitudinal analysis [34], [35], [37], [38], [39], [41], [99], [100], [101], [104], [129].

E. SUMMARY AND LESSONS LEARNED

Across driving and UAV domains, the surveyed simulation platforms and benchmarks form a heterogeneous but increasingly interconnected infrastructure for studying LLM-based autonomous systems. Driving and UAV simulators provide controllable, physics-aware environments, synthetic generators enrich data diversity and long-tail coverage, and emerging V2V/V2I and 6G-aware testbeds begin to expose coordination and communication effects. At the same time, most tools privilege either realism, controllability, or reasoning coverage, but rarely all three, leading to fragmented evaluations that are difficult to compare across domains and autonomy stacks. Current datasets and benchmarks have evolved from static, rule-based tests toward multimodal, CoT-oriented, and agent-centric evaluations, yet they remain largely decoupled from closed-loop, real-world deployments. A key lesson is that future progress will hinge on cross-platform, cross-domain validation pipelines with standardized state abstractions and safety metrics, enabling reproducible assessment of reasoning, agency, communication, and robustness within a unified view of autonomous systems.

V. SECURITY CHALLENGES AND FUTURE DIRECTION

The integration of LLMs and AI agents into autonomous systems introduces unprecedented opportunities for reasoning, adaptability, and multimodal decision-making. However, these advances also expand the attack surface, raising new security and reliability risks across the entire autonomy pipeline from scenario generation and decision-making to perception, human alignment, and swarm-based coordination. Unlike traditional modular architectures, LLM-driven systems are inherently vulnerable to adversarial prompt injection, data poisoning, multimodal perturbations, and privacy leakage, making their safe deployment in safety-critical environments an open research challenge. To provide a structured view of these risks, Table 11 summarizes key domains, representative systems, security challenges, adversary actions, consequences, and future research directions.

In this section, we categorize the security landscape and forward-looking directions for LLM-enabled autonomy. We synthesize adversarial risks in scenario generation, vulnerabilities across decision-making and planning, threats to perception and multimodal representation, human-alignment and interaction risks, and challenges unique to UAV/swarm autonomy. We then outline security-oriented evaluation needs and future deployment pathways spanning onboard, edge, and cloud to guide robust, trustworthy, and practical LLM integration in autonomous systems.

A. ADVERSARIAL RISKS IN SCENARIO GENERATION AND SIMULATION

LLM-driven scenario generation and driving environment simulation have greatly advanced the evaluation of autonomous systems, but they also raise significant security concerns. Automated scenario frameworks, such as CARLA-based generation [9], LEADE evolutionary search [10],

and adversarial setups like LLM-Attacker [59], demonstrate how LLMs can expose safety-critical edge cases, yet they also open avenues for malicious exploitation where adversaries deliberately craft hazardous situations to degrade system performance. Diffusion-based approaches, such as LD-Scene [42], introduce additional risks due to controllability and code-guided adversarial losses, which can be manipulated for poisoning or injection attacks. Likewise, environment-focused systems like DriveDreamer-2 [39] and OmniDrive [40] highlight new threats, including denial-of-service (DoS) vulnerabilities from computationally intensive video rendering, distributional shifts when relying on synthetic datasets, and inconsistencies in counterfactual reasoning that could lead to unsafe decision-making. Beyond these, multimodal adversarial techniques such as MMA-Diffusion [134] can bypass prompt filters and safety checkers, generating adversarial content within simulation pipelines. Moreover, natural language-driven 3D editing frameworks such as ChatSim increase the attack surface by integrating external assets, thereby making them susceptible to supply-chain and data-integrity threats. These challenges underscore the urgent need for cross-platform validation, robustness mechanisms against adversarial attacks, and secure simulation pipelines to ensure that LLM-based driving scenarios and environments remain trustworthy tools for safety evaluation in autonomous systems.

B. VULNERABILITIES IN DECISION-MAKING AND PLANNING PIPELINES

LLM-driven decision-making and planning architectures promise human-like reasoning and adaptability in autonomous systems, yet they expose a new layer of security vulnerabilities. Reasoning-based frameworks that integrate LLMs into closed-loop control [60] raise concerns over hallucinations or unsafe reasoning chains that could propagate critical errors into real-world actions. Dual-rate systems such as LeAD [12], which combine high-frequency perception modules with low-frequency LLM reasoning, are vulnerable to latency exploitation and adversarial desynchronization, where attackers can deliberately exploit timing mismatches between subsystems. Learning integration methods, including RLHF-guided frameworks [61] and teacher–student strategies like TeLL-Drive [13], introduce vulnerabilities in human-in-the-loop settings, such as poisoning feedback signals or corrupting preference data, undermining safety alignment. Similarly, hierarchical reinforcement learning approaches [63] expose risks in reward-model manipulation and semantic goal misinterpretation, where adversaries could craft deceptive textual prompts or adversarial scenarios to induce unsafe meta-actions. Distillation-based systems, such as RAPID [64], risk inheriting and amplifying biases or vulnerabilities embedded in LLM-generated knowledge, and pose additional threats when distilled policies are deployed at scale through policy extraction attacks. Moreover, environment injection attacks (AEIA) [109] highlight how adversaries may insert adversarial cues into operational contexts, manipulating both reasoning chains and planning outcomes. These challenges highlight the critical need for secure reasoning validation, robust preference alignment mechanisms, and adversarially resilient planning pipelines to ensure that LLM-enhanced decision-making frameworks maintain safety and trustworthiness in complex driving environments.

C. THREATS TO PERCEPTION AND MULTIMODAL REPRESENTATION

Integrating LLMs with perception and representation pipelines offers richer spatial reasoning and multimodal alignment for autonomous driving, but it also introduces critical security vulnerabilities. 3D tokenization frameworks, such as Atlas [15], improve spatial grounding; however, they also heighten exposure to adversarial point cloud attacks and dataset poisoning, which could distort geometric priors. Occupancy-based reasoning systems, such as Occ-LLM [66], face challenges distinguishing between dynamic and static elements, as tampered occupancy maps or manipulated motion signals can lead to unsafe trajectory predictions. Multimodal approaches that fuse object-level vectors with LLM reasoning [14] are particularly susceptible to cross-modal adversarial perturbations, where inconsistencies between numeric and language inputs can be exploited to mislead decision-making. Similarly, situational awareness frameworks, such as SenserAG [16], which rely on real-time sensor fusion with retrieval-augmented generation (RAG), expose new attack surfaces through data injection, synchronization delays, and RAG poisoning, which could degrade responsiveness in critical moments. Membership inference and datastore leakage risks in multimodal RAG pipelines further raise concerns over privacy and confidentiality [157]. Finally, trajectory generation systems like Traj-LLM [72] amplify the risks of synthetic data poisoning, where adversaries might inject biased or unrealistic trajectories into training corpora, ultimately degrading the reliability of downstream predictions. These challenges underscore the necessity of secure sensor pipelines, adversarially robust multimodal fusion, and verifiable trajectory datasets to ensure that perception-driven LLM integration enhances rather than undermines the safety of autonomous systems.

D. PROPAGATION OF LLM-CENTRIC VULNERABILITIES INTO PHYSICAL AND COOPERATIVE AUTONOMY

While large language models enable high-level reasoning and coordination in autonomous systems, their integration into closed-loop planning and control pipelines introduces new classes of security risks that are fundamentally different from traditional cyber-physical attacks. In particular, vulnerabilities originating at the LLM or agent reasoning layer can directly influence physical actuation and propagate across cooperative multi-agent networks.

TABLE 14. Deploying LLMs on Drones, Edge, and cloud.

Model	Scale (Params)	Typical GPU/Hardware	Rationale for Feasibility
Onboard-feasible (≤7–8B)			
LLaMA 2–7B / 3–7B [135]	7B	Jetson Orin Nano 8–16GB / Snapdragon NPU [136], [137]	Quantized (∼3.5GB int4) fits on embedded GPUs; runs 10–15 tokens/s onboard. Practical for lightweight reasoning and NL interfaces.
Mistral 7B / 8B [138]	7–8B	Jetson Orin AGX / Qualcomm RB5 [136], [137]	Efficient dense models. Benchmarked at 14–28 tokens/s on Jetson Orin Nano. Suitable for onboard flight reasoning and instructions.
Qwen 2.5–7B [139]	7B	Jetson Orin [136]	Compact open-weight model; easily deployable with quantization on edge NPUs.
GPT-J 6B [140] / Dolly 6B [141]	6B	Jetson Xavier NX / Snapdragon Flight Pro [136], [137]	Legacy lightweight models. Already demonstrated on small NPUs and Jetson-class boards. Feasible for onboard real-time reasoning.
Cloud-connected via Edge GPUs (10–30B)			
LLaMA 3 13B [135]	13B	RTX 4090 (24GB) / A100 40GB [142], [143]	Quantized model fits 6–7GB VRAM, but drones cannot host directly. Can run on a ground station GPU and serve drones via Wi-Fi.
Vicuna 13B [144] / MPT-30B [145]	13B / 30B	RTX 4090 / A100 [142], [143]	Similar mid-size models. Typically requires a high-end single GPU. Drones connect to edge servers for access.
Cloud-only (≥70B and Frontier Models)			
LLaMA 3 70B [135]	70B	A100 80GB / H100 80GB / H200 [143], [146], [147]	Very large dense model. Quantized ∼35GB. Requires datacenter GPUs; not feasible for drones.
Bloom 176B [148]	176B	Multi-node A100 cluster [143]	Extremely large dense model. Too resource-intensive for edge or drones. Cloud inference only.
DeepSeek-V3 [149]	671B total (37B active)	8×H100 80GB [146], H200 [147]	MoE model with 37B active parameters per query. Too large for edge or onboard deployment. Cloud-only.
Mixtral-8×22B [150]	176B total (44B active)	4–8×H100 80GB / H200 [147]/ TPU v5 [146], [151]	Sparse MoE architecture; 44B active per input. Still well beyond drone hardware. Requires datacenter clusters.
Claude 3.5 Sonnet [152]	∼175B	8×A100 80GB / TPU v4 pods [143], [151]/ H200 [147]	Closed-source large model. Requires datacenter inference; API access only.
GPT-4 / GPT-4o [153]	∼200B–1T	Multi-GPU A100/H100/H200 [143], [146], [147]	Proprietary OpenAI models. Cannot be deployed locally. Cloud-only.
Kimi K2 [154]	1T (32B active)	8×H100 80GB [146]	MoE with ∼32B active params. Still requires large GPU clusters. Cloud-only.
Gemini Ultra [155]	∼1.5T (MoE)	TPU [151]	Google's top Gemini model. Trillion+ parameters. Datacenter-only.
GPT-5 (Orion) [156]	∼3.5T	Multi-node H100 / TPU v5 [146], [151]	Next-gen frontier model. Requires massive infrastructure. Cloud-only.

TABLE 15. Cross-layer propagation of LLM-centric threats into physical control and cooperative autonomy.

Threat Layer	Representative LLM-Centric Threats	Physical Control Impact	Multi-Agent Propagation Effect	Security Dimension
LLM Agent	Prompt injection, hallucinated actions, memory poisoning	Unsafe planning decisions entering closed-loop control	Propagation of corrupted intent or plans across agents	Trust, Security
Planning	Mission context forgery, goal manipulation	Constraint violations, unsafe trajectories	Coordinated mission drift in swarms or V2X systems	Security, Resilience
Control	Tool-call injection, failsafe suppression	Actuator saturation, loss of stability	Leader–follower instability and cascading failures	Safety, Resilience
Network / 6G Control Plane	Semantic message poisoning, URLLC micro-delays	Timing violations affecting control loops	Swarm-wide desynchronization and consensus failure	Resilience, Trust
Cross-Layer	Agent-to-agent prompt injection, long-horizon deception	Delayed or unsafe degradation responses	Amplified system-wide safety violations	Security, Resilience, Trust

Table 15, derived from the layered threat taxonomy and empirical findings of α^3-SecBench [101], explicitly illustrates how LLM-centric vulnerabilities propagate from semantic reasoning layers into physical control loops and cooperative multi-agent autonomy. Unlike traditional cyber-physical attacks that primarily target sensing or actuation, LLM-specific threats originate at the cognitive interface, which is responsible for goal interpretation, tool orchestration, and intent exchange. Once injected into planning or control pipelines, these semantic corruptions manifest as unsafe trajectories, constraint violations, or delayed safe-degradation behaviors. Moreover, in cooperative settings such as V2X systems and UAV swarms, compromised reasoning in a single agent can propagate through semantic communication and coordination protocols, amplifying local failures into system-wide safety risks.

E. RISKS IN HUMAN ALIGNMENT AND DRIVER INTERACTION

Efforts to align autonomous driving agents with human driving styles and interaction patterns pose critical challenges for safety, privacy, and robustness. Frameworks for driving style alignment [73] and generative agents trained on verbalized driver thought processes [53] risk amplifying subjective bias, as adversaries could poison demonstration datasets or manipulate verbal reports to misalign agent behavior with unsafe practices. Similarly, evaluation frameworks that integrate LLMs for assessing driving intelligence [58] depend heavily on subjective human judgments, making them vulnerable to biased scoring, data leakage, or adversarial manipulation of evaluation prompts. The use of conversational assistants for driver support, such as Driver Mate [74], introduces attack surfaces for distraction-based adversarial prompts,

malicious voice injection, or privacy breaches through continuous collection of physiological and behavioral data. Beyond technical vulnerabilities, the reliance on naturalistic driving experiments and self-reported data raises ethical and security concerns regarding informed consent, data ownership, and exposure of sensitive cognitive or behavioral information. Collectively, these risks underscore the need for privacy-preserving alignment protocols, adversarially robust evaluation methods, and secure safeguards for conversational AI to ensure that human-aligned LLM agents foster safety and trust rather than introduce new vectors of manipulation in autonomous driving systems.

F. SECURITY CHALLENGES IN UAV AND SWARM AUTONOMY

LLM-enabled UAV and swarm systems promise enhanced autonomy and collaboration, yet they introduce unique security and reliability risks. Collaborative inference frameworks, such as SwarmChain [20], distribute reasoning workloads across UAVs; however, adversaries can exploit communication bottlenecks, inject poisoned updates, or trigger denial-of-service (DoS) attacks in resource-constrained devices. Role-adaptive coordination models, such as RALLY [21], rely on structured natural-language exchanges between agents, creating vulnerabilities to semantic poisoning: adversarial prompts or corrupted role assignments can destabilize swarm behavior. LLM-assisted routing algorithms, such as LAURA [22], introduce new attack surfaces in evolutionary optimization, where adversarial sensor data or manipulated network conditions can mislead routing decisions and compromise information freshness. Similarly, wildfire monitoring systems such as FRSICL [23], which leverage in-context learning for real-time scheduling, are vulnerable to adversarial task injection or hallucination-driven errors under time-sensitive conditions. Tasking frameworks like LLVM-Drone [24], which translate natural language prompts into UAV executable code, expose mission-critical pipelines to prompt injection, adversarial instruction ambiguity, and sensor–language misalignment, thereby increasing the risk of unsafe or malicious task execution. Beyond these, federated and multi-agent learning settings [113] expose UAV swarms to contagious blocking or poisoning attacks that propagate across topologies. These challenges underscore the need for resilient distributed trust mechanisms, secure natural-language communication protocols, and adversarially robust task-execution frameworks to ensure that UAV and swarm systems can safely leverage LLM reasoning in dynamic, mission-critical environments.

G. SECURITY-ORIENTED EVALUATION OF LLM AGENTS

Despite rapid progress in driving theory MCQs and multimodal reasoning datasets, a critical gap remains: we lack security-centric benchmarks that directly evaluate LLM reasoning and agent behavior under adversarial and safety-critical conditions. Existing MCQ suites establish factual competence, but rarely probe threat-model awareness (e.g., sensor spoofing, poisoned trajectories, V2V misinformation), attack surface mapping, or policy robustness to red-team prompts and distribution shifts [94]. Edge and remote AD evaluations add useful cost and latency constraints, but still focus on static knowledge rather than defensive decision-making under time pressure [95]. Likewise, CoT-oriented and spatio-temporal benchmarks advance interpretability and temporal grounding, yet they seldom stress-test security reasoning chains (detect, diagnose, mitigate, recover) with graded severity and partial observability [34], [35].

For multi-agent autonomy, we require cooperative tasks in which models must identify, localize, and contain adversarial V2V signals, and negotiate safe plans under contested communication capabilities not currently captured by cooperative QA settings [17]. Protocol-level risks, such as speculative decoding side-channels or Web3 context manipulation [52], remain outside the scope of existing benchmarks. We therefore propose a next-generation benchmark family: (i) security-MCQ tracks spanning threat taxonomy, incident response, and standards compliance (with explanation audits); (ii) interactive agent-in-the-loop trials that inject adversarial events into simulators and require real-time mitigation; (iii) multimodal security items (camera, LiDAR, RADAR, V2X logs) paired with causal CoT rubrics; (iv) robustness scorecards that jointly measure safety, latency, compute, and cost under attack; and (v) reproducible red-team suites (prompt injections, sensor corruptions, communications jamming) with graded difficulty and sim-to-real diagnostics. Such benchmarks would let the community quantify and compare the security reasoning and defensive competence of LLMs and AI agents alongside traditional perception and planning metrics.

H. FUTURE DIRECTIONS AND CHALLENGES IN DRONE-LLM DEPLOYMENT

Table 14 presents a comparative view of large language models and their feasibility for deployment on drones, at the edge, or in the cloud. Compact models such as LLaMA 3–7B [135], Mistral 7B [138], Qwen 7B [139], GPT-J 6B [140], and Dolly 6B [141] can be deployed directly on-board drones or edge devices equipped with embedded accelerators like NVIDIA Jetson Orin [136] or Qualcomm RB5 [137]. With quantization techniques, these models achieve acceptable inference throughput while remaining within the power and memory budgets of UAV platforms. This enables offline reasoning for mission-critical tasks, such as navigation, obstacle avoidance, and emergency response, without relying on external connectivity.

By contrast, mid-size models such as Vicuna 13B [144], MPT-30B [145], and Mixtral 22B [150], as well as large-scale systems including BLOOM 176B [148], DeepSeek-V3 [149], Claude 3.5 [152], GPT-4 [153], Gemini Ultra [155], Kimi K2 [154], and GPT-5 [156], require more powerful hardware such as H200 [147], A100 [143], H100 [146], or Google TPU v5 [151]. These models typically run in the cloud and connect to drones via Wi-Fi or 5G for real-time reasoning

and decision-making. Although cloud-based deployment provides access to state-of-the-art reasoning capabilities, it introduces dependencies on connectivity and increases latency, which may be critical in highly dynamic or adversarial environments. As a result, future research should explore hybrid deployments that combine lightweight LLMs onboard for safety-critical autonomy with cloud-scale reasoning for complex planning and multimodal perception.

VI. CONCLUSION AND LESSONS LEARNED

In this paper, we present a comprehensive survey of the role of LLMs and AI agents in autonomous systems, focusing on applications, datasets, simulation platforms, and security challenges. By examining advances in autonomous driving and unmanned aerial vehicles, we demonstrated how LLMs are increasingly applied to scenario generation, decision-making, interpretability, multimodal reasoning, cooperative interactions, and swarm management. These findings highlight the potential of LLMs to bridge natural language intent with executable control tasks, offering a new paradigm for adaptive and human-aligned autonomy.

Our review of simulation platforms and benchmarks underscored their importance for reproducible experimentation and for exploring edge cases. Tools such as CARLA and AirSim, as well as synthetic environments like DriveDreamer-2, OmniDrive, and AeroVerse, have enabled researchers to validate reasoning, perception, and planning in diverse yet controlled conditions. Nevertheless, most platforms remain simulation-centric and insufficiently connected to real-world dynamics, underscoring that the future of LLM-based autonomy depends on hybrid frameworks that combine realism, scalability, and multi-agent negotiation across simulated and physical environments.

Another key lesson learned relates to the vulnerabilities and security risks inherent in LLM-driven autonomy. Our analysis showed that hallucination-induced control failures, adversarial perception attacks, privacy risks, and weaknesses in cooperative communication present significant barriers to safe deployment. This highlights the necessity of embedding verification, adversarial resilience, and secure communication protocols directly into the design pipeline. Equally important is the realization that Drone-LLM deployment whether on-board, at the edge, or in the cloud faces critical trade-offs between latency, energy efficiency, scalability, and security. Addressing these challenges requires carefully engineered architectures that balance performance and trustworthiness. Looking ahead, we envision a future where LLM-driven autonomous systems evolve into secure, trustworthy, and human-aligned collaborators capable of reasoning across modalities, coordinating seamlessly in multi-agent settings, and operating safely in real-world environments. Realizing this vision will not only advance the state of autonomy but also redefine the relationship between humans and intelligent machines.

REFERENCES

[1] Z. Guo, F. Tang, L. Luo, M. Zhao, and N. Kato, "A survey on applications of large language model-driven digital twins for intelligent network optimization," *IEEE Commun. Surveys Tuts.*, vol. 28, pp. 3388–3411, 2025.

[2] Y. Chen et al., "When large language models meet UAV projects: An empirical study from developers' perspective," 2025, *arXiv:2509.12795*.

[3] A. Plaat, A. Wong, S. Verberne, J. Broekens, N. van Stein, and T. Back, "Multi-step reasoning with large language models, a survey," 2024, *arXiv:2407.11511*.

[4] N. Yang, M. Fan, W. Wang, and H. Zhang, "Decision-making large language model for wireless communication: A comprehensive survey on key techniques," *IEEE Commun. Surveys Tuts.*, vol. 28, pp. 3055–3088, 2026.

[5] C. Cui et al., "A survey on multimodal large language models for autonomous driving," in *Proc. IEEE/CVF winter Conf. Appl. Comput. Vis.*, Jan. 2024, pp. 958–979.

[6] S. Fourati, W. Jaafar, N. Baccar, and S. Alfattani, "XLM for autonomous driving systems: A comprehensive review," 2024, *arXiv:2409.10484*.

[7] H. Zhou et al., "Large language model (LLM) for telecommunications: A comprehensive survey on principles, key techniques, and opportunities," *IEEE Commun. Surveys Tuts.*, vol. 27, no. 3, pp. 1955–2005, Jun. 2025.

[8] G. Qu, Q. Chen, W. Wei, Z. Lin, X. Chen, and K. Huang, "Mobile edge intelligence for large language models: A contemporary survey," *IEEE Commun. Surveys Tuts.*, vol. 27, no. 6, pp. 3820–3860, Dec. 2025.

[9] Y. Fu, R. Zha, P. Tian, and X. Di, "LLM-based realistic safety-critical driving video generation," 2025, *arXiv:2507.01264*.

[10] H. Tian et al., "An LLM-enhanced multi-objective evolutionary search for autonomous driving test scenario generation," 2024, *arXiv:2406.10857v1*.

[11] N. Petrovic et al., "LLM-driven testing for autonomous driving scenarios," in *Proc. 2nd Int. Conf. Found. Large Lang. Models (FLLM)*, Nov. 2024, pp. 173–178.

[12] Y. Zhang, J. Liu, C. Xu, P. Hang, and J. Sun, "LeAD: The LLM enhanced planning system converged with End-to-end autonomous driving," 2025, *arXiv:2507.05754*.

[13] C. Xu et al., "TeLL-drive: Enhancing autonomous driving with teacher LLM-guided deep reinforcement learning," 2025, *arXiv:2502.01387*.

[14] L. Chen et al., "Driving with LLMs: Fusing object-level vector modality for explainable autonomous driving," in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, May 2024, pp. 14093–14100.

[15] Y. Bai et al., "Is a 3D-tokenized LLM the key to reliable autonomous driving?" 2024, *arXiv:2405.18361*.

[16] X. Luo et al., "SenseRAG: Constructing environmental knowledge bases with proactive querying for LLM-based autonomous driving," in *Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis. Workshops (WACVW)*, Feb. 2025, pp. 899–906.

[17] H.-K. Chiu, R. Hachiuma, C.-Y. Wang, S. F. Smith, Y.-C. F. Wang, and M.-H. Chen, "V2V-LLM: Vehicle-to-vehicle cooperative autonomous driving with multimodal large language models," 2025, *arXiv:2502.09980*.

[18] C. Liu, G. Liu, Z. Wang, J. Yang, and S. Chen, "CoLMDriver: LLM-based negotiation benefits cooperative autonomous driving," 2025, *arXiv:2503.08683*.

[19] Z. Yan, H. Zhou, H. Tabassum, and X. Liu, "Hybrid LLM-DDQN-based joint optimization of V2I communication and autonomous driving," *IEEE Wireless Commun. Lett.*, vol. 14, no. 4, pp. 1214–1218, Apr. 2025.

[20] B. Han, Y. Chen, J. Li, J. Li, and J. Su, "SwarmChain: Collaborative LLM inference for UAV swarm control," *IEEE Internet Things Mag.*, vol. 8, no. 5, pp. 64–71, Sep. 2025.

[21] Z. Wang et al., "RALLY: Role-adaptive LLM-driven yoked navigation for agentic UAV swarms," 2025, *arXiv:2507.01378*.

[22] B. Wei, R. Zhang, R. Jiang, M. Peng, and D. Niyato, "LAURA: LLM-assisted UAV routing for AoI minimization," 2025, *arXiv:2503.23132*.

[23] Y. Emami, H. Zhou, M. G. Gaitan, K. Li, and L. Almeida, "FRSICL: LLM-enabled in-context learning flight resource allocation for fresh data collection in UAV-assisted wildfire monitoring," 2025, *arXiv:2507.10134*.

[24] Y. Hu et al., "LLVM-drone: A synergistic framework integrating large language models and vision models for visual tasks in unmanned aerial vehicles," *Knowl.-Based Syst.*, vol. 327, Oct. 2025, Art. no. 114190.

[25] S. Javaid, H. Fahim, B. He, and N. Saeed, "Large language models for UAVs: Current state and pathways to the future," *IEEE Open J. Veh. Technol.*, vol. 5, pp. 1166–1192, 2024.

[26] S. Long et al., "A survey on intelligent network operations and performance optimization based on large language models," *IEEE Commun. Surveys Tuts.*, vol. 27, no. 6, pp. 3915–3949, Dec. 2025.

[27] A. Dosovitskiy, G. Ros, F. Codevilla, A. M. López, and V. Koltun, "CARLA: An open urban driving simulator," in *Proc. Conf. Robot Learn.*, 2017, pp. 1–16.

[28] *Apollo: Open Source Autonomous Driving Platform*. Accessed: Sep. 26, 2025. [Online]. Available: https://apollo.auto/

[29] S. Shah, D. Dey, C. Lovett, and A. Kapoor, "AirSim: High-fidelity visual and physical simulation for autonomous vehicles," in *Proc. 11th Int. Conf. Field Service Robot.*, 2017, pp. 621–635.

[30] *Software in the Loop (SITL) Simulation*. Accessed: Sep. 26, 2025. [Online]. Available: https://ardupilot.org/dev/docs/sitl-simulator-software-in-the-loop.html

[31] H. Caesar et al., "NuScenes: A multimodal dataset for autonomous driving," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 11618–11628.

[32] T. Qian, J. Chen, L. Zhuo, Y. Jiao, and Y.-G. Jiang, "NuScenes-QA: A multi-modal visual question answering benchmark for autonomous driving scenario," in *Proc. AAAI Conf. Artif. Intell.*, 2023, pp. 4542–4550.

[33] Y. Wei et al., "Editable scene simulation for autonomous driving via collaborative LLM-agents," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 15077–15087.

[34] C. Fruhwirth-Reisinger, D. Malić, W. Lin, D. Schinagl, S. Schulter, and H. Possegger, "STSBench: A spatio-temporal scenario benchmark for multi-modal large language models in autonomous driving," 2025, *arXiv:2506.06218*.

[35] Z. Wei, C. Qiang, B. Jiang, X. Han, X. Yu, and Z. Han, "AD^2-bench: A hierarchical CoT benchmark for MLLM in autonomous driving under adverse conditions," 2025, *arXiv:2506.09557*.

[36] C. Sima et al., "DriveLM: Driving with graph visual question answering," in *Proc. Eur. Conf. Comput. Vis.*, 2023, pp. 256–274.

[37] J. Xiao et al., "UAV-ON: A benchmark for open-world object goal navigation with aerial agents," 2025, *arXiv:2508.00288*.

[38] M. Guo, M. Wu, J. He, S. Li, H. Li, and C. Tao, "BEDI: A comprehensive benchmark for evaluating embodied agents on UAVs," 2025, *arXiv:2505.18229*.

[39] G. Zhao et al., "DriveDreamer-2: LLM-enhanced world models for diverse driving video generation," in *Proc. AAAI Conf. Artif. Intell.*, 2025, vol. 39, no. 10, pp. 10412–10420.

[40] S. Wang et al., "OmniDrive: A holistic LLM-agent framework for autonomous driving with 3D perception, reasoning and planning," 2024, *arXiv:2405.01533v1*.

[41] F. Yao, Y. Yue, Y. Liu, X. Sun, and K. Fu, "AeroVerse: UAV-agent benchmark suite for simulating, pre-training, finetuning, and evaluating aerospace embodied world models," 2024, *arXiv:2408.15511*.

[42] M. Peng, Y. Xie, X. Guo, R. Yao, H. Yang, and J. Ma, "LD-scene: LLM-guided diffusion for controllable generation of adversarial safety-critical driving scenarios," 2025, *arXiv:2505.11247*.

[43] K. Lebioda, N. Petrovic, F. Pan, V. Zolfaghari, A. Schamschurko, and A. Knoll, "Are requirements really all you need? Using LLMs to generate configuration code: A case study in automotive simulations," *IEEE Access*, vol. 13, pp. 145115–145126, 2025.

[44] Y. Cheng, W. Zhang, Z. Zhang, C. Zhang, S. Wang, and S. Mao, "Toward federated large language models: Motivations, methods, and future directions," *IEEE Commun. Surveys Tuts.*, vol. 27, no. 4, pp. 2733–2764, Aug. 2025.

[45] Z. Yang, X. Jia, H. Li, and J. Yan, "LLM4Drive: A survey of large language models for autonomous driving," 2023, *arXiv:2311.01043*.

[46] Z. Yang et al., "AI-driven safety and security for UAVs: From machine learning to large language models," *Drones*, vol. 9, no. 6, p. 392, May 2025.

[47] Y. Zhu et al., "A survey on large language model-powered autonomous driving," *Engineering*, Aug. 2025.

[48] A. Stefanidou, E. Politi, and G. Dimitrakopoulos, "Foundation models in autonomous driving: A review of current tasks and applications," *IEEE Open J. Intell. Transp. Syst.*, vol. 6, pp. 1522–1538, 2025.

[49] M. F. A. M. S. Anuar, F. H. K. Zaman, S. A. B. C. Abdullah, K. M. Ng, K. N. Vijyakumar, and S. K. Ng, "Transformer architectures for distracted driving behavior detection: A comprehensive review of vision-based approaches," *IEEE Open J. Intell. Transp. Syst.*, vol. 7, pp. 233–267, 2026.

[50] R. Song, M. O. Ozmen, H. Kim, A. Bianchi, and Z. B. Celik, "Enhancing LLM-based autonomous driving agents to mitigate perception attacks," 2024, *arXiv:2409.14488*.

[51] X. Kong, T. Braunl, M. Fahmi, and Y. Wang, "A superalignment framework in autonomous driving with large language models," in *Proc. IEEE Intell. Vehicles Symp. (IV)*, Jun. 2024, pp. 1715–1720.

[52] M. A. Ferrag, N. Tihanyi, D. Hamouda, L. Maglaras, A. Lakas, and M. Debbah, "From prompt injections to protocol exploits: Threats in LLM-powered AI agents workflows," 2025, *arXiv:2506.23260*.

[53] Y. Jin et al., "SurrealDriver: Designing LLM-powered generative driver agent framework based on human drivers' driving-thinking data," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst. (IROS)*, Oct. 2024, pp. 966–971.

[54] Z. Chen et al., "HCRMP: A LLM-hinted contextual reinforcement learning framework for autonomous driving," 2025, *arXiv:2505.15793*.

[55] Y. Gong, J. Fan, R. Zhang, D. Niyato, Y. Yao, and X. Chang, "Safe and economical UAV trajectory planning in low-altitude airspace: A hybrid DRL-LLM approach with compliance awareness," 2025, *arXiv:2506.08532*.

[56] L. Zheng, J. He, S. Yu Chang, Y. Shen, and D. Niyato, "LLM meets the sky: Heuristic multi-agent reinforcement learning for secure heterogeneous UAV networks," 2025, *arXiv:2507.17188*.

[57] T. Cai et al., "Driving with regulation: Trustworthy and interpretable decision-making for autonomous driving with retrieval-augmented reasoning," 2024, *arXiv:2410.04759*.

[58] S. You, X. Luo, X. Liang, J. Yu, C. Zheng, and J. Gong, "A comprehensive LLM-powered framework for driving intelligence evaluation," 2025, *arXiv:2503.05164*.

[59] Y. Mei, T. Nie, J. Sun, and Y. Tian, "LLM-attacker: Enhancing closed-loop adversarial scenario generation for autonomous driving with large language models," 2025, *arXiv:2501.15850*.

[60] D. Fu et al., "Drive like a human: Rethinking autonomous driving with large language models," in *Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis. Workshops (WACVW)*, Jan. 2024, pp. 910–919.

[61] Y. Sun, N. Salami Pargoo, P. Jin, and J. Ortiz, "Optimizing autonomous driving for safety: A human-centric approach with LLM-enhanced RLHF," in *Proc. Companion ACM Int. Joint Conf. Pervasive Ubiquitous Comput.*, Oct. 2024, pp. 76–80.

[62] D. Xu et al., "LEAD: LLM-enhanced deep reinforcement learning for stable decision-making in critical autonomous driving scenarios," *Neurocomputing*, vol. 657, Dec. 2025, Art. no. 131619.

[63] L. Li, R. Tan, J. Fang, J. Xue, and C. Lv, "LLM-augmented hierarchical reinforcement learning for human-like decision-making of autonomous driving," *Expert Syst. Appl.*, vol. 294, Dec. 2025, Art. no. 128736.

[64] S. Wu et al., "Robust RL with LLM-driven data synthesis and policy adaptation for autonomous driving," 2024, *arXiv:2410.12568*.

[65] G. Li and B. Zhang, "Automated valet parking trajectory planning with dual-branch conditional transformer," *IEEE Open J. Intell. Transp. Syst.*, vol. 7, pp. 434–444, 2026.

[66] T. Xu, H. Lu, X. Yan, Y. Cai, B. Liu, and Y. Chen, "Occ-LLM: Enhancing autonomous driving with occupancy-based large language models," 2025, *arXiv:2502.06419*.

[67] D. P. Bavirisetti, H. R. Martinsen, G. H. Kiss, and F. Lindseth, "A multi-task vision transformer for segmentation and monocular depth estimation for autonomous vehicles," *IEEE Open J. Intell. Transp. Syst.*, vol. 4, pp. 909–928, 2023.

[68] Z. Wang, J. Guo, Z. Hu, H. Zhang, J. Zhang, and J. Pu, "Lane transformer: A high-efficiency trajectory prediction model," *IEEE Open J. Intell. Transp. Syst.*, vol. 4, pp. 2–13, 2023.

[69] M. J. Abyaneh and J. Jang, "Transformer-based trajectory prediction using LiDAR data for situational awareness in complex urban environments," *IEEE Open J. Intell. Transp. Syst.*, vol. 7, pp. 16–28, 2026.

[70] S. Park et al., "VLAAD: Vision and language assistant for autonomous driving," in *Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis. Workshops (WACVW)*, Jan. 2024, pp. 980–987.

[71] Y. Huang, J. Sansom, Z. Ma, F. Gervits, and J. Chai, "DriVLMe: Enhancing LLM-based autonomous driving agents with embodied and social experiences," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst. (IROS)*, Oct. 2024, pp. 3153–3160.

[72] K. Yang et al., "Back arrowGo to ICLR 2025 conference homepage trajectory-LLM: A language-based data generator for trajectory prediction in autonomous driving," in *Proc. 13th Int. Conf. Learning Representations (ICLR)*, 2025.

[73] R. Yang, X. Zhang, A. Fernandez-Laaksonen, X. Ding, and J. Gong, "Driving style alignment for LLM-powered driver agent," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst. (IROS)*, Oct. 2024, pp. 11318–11324.

[74] S. Huang, X. Zhao, D. Wei, X. Song, and Y. Sun, "Chatbot and fatigued driver: Exploring the use of LLM-based voice assistants for driving fatigue," in *Proc. Extended Abstr. CHI Conf. Hum. Factors Comput. Syst.*, May 2024, pp. 1–8.

[75] Q. Zhou, J. Wu, M. Zhu, Y. Zhou, F. Xiao, and Y. Zhang, "LLM-QL: A LLM-enhanced Q-learning approach for scheduling multiple parallel drones," *IEEE Trans. Knowl. Data Eng.*, vol. 37, no. 9, pp. 5393–5406, Sep. 2025.

[76] X. Hou et al., "DriveAgent: Multi-agent structured reasoning with LLM and multimodal sensor fusion for autonomous driving," 2025, *arXiv:2505.02123*.

[77] M. Arslan, L. Mahdjoubi, and S. Munawar, "Driving sustainable energy transitions with a multi-source RAG-LLM system," *Energy Buildings*, vol. 324, Dec. 2024, Art. no. 114827.

[78] C. Chakraborttii and L. Cheng, "TransPark: Leveraging transfer learning with transformers for intelligent parking systems," *IEEE Open J. Intell. Transp. Syst.*, vol. 7, pp. 166–189, 2026.

[79] D. C. Larese, A. B. Cerrada, G. D. Tomei, A. Guerrero-López, P. M. Olmos, and M. J. G. García, "Transformer vibration forecasting for advancing rail safety and maintenance 4.0," *IEEE Open J. Intell. Transp. Syst.*, vol. 6, pp. 1612–1624, 2025.

[80] J. Zhou, Y. Zhao, A. Yang, and A. Eichberger, "Benchmarking large language models for motorway driving scenario understanding," in *Proc. SAE Tech. Paper Ser.*, vol. 1, Feb. 2025.

[81] H.-Y. Jung, D.-H. Paek, and S.-H. Kong, "Open-source autonomous driving software platforms: Comparison of autoware and Apollo," 2025, *arXiv:2501.18942*.

[82] *Waymo Open Dataset: Autonomous Driving Data*. Accessed: Sep. 26, 2025. [Online]. Available: https://waymo.com/open/

[83] F. Jiang, C. Pan, L. Dong, K. Wang, O. A. Dobre, and M. Debbah, "From large AI models to agentic AI: A tutorial on future intelligent communications," 2025, *arXiv:2505.22311*.

[84] Z. Qu, W. Wang, Z. Yu, B. Sun, Y. Li, and X. Zhang, "LLM enabled multi-agent system for 6G networks: Framework and method of dual-loop edge-terminal collaboration," 2025, *arXiv:2509.04993*.

[85] K. Team et al., "Kimi k2: Open agentic intelligence," 2025, *arXiv:2507.20534*.

[86] *Introduction to MCP*. Accessed: Feb. 1, 2026. [Online]. Available: https://modelcontextprotocol.io/introduction

[87] M. Seegert, K. Moller, and J. Betz, "Modular autonomy with conversational interaction: An LLM-driven framework for decision making in autonomous driving," 2026, *arXiv:2601.05806*.

[88] X. Hou, Y. Zhao, S. Wang, and H. Wang, "Model context protocol (MCP): Landscape, security threats, and future research directions," 2025, *arXiv:2503.23278*.

[89] *A2A: A New Era of Agent Interoperability*. Accessed: Feb. 1, 2026. [Online]. Available: https://developers.googleblog.com/en/a2a-a-new-era-of-agent-interoperability/

[90] K. Xiao, H. Zhang, R. Mao, H. Li, and Z. X. Chen, "Towards LLM-enabled autonomous combustion research: A literature-aware agent for self-corrective modeling workflows," 2026, *arXiv:2601.01357*.

[91] S. Guo et al., "A survey on semantic communication networks: Architecture, security, and privacy," *IEEE Commun. Surveys Tuts.*, vol. 27, pp. 2860–2894, 2024.

[92] Y. Wang et al., "Internet of Agents: Fundamentals, applications, and challenges," 2025, *arXiv:2505.07176*.

[93] Y. Yang et al., "Agentic web: Weaving the next web with AI agents," 2025, *arXiv:2507.21206*.

[94] D. Pei et al., "Methodology and benchmark for automated driving theory test of large language models," *IEEE Trans. Intell. Transp. Syst.*, vol. 26, no. 10, pp. 17013–17026, Oct. 2025.

[95] Z. Tang, J. He, D. Pe, K. Liu, T. Gao, and J. Zheng, "Test large language models on driving theory knowledge and skills for connected autonomous vehicles," in *Proc. Workshop Mobility Evolving Internet Archit.*, Nov. 2024, pp. 1–6.

[96] B. Zhao et al., "UrbanVideo-bench: Benchmarking vision-language models on embodied intelligence with video data in urban spaces," 2025, *arXiv:2503.06157*.

[97] A.-M. Marcu et al., "LingoQA: Visual question answering for autonomous driving," in *Proc. Eur. Conf. Comput. Vis.*, 2023, pp. 252–269.

[98] X. Cao et al., "MAPLM: A real-world large-scale vision-language benchmark for map and traffic scene understanding," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2024, pp. 21819–21830.

[99] Y. Ma et al., "LaMPilot: An open benchmark dataset for autonomous driving with language model programs," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 15141–15151.

[100] M. Amine Ferrag, A. Lakas, and M. Debbah, "α^3-bench: A unified benchmark of safety, robustness, and efficiency for LLM-based UAV agents over 6G networks," 2026, *arXiv:2601.03281*.

[101] M. Amine Ferrag, A. Lakas, and M. Debbah, "α^3-SecBench: A large-scale evaluation suite of security, resilience, and trust for LLM-based UAV agents over 6G networks," 2026, *arXiv:2601.18754*.

[102] Y. Yao et al., "AGENTS-LLM: Augmentative GENeration of challenging traffic scenarios with an agentic LLM framework," 2025, *arXiv:2507.13729*.

[103] X. Wang et al., "Towards realistic UAV vision-language navigation: Platform, benchmark, and methodology," 2024, *arXiv:2410.07087*.

[104] M. A. Ferrag, A. Lakas, and M. Debbah, "UAVBench: An open benchmark dataset for autonomous and agentic AI UAV systems via LLM-generated flight scenarios," 2025, *arXiv:2511.11252*.

[105] I. Nakash, G. Kour, K. Lazar, M. Vetzler, G. Uziel, and A. Anaby-Tavor, "Effective red-teaming of policy-adherent agents," 2025, *arXiv:2506.09600*.

[106] M. Pavlova et al., "Automated red teaming with GOAT: The generative offensive agent tester," 2024, *arXiv:2410.01606*.

[107] A. Zhou et al., "AutoRedTeamer: Autonomous red teaming with lifelong attack integration," 2025, *arXiv:2503.15754*.

[108] S. Schoepf et al., "MAD-MAX: Modular and diverse malicious attack MiXtures for automated LLM red teaming," 2025, *arXiv:2503.06253*.

[109] Y. Chen, X. Hu, K. Yin, J. Li, and S. Zhang, "AEIA-MN: Evaluating the robustness of multimodal LLM-powered mobile agents against active environmental injection attacks," 2025, *arXiv:2502.13053*.

[110] Y. Wang, Y. Mo, Y. Wang, and Z. Wei, "Fight back against jailbreaking via prompt adversarial tuning," in *Proc. Adv. Neural Inf. Process. Syst.*, 2024, pp. 64242–64272.

[111] Y. Liu, C. Cai, X. Zhang, X. Yuan, and C. Wang, "Arondight: Red teaming large vision language models with auto-generated multi-modal jailbreak prompts," in *Proc. 32nd ACM Int. Conf. Multimedia*, Oct. 2024, pp. 3578–3586.

[112] C. Jie Chong, C. Hou, Z. Yao, and S. M. S. Talebi, "Casper: Prompt sanitization for protecting user privacy in Web-based large language models," 2024, *arXiv:2408.07004*.

[113] Z. Zhou et al., "CORBA: Contagious recursive blocking attacks on multi-agent systems based on large language models," 2025, *arXiv:2502.14529*.

[114] M. Kumar, J. Xue, M. Zheng, and Q. Lou, "TFHE-coder: Evaluating LLM-agentic fully homomorphic encryption code generation," 2025, *arXiv:2503.12217*.

[115] N. Tihanyi et al., "Dynamic intelligence assessment: Benchmarking LLMs on the road to AGI with a focus on model confidence," in *Proc. IEEE Int. Conf. Big Data (BigData)*, Dec. 2024, pp. 3313–3321.

[116] C. Cui et al., "LLM4AD: Large language models for autonomous driving - concept, review, benchmark, experiments, and future trends," 2024, *arXiv:2410.15281*.

[117] Z. Xu et al., "DriveGPT4: Interpretable end-to-end autonomous driving via large language model," *IEEE Robot. Autom. Lett.*, vol. 9, no. 10, pp. 8186–8193, Oct. 2024.

[118] A. Alhammadi, A. Abraham, and Q. Zhao, "LLM-powered UAV automations for city-wide operations," in *Internet of Vehicles and Computer Vision Solutions for Smart City Transformations*. Cham, Switzerland: Springer, 2025, pp. 69–83.

[119] T. M. Nguyen, V. T. Truong, and L. B. Le, "Agentic AI meets edge computing in autonomous UAV swarms," 2026, *arXiv:2601.14437*.

[120] Z. Yan, H. Zhou, J. Pei, and H. Tabassum, "Hierarchical and collaborative LLM-based control for multi-UAV motion and communication in integrated terrestrial and non-terrestrial networks," 2025, *arXiv:2506.06532*.

[121] J. Wei et al., "Chain-of-thought prompting elicits reasoning in large language models," in *Proc. Adv. neural Inf. Process. Syst.*, vol. 35, 2022, pp. 24824–24837.

[122] S. Yao et al., "ReAct: Synergizing reasoning and acting in language models," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2023.

[123] M. Mohammadi, Y. Li, J. Lo, and W. Yip, "Evaluation and benchmarking of LLM agents: A survey," in *Proc. 31st ACM SIGKDD Conf. Knowl. Discovery Data Mining*, Aug. 2025, pp. 6129–6139.

[124] H. Jiang et al., "LongLLMLingua: Accelerating and enhancing LLMs in long context scenarios via prompt compression," in *Proc. 62nd Annu. Meeting Assoc. Comput. Linguistics (Long Papers)*, 2024, pp. 1658–1677.

[125] Z. Zhou et al., "ResMAS: Resilience optimization in LLM-based multi-agent systems," 2026, *arXiv:2601.04694*.
[126] H. Luo et al., "Toward edge general intelligence with multiple-large language model (multi-LLM): Architecture, trust, and orchestration," *IEEE Trans. Cognit. Commun. Netw.*, vol. 11, no. 6, pp. 3563–3585, Dec. 2025.
[127] Y. Gao et al., "Foundation models in autonomous driving: A survey on scenario generation and scenario analysis," 2025, *arXiv:2506.11526*.
[128] Q. Song, H. Ye, M. Harman, and F. Sarro, "Generative AI for testing of autonomous driving systems: A survey," 2025, *arXiv:2508.19882*.
[129] T. Zhang et al., "Bench2ADVLM: A closed-loop benchmark for vision-language models in autonomous driving," 2025, *arXiv:2508.02028*.
[130] P. Wang et al., "Qwen2-VL: Enhancing vision-language model's perception of the world at any resolution," 2024, *arXiv:2409.12191*.
[131] J. Liu et al., "Language-driven policy distillation for cooperative driving in multi-agent reinforcement learning," *IEEE Robot. Autom. Lett.*, vol. 10, no. 5, pp. 4292–4299, May 2025.
[132] Y. Zhang et al., "Towards efficient LLM grounding for embodied multi-agent collaboration," in *Proc. Findings Assoc. Comput. Linguistics ACL*, 2025, pp. 1663–1699.
[133] J. Betz et al., "Autonomous vehicles on the edge: A survey on autonomous vehicle racing," *IEEE Open J. Intell. Transp. Syst.*, vol. 3, pp. 458–488, 2022.
[134] Y. Yang, R. Gao, X. Wang, T.-Y. Ho, N. Xu, and Q. Xu, "MMA-diffusion: MultiModal attack on diffusion models," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 7737–7746.
[135] M. AI. *Llama 3*. Accessed: Oct. 2, 2025. [Online]. Available: https://www.llama.com/models/llama-3/
[136] NVIDIA. *Nvidia Jetson Orin*. Accessed: Oct. 2, 2025. [Online]. Available: https://developer.nvidia.com/embedded/jetson-orin
[137] Qualcomm. *Qualcomm Flight Rb5 5G Platform*. Accessed: Oct. 2, 2025. [Online]. Available: https://www.qualcomm.com/products/internet-of-things/robotics-processors/flight-rb5-platform
[138] M. AI. *Mistral 7b*. Accessed: Oct. 2, 2025. [Online]. Available: https://huggingface.co/mistralai/Mistral-7B-v0.1
[139] B. Hui et al., "Qwen2.5-coder technical report," 2024, *arXiv:2409.12186*.
[140] B. Wang and A. Komatsuzaki. *GPT-J-6B: A 6 Billion Parameter Autoregressive Language Model*. Accessed: Oct. 2, 2025. [Online]. Available: https://github.com/kingoflolz/mesh-transformer-jax
[141] Databricks. *Dolly 6b*. Accessed: Oct. 2, 2025. [Online]. Available: https://huggingface.co/databricks/dolly-v1-6b
[142] NVIDIA. (4090). *Nvidia Rtx 4090*. Accessed: Oct. 2, 2025. [Online]. Available: https://www.nvidia.com/en-us/geforce/graphics-cards/40-series/rtx-4090
[143] NVIDIA. *Nvidia A100 Tensor Core Gpu*. Accessed: Oct. 2, 2025. [Online]. Available: https://www.nvidia.com/en-us/data-center/a100
[144] L. Org. *Vicuna 13b*. Accessed: Oct. 2, 2025. [Online]. Available: https://huggingface.co/lmsys/vicuna-13b-v1.3
[145] MosaicML. *Mpt-30b*. Accessed: Oct. 2, 2025. [Online]. Available: https://huggingface.co/mosaicml/mpt-30b
[146] NVIDIA. *Nvidia H100 Tensor Core Gpu*. Accessed: Oct. 2, 2025. [Online]. Available: https://www.nvidia.com/en-us/data-center/h100
[147] NVIDIA. *Nvidia H200 Tensor Core Gpu*. Accessed: Oct. 2, 2025. [Online]. Available: https://www.nvidia.com/en-us/data-center/h200/
[148] B. Workshop. *Bloom 176b*. Accessed: Oct. 2, 2025. [Online]. Available: https://huggingface.co/bigscience/bloom
[149] A. Liu et al., "DeepSeek-V3 technical report," 2024, *arXiv:2412.19437*.
[150] M. AI. *Mixtral 8x22b*. Accessed: Oct. 2, 2025. [Online]. Available: https://huggingface.co/mistralai/Mixtral-8x22B-v0.1
[151] Google. *Google Cloud TPU V5*. Accessed: Oct. 2, 2025. [Online]. Available: https://cloud.google.com/tpu
[152] Anthropic. *Claude 3.5 Sonnet*. Accessed: Oct. 2, 2025. [Online]. Available: https://www.anthropic.com/news/claude-3-5-sonnet
[153] OpenAI. *GPT-4 and GPT-4O*. Accessed: Oct. 2, 2025. [Online]. Available: https://openai.com/index/gpt-4-1/
[154] M. AI. *Kimi K2*. Accessed: Oct. 2, 2025. [Online]. Available: https://moonshotai.github.io/Kimi-K2/
[155] G. DeepMind. *Gemini Ultra*. Accessed: Oct. 2, 2025. [Online]. Available: https://deepmind.google/models/gemini/
[156] OpenAI. *GPT-5*. Accessed: Oct. 2, 2025. [Online]. Available: https://openai.com/gpt-5/
[157] Y. Li, G. Liu, Y. Yang, and C. Wang, "Seeing is believing: Black-box membership inference attacks against retrieval augmented generation," 2024, *arXiv:2312.08207*.



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